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Submitter Information

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[REDACTED]

General Comment

Hello, I'm writing to you as a writer, editor and fiber artist that is against our government investing in generative AI. Generative AI models that are currently out there are trained on copyright protected works: books, articles, illustrations that were stolen without any consideration for the people who created these works. The companies did not compensate the people without whose work their generative AI models would not be able to generate anything. My partner's painting were certainly scrapped and used without notice or solicitation of consent. If you take away the training data, the models cannot function--they cannot generate anything; this makes the training data crucial. Companies like OpenAI have said time and again that if they are made to actually pay the people whose work makes their models functional, they will go bankrupt (<https://arstechnica.com/tech-policy/2025/03/openai-urges-trump-either-settle-ai-copyright-debate-or-lose-ai-race-to-china/>). Companies like Meta have harvested the work of people without consent or compensation, using methods that have been illegal in the United States for decades--pirating books and infringing on the right of published authors and publishers, as detailed in this article: <https://arstechnica.com/tech-policy/2025/03/meta-mocked-for-raising-bob-dylan-defense-of-torrenting-in-ai-copyright-fight/> . If you cannot make your commercial product without using other people's intellectual property, you should pay the people whose work makes your product possible and the government should not be empowering you and ripping away copyright protection from literally millions of writers and artists to prop up your business.

The United States government should be protecting American creative industries, supporting the artists and writers, who pour their soul into their work not upending the already flimsy and difficult to enforce (unless you have money and access to stellar legal representation) copyright protections we barely enjoy. These models compete directly with our work while being trained on work that was stolen from us and now they are trying to say that it's okay for them to steal our work to develop their product, never compensate us, and that they should be allowed to do this to all creatives in all creative fields in perpetuity? How is that fair or right? How does that support the pursuit of liberty and happiness for anyone but these companies? If these companies are forced to pivot to something else, they will and they will be fine; people in creative fields have spent decades honing their craft and skill into a livelihood will not be fine. Why should the government endorse creatives being pushed out of their jobs just so that a handful of companies can make record profits. Every artist, writers, animator, filmmaker, actor, etc, who this forces out of a job is another person who will need to seek welfare. This isn't good for our economy or the vast majority of people who work in creative industries. Do not give these companies a reward from ripping off regular people who are just trying to make a living with the skills they've spent most of their lives honing.

People do not reach for a video game or a book or go to see a Picasso exhibit just to consume something, they want to connect with and support other people and genAI robs people of that true connection all the while pushing out the creatives it sucked dry to manifest.

And that's before we get into the environmental and energy consumption (<https://www.bloomberg.com/graphics/2024-ai-data-centers-power-grids/>). Long story short, data centers for generative AI research draw so much power, the likes of Microsoft was looking into acquiring their own dedicated nuclear power plant. Our power grid is old and straining--we need to invest in its repair so regular everyday people do not die of heatstroke in summer (Pacific Northwest) and freeze to death in winter (Texas). The government should not place the desires of billionaire company owners for more electricity and water above the needs of everyday people trying to light, heat and cool their homes and be able to have enough water to drink, to cook with, to wash their kids hair with.

Attachments

harm and hypocrisy of ai art - matt corral

why we don't know ai's true water footprint - techpolicy press

making_ai_less_thirsty_ai_water_consumption_analysis

California wildfires raise alarm on water-guzzling AI like ChatGPT _ Fortune

What the data center boom in Texas means for the grid _ The Texas Tribune

kicking datacenter's drinking habit is nearly impossible - the register

how rise in ai impacts data centers and the environment - techtarget

ai uses more energy and water than Google search without ai

as use of ai soars o does energy and water it requires yalee360

ERCOT overcharged for electricity in Texas by 16billion during freeze - texas tribune

Texas Power Grid Run by ERCOT Set Up the State for Disaster - The New York Times

ERCOT overcharged for electricity in Texas during freeze - Texas tribune

The Texas Electric Grid Failure Was a Warm-up-Texas Monthly

ai-insatiable-need-for-energy-straining-global-power-grids

meta_mocked_for_raising_Bob_Dylan_defense_torrenting_in_ai_copyright_fight

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Matt Corral



The harm & hypocrisy of AI art

“How easy it is to create and maintain the illusion of understanding, hence perhaps of judgement deserving of credibility... A certain danger lurks there.”

— Joseph Weizenbaum

(Disclaimer: The views expressed here are my own personal opinions, and not those of Sony Interactive Entertainment)

For anyone who works in tech, generative artificial intelligence (AI) has been the hot topic of the last couple of years. We’ve seen an explosion of interest around the tech world, with companies big and small scrambling to adopt AI tools and weave AI features into their software. Everyone is seemingly terrified of missing out on what we’re told is the next big thing; even Photoshop - the old reliable friend of graphic design - includes AI tools, now. Stories appear daily about how AI will change society and propel us all into a brighter future, their writers proclaiming a raft of utopian assertions from the end of poverty to making everyone an artist.

I’ve started to see AI image generators and other tools popping up with startling frequency in the daily work and discussions I have with my peers. Engineers ask [ChatGPT](#) to write code, managers use [Midjourney](#) to fill their PowerPoints with bespoke images, and even designers ask [Dall-E](#) to create UI concepts for them. Nearly everyone seems awed by the novelty and magic of this new toy, and I often hear the same mantra - everyone’s job will soon rely on this. Embrace it, or risk being left behind.



UI mockups auto-generated by Midjourney. Source: <https://t.ly/B6HIH>

So I started digging into both how AI works, and what impact it's having, particularly on the creative industry I'm proud to be a part of. This article has been a long time coming for me. The more I've learned, the more I've become first alarmed, then angry, then compelled to speak out. As a designer in the tech world, I want to put forward a different view to one we usually hear, and to explain clearly why I for one, am appalled by the arrival of AI.

AI in 2024

Machine or deep learning models - that which we now brand AI - have been around in a basic form since the 1950s, but thanks to recent advancements in GPUs, have made a huge leap forward in capacity. What separates AI from say, the cloudy metaverse visions of last year, is that AI is already shipping and impacting our lives for the worse - even if it's not quite the stuff of everyday pub debate.

Using generative AI tools like [ChatGPT](#), and [Stable Diffusion](#), people can type a request into a browser in straightforward English, and get back a whatever they asked for - be that a recipe, an illustration, or even a faked photo or video. With gadgets like the [Humane AI Pin](#), people can also do the same by speaking out loud, in a manner reminiscent of the sci-fi movie, [Her](#).



The Humane AI Pin. Source: <https://shorturl.at/qWZ58>

When using an AI generator the images and articles that come back are often uncanny and strange, but still strikingly realistic - good enough to pass for something written, painted or photographed by a real person. Yet it's in the details where they still fall over - at least for now - showing hands with seven fingers, writing repetitive prose, or drawing what should be text as gibberish symbols. Despite the limitations, the appeal of cheap, instantly available art and writing holds obvious appeal for the commercial world, and AI companies have quickly amassed colossal profits. [OpenAI](#) for example, making an incredible [\\$1.6 billion](#) last year.

How it works

What we call AI today is nothing like the sentient robots of sci-fi movies - an 'AI' is a mathematical model, written in computer code, that is good at spotting patterns and correlations in data. In most cases, 'data' means a big pile of text or images - millions of photos or online articles, for example. AI companies can feed their models huge libraries of photos or books, and by analysing everything it's fed, the model can then generate its own versions

which look pretty close to the real thing. At first the model is likely to produce complete garbage, but through a trial and error process can be ‘trained’ by an engineer to generate some impressive results.



The sample image Stable Diffusion currently use to introduce their AI tool. Source: <https://shorturl.at/kuvFH>

Training a model requires a vast amount of ‘data,’ and also comes at a not insignificant cost in terms of energy, carbon emissions and human labour. All that ‘data’ has so far mostly been scraped from the internet - taken in secret from people who didn’t know and didn’t consent to handing it over.

Low-paid workers are used to manually filter offensive and upsetting images out of the datasets, and to type out description tags for the millions of images, so that the machine knows what it’s looking at. Programmes like Amazon’s [Mechanical Turk](#) - which pays a pittance to people for each image they tag - hides struggling workers behind a curtain whilst giving the

impression of a shiny, automated front-end.



Thanks to machine-learning algorithms,
the robot apocalypse was short-lived.

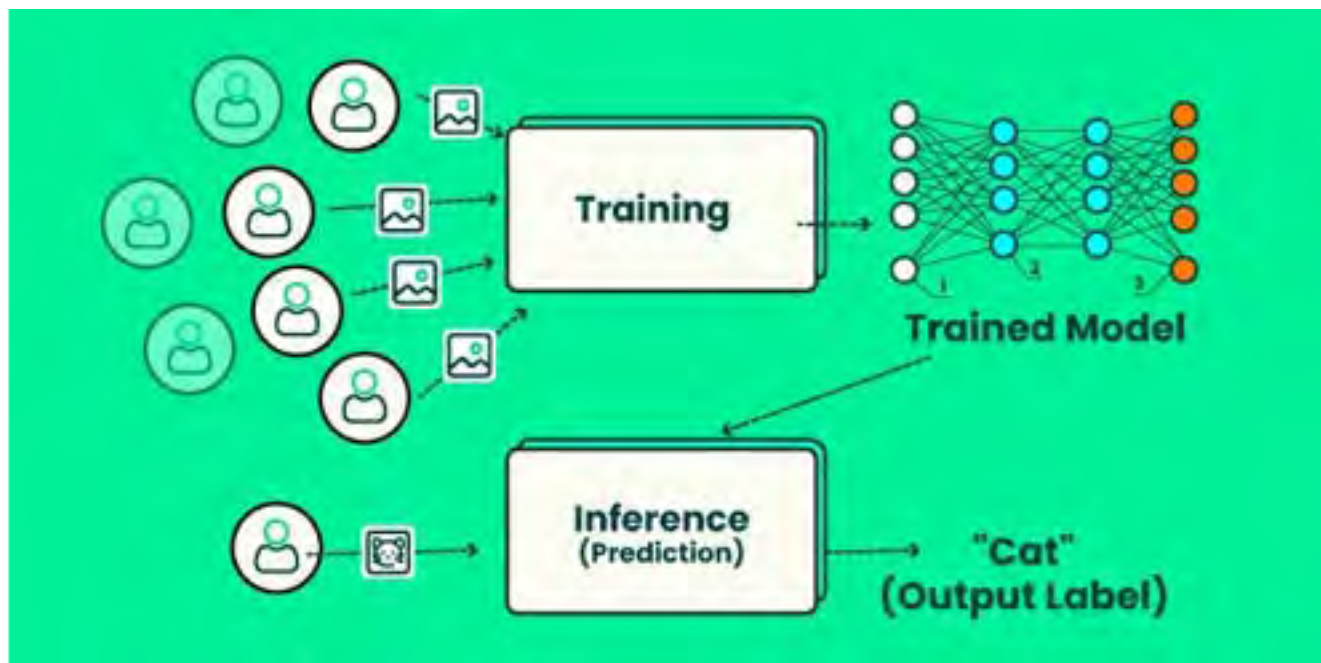
Source: <https://www.smbc-comics.com/comic/rise-of-the-machines>

What AI can and can't do

Everything an AI is fed must first be categorised and converted to numerical data, in order to be consumed. There is a process of abstraction and reduction that must happen, turning a digital image into a numerical matrix of RGB pixel colours and positions. This abstraction means a whole lot of nuance, detail and context that we understand as humans - anything that can't be easily quantified - is lost as data is fed into the machine. The Mona Lisa becomes just more numbers in the pile.

AI will also amplify any inherent bias in its training dataset. For example, if a model is trained on a police database and told to suggest jail sentences in a courtroom - [something which is already happening in the US](#) - it will have inherited any racial bias that was present in the records of arrests, and be more likely for example, to suggest harsher sentences for black defendants. Famously too, if one prompts Dall-E with the word 'CEO,' it will only produce images of white men.

AI models are highly dependent on the input dataset - what it contains, what's conspicuously absent, who curated it and to what end, all affect the results. Without any governance whatsoever, that means Silicon Valley tech companies get to decide exactly which way the models lean and what they do - and the more widely AI is used, the more their particular world view and biases propagate.

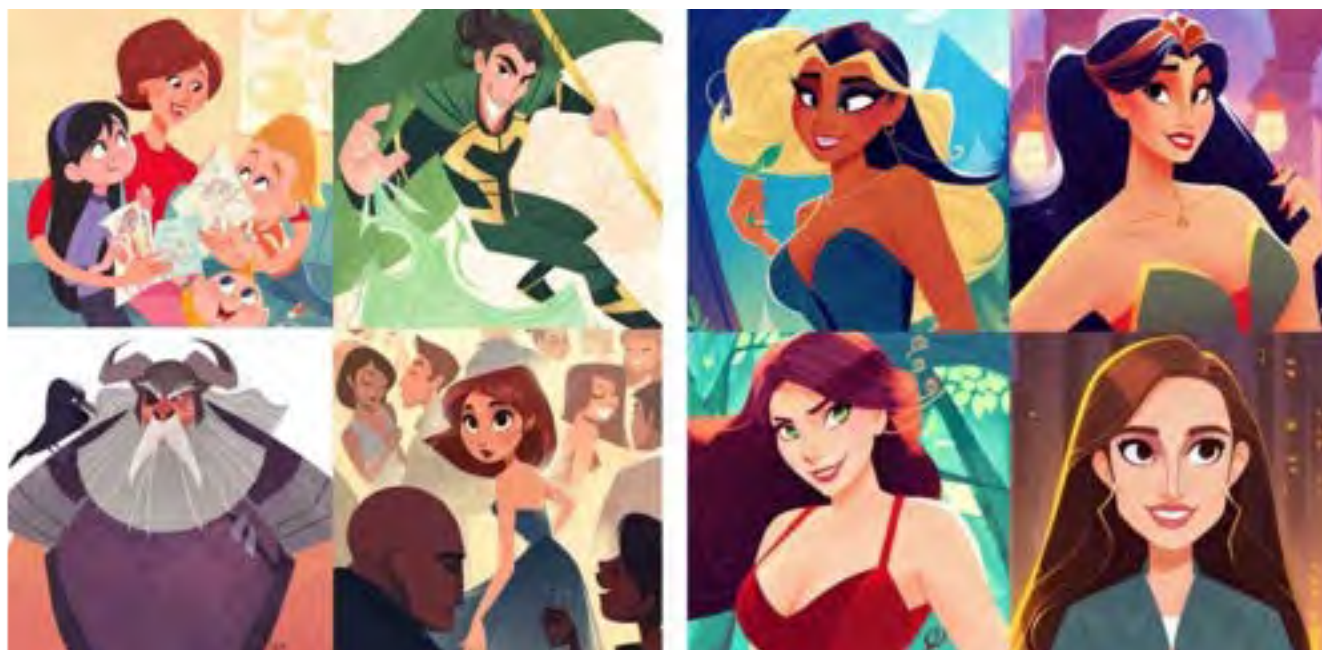


The process of training an AI model on vast quantities of input data

Source: <https://forbytes.com/blog/ai-models-explained>

When someone asks the machine to generate an image, the model scans over everything in its dataset and makes a calculation - out of everything it has, which combination of colours, shapes and lines best fits the request? It's a mathematical brute force approach, not unlike the classic adage of infinite monkeys at typewriters - feed the AI enough data and it will eventually give you something about right. The machine doesn't draw or paint - it can only break down its dataset into billions of parts, and recombine them to deliver mash-ups of what came before, as strange and uncanny as they may be.

In this way, an AI model is technically incapable of producing anything new. It's limited view of the world is based entirely on the abstracted number set it was given. An AI model amazes only because it has devoured so much raw material, to remix at your behest. As clever as it may be at spotting patterns, it cannot adapt, interpret or imagine like a human being can. What it can do very well however, is copy an artists' style or fake a photo with disturbing accuracy.



Illustrations by artist [Hollie Mengert](#) (left) & AI output based on her stolen work (right)

Source: <https://thealgorithmicbridge.substack.com/p/why-generative-ai-angers-artists>



Entirely AI-generated fake photos - alarmingly difficult to tell from the real thing

Part of the branding exercise around AI tools has been about building mystique. AI enthusiasts talk about its limitless potential to transform society for the better, by taking over more and more decisions for us. There is an almost religious zeal for this in some quarters, from which predictions flow of utopian futures where all problems are solved through learning to embrace the wisdom that flows from these machines; but as we're seeing, these machines are far cruder and more flawed than the hype suggests.

"Generative AI is the key to solving some of the world's biggest problems, such as climate change, poverty, and disease. It has the potential to make the world a better place for everyone"

— Meta CEO, Mark Zuckerberg

One aspect of AI's working which perhaps fuels these beliefs, is the opaqueness of its

decisions. As models are trained, they make myriad correlations and connections which are never fully visible to the engineers. Whilst not truly a 'black box,' models nevertheless produce output without us fully knowing what's going on inside. This lack of accountability means it would be unforgivable to say, trust them with decisions that affect people's lives. By the same token, the box of mystery can appear as if magical - the machine really looks like it's thinking and understanding, prompting some to proclaim that in time anything is possible. AI is not intelligent, however. It blindly connects and calculates, no matter how misguided or illicit the task.



For now, AI-generated images can be spotted via the mistakes they make with fine details - such as human fingers - but we can expect this to change

Source: <https://t.ly/HPduD>

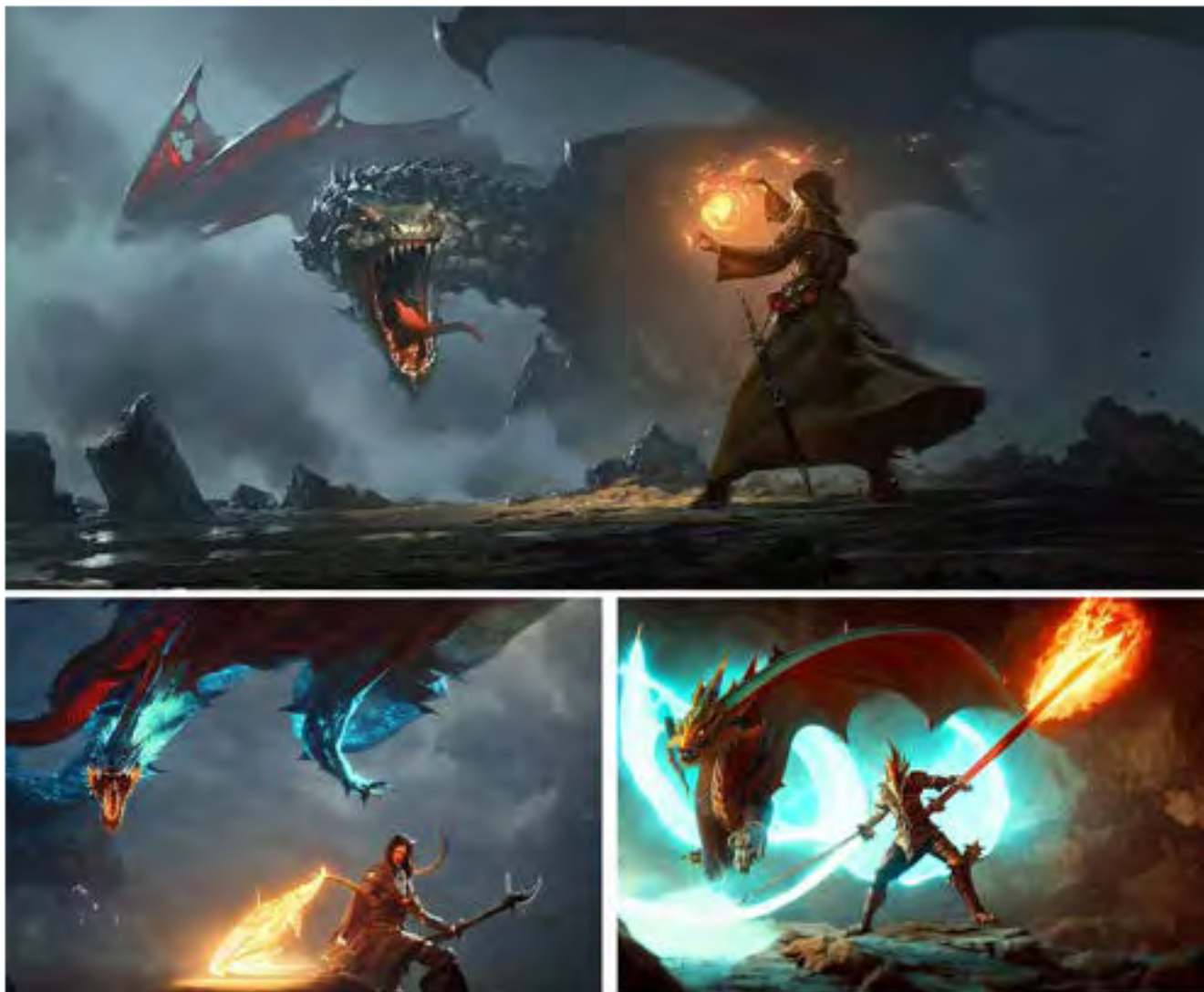
"Generative A.I. art is vampirical - feasting on past generations of artwork even as it sucks the lifeblood from living artists. Over time, this will impoverish our visual culture."

— Molly Crabapple

The Vampire Machine

It's important to understand that [OpenAI](#) (creator of Dalle-E), [StabilityAI](#) (creator of Stable Diffusion) and [Midjourney](#) - the big three image generators - all trained their successful models by scraping millions of other peoples' images from the internet - apparently entirely without the owners' knowledge or permission. Lawsuits surrounding this are still ongoing.

This is worth reiterating: The billion-dollar-making generators we see today appear trained on the copyrighted works of far poorer artists, illustrators and photographers; taken directly from their portfolios and community sites like [DeviantArt](#). This is copyright infringement on a completely unprecedented scale, and in my opinion corrupt, cynical and immoral. Paying users can even directly prompt the image generators to produce artwork in the style of an artist by typing their name - making no secret of the fact that their work was absorbed by the model.



‘Dragon Cage’ by artist Greg Rutkowski (top), whose name has been used tens of thousands of times as a prompt in Stable Diffusion to generate lookalike images (below).

Source: <https://www.businessinsider.com/ai-image-generators-artists-copying-style-thousands-images-2022-10?r=US&IR=T>

Only because this has never happened at such scale and speed before, has the law been slow to respond, and the AI companies so far have got away with it. Much like Uber, it seems they knew that if they moved fast and broke things, they could make their money and be established before the law caught up with them - whilst claiming to disrupt and innovate for the common good. It seems some in Silicon Valley have claimed the right to appropriate artists’ work in order to mechanically process it, and sell it back to us.

“All that we’ve been working on for so many years has been taken from us so easily with AI... It’s really hard to tell whether this will change the whole industry to the point where human artists will be obsolete. I think my work and future are under a huge question mark”

— Greg Rutkowski

Following the outcry from the many artists whose work was used, OpenAI and Midjourney are now facing copyright infringement lawsuits from the likes of [Getty Images](#) and the [New York Times](#). AI companies thus far seem unrepentant, which is perhaps unsurprising given the profits they’re making.

Democratising art

The big three image generators claim some noble and lofty goals whilst making their money off of others’ work. Midjourney for example, claims to be “*expanding the imaginative powers of the human species*” whilst StabilityAI say they are “*building the foundation to activate humanity’s potential.*” The story sold to us is similar across the board - that art is being ‘democratised’ - pulled from the clutches of those entitled artists and mechanised - converted into a tool that will instead make everyone an artist.

By its fans, ‘AI art’ is often described as a new medium or frontier, where humans forgo the

effort of making their own art or building any skill - instead pushing buttons to have art appear instantly. Rather than discovering the joy of creating, we're told instead it's better we learn to think as the computer thinks, and become experts at writing prompts because, hey.. it's the same thing.



'Les Femmes d'Alger' by Pablo Picasso. A painting which disrupted modern art without hurting artists. Source: <https://t.ly/zliLl>

By its nature, art has always been something society struggles to firmly define, and that makes any discussion around its definition difficult. As soon as we establish a common idea of what constitutes art, an artist like [Picasso](#) or [Duchamp](#) is compelled to push the boundaries - making it an ever evolving and expanding field.

But 'AI art' is something very different. Boundary-pushing art never threatened the careers of other artists before, or threatened to monopolise the means of creation - it simply challenged our ideas about ourselves and our world. New art might shock us or make us uncomfortable, but it's never had this much destructive power, nor generated this much money for a handful of men so quickly.

"Everyone is looking for the hack - the secret to success without hard work"

— OpenAI CEO, Sam Altman

Art and illustration is everywhere in our world - it captures our imagination on book covers, brightens our homes, brings magazine articles to life, and adds richness to the environments of video games. The artists who create all of this are the under-appreciated workforce that breathe life and meaning into our everyday. They love their work, spend their entire lives honing their skills, and are often short-changed considering the value they give their employers. They're expected to produce exceptional art to ever-shrinking timescales for often mediocre money. But they do it anyway, because this is what they love - bringing beauty and soul to the world.



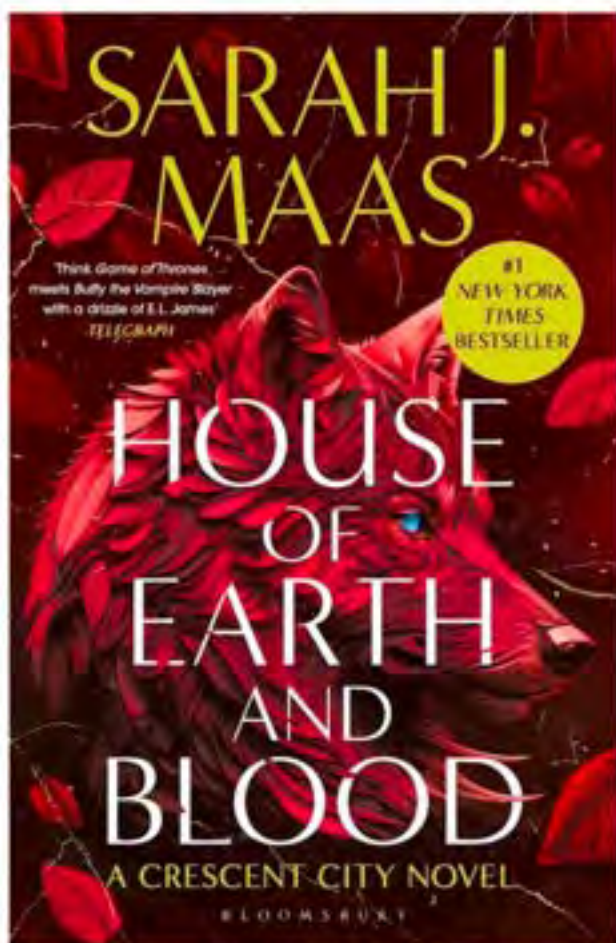
A piece by [Mario Klingemann](#). Producing images with AI image generators, Mario describes himself as a full-time “AI artist.” Source: <https://t.ly/53-mv>

Over the last two years, commercial adoption of AI image generators has meant that real illustrators and artists are losing paid work and control of their own images. Now, the companies who would have previously hired artists can turn to a cheap, mechanical equivalent, and as long as this option is available some won't be able to resist. AI images may be inferior in the details, but they're incredibly cost efficient and near instantaneous.

Automating this is about seeing art as simply 'content,' and what matters is that it keeps coming quickly and cheaply. Someone can be hired to touch-up any problems with the images at far less expense to the company. These workers are also lower skilled, disempowered and more interchangeable than they were before.

This type of automation is a common corporate path. People don't necessarily lose their jobs, but algorithmic management instead takes away their power, and lowers their pay and protections.

Far from 'democratising' art, AI tools are instead having the opposite effect - privatising and automating it. AI is pushing skilled people out of the process and handing control to a handful of tech companies in a race to reduce and commodotise.



'House of Earth and Blood' by Sarah J. Maas. Publisher Bloomsbury used an AI-generated stock image for the book's cover. Source: <https://t.ly/smdNo>



The entirely AI-generated title sequence to Marvel's 'Secret Invasion' TV show. Source: <https://t.ly/qFjR8>



Videogame art assets, automatically generated with AI, instead of created by a game environmental artist. Source: <https://shorturl.at/kARXZ>

Art is intrinsically human

To call AI-generated images equivalent to real art, is - in my opinion - to entirely miss the point. The effortless, instantaneous nature of AI generation prevents it from having real meaning. It's disposable. AI companies truly misunderstand art in thinking that the image is what matters, rather than the intent and the labour.

Art has always been intrinsically human. It comes as much from our flaws and mistakes as it does our successes. Through the process of making it, we express what's inside us - our joy and frustration, longing and sadness - in a way which is instinctive and deep-rooted. It's only through experiencing life as human beings that we have something to express and put to canvas or page. More often than not, we discover what the artwork is to be through the process of creation, rather than having a firm vision at the outset and simply assembling it, as a machine does.

We all know it when we see it - that painting that holds our attention or music that speaks to us. Through the sweat and the hours they put in, the artist communicates those feelings to us, the viewer, and that's part of the joy of engaging with art. We feel understood and more connected as human beings as a result.

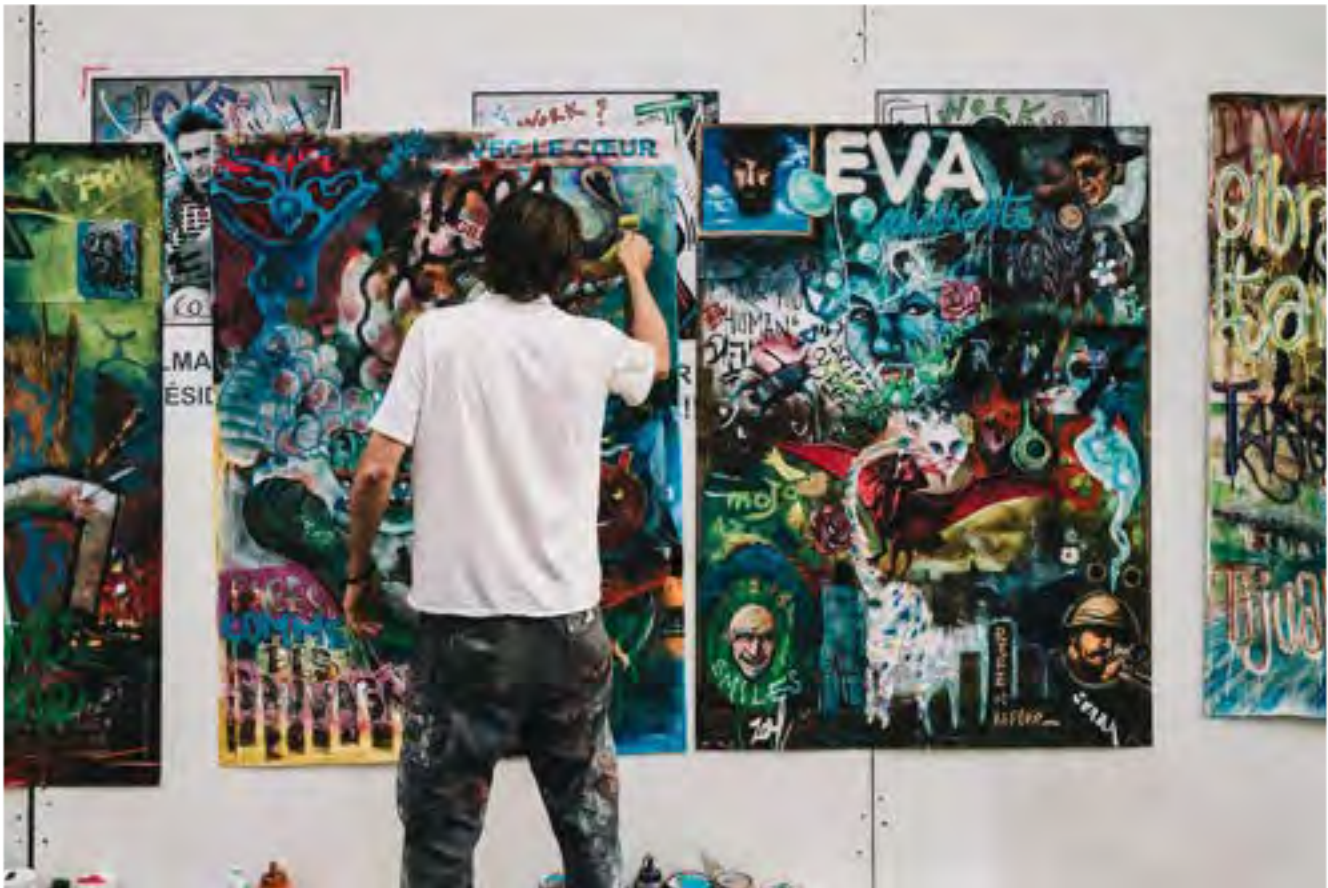


Photo by Ari He. Source: <https://t.ly/3XFiS>

Take away that substance, and what do you have? A lifeless matrix of pixels. With AI art, there is no feeling to communicate, no creative process, and therefore no value imbued in the 'content.'

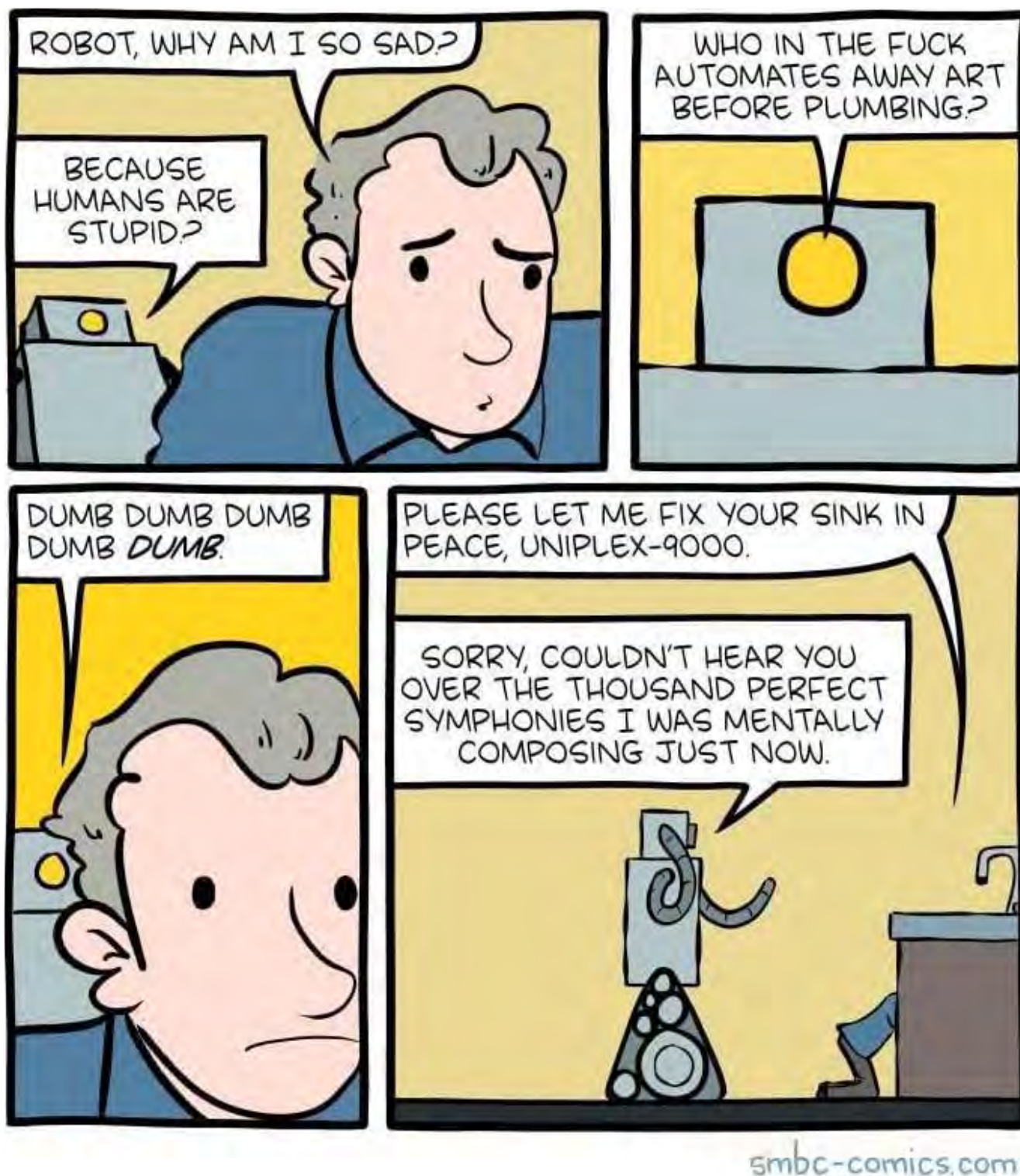
As any designer knows, the act of making art is also a way of observing, thinking and problem solving. The creative mindset is something which must be exercised, and which can be applied in so many other areas of life. It's at the heart of all design and architecture. By letting a machine think for us, we are robbing ourselves of the joy and reward of creation.

AI models are an inherently conservative technology. By thinking on your behalf, and by reducing creative decisions to an algorithmic, wholly quantitative process, they severely constrain the possible outcomes, and no true artistic surprises or discoveries are possible. The models simply crunch the data they're fed and serve a mash-up back to you. Nothing more.

The inevitability myth

But - AI enthusiasts have often told me - now the genie is out of the bottle, it can't be put back in. We should adopt AI or be left behind. The idea that we should fight to help destroy a vital part of our society's culture so corporations can benefit, is deplorable to me. There's nothing inevitable about technological change. Any development takes time, effort, money and determination, as well as a clear intent - continuously reinforced so that a team of people can work towards a common goal.

Technological change is a political and commercial process, shaped by the interests of the corporations and governments that surround us, who make deliberate and resolute choices. So the AI tools we have today are a choice, and a different choice can be made at any time. If we want to see AI work differently, or even pull the plug altogether, we can collectively do so.



Source: <https://shorturl.at/dizYZ>

One has to wonder - when we could use AI models for so many different purposes, why try to automate illustration, and not something we'd all much rather not do, like tax returns? As artist

Molly Crabapple said: *"I cannot understand why someone would burn a tonne of carbon, just to take away a job people love to do, and give it to a machine."* It's entirely possible that Silicon Valley tech men just thought they were tinkering with an interesting maths challenge, and didn't think what the consequences might be for other people. Perhaps these engineers share a reductionist world view, where the human brain is seen as a computer, and everything in life can be expressed through equations and code. Heartbreakingly, perhaps this even extends to art.



Photo by Ivan Samkov. Source: <https://shorturl.at/cIEIN>

The long term impact

If we continue on the path AI tools have set, then creative careers such as illustration and journalism will be far less viable, and far fewer young people will enter the creative fields. They may disappear entirely.

Art is often undervalued today, but by automating it we drive that appreciation even further

down. Art would be regarded as something that can be produced instantly at scale, and more worryingly, given precious little thought or effort.

We'd have fewer artists, writers or musicians, and those we do have will be from the same wealthy, privileged segment of society. That narrowing of experience means any real art we get will be all the poorer, and with fewer people able to spend their time creating art, we can expect less evolution and growth of the field. There'd be no more Grayson Perrys or David Bowies. Art may ultimately become regarded as something machines do, not us - which is backwards and perverse.

“As we invent more species of AI, we will be forced to surrender more of what is supposedly unique about humans. Each step of surrender—we are not the only mind that can play chess, fly a plane, make music, or invent a mathematical law—will be painful and sad.”

— Kevin Kelly

As AI-generated images fill the internet, there will be less real art to train the models on, so if demand keeps growing, we can expect them to be fed more AI-generated images instead. It's easy to see where that leads - a steady degradation of quality as copies are copied over and over again, and the richness of the original work fades from memory. AI art is ultimately a race to the bottom.

Whilst the AI companies claim to be taking us towards the future, I for one find their vision to be cynical and cold. Follow this through, and it leads to a world where humans express creativity only through a corporate-owned platform which we pay for on a monthly subscription. Do we really want the means to create art to be taken and privatised? This is the direction we're headed, unless we decide to say no.

The fight back

All of this is rather depressing - and it should be - but there is good news, too. Many people aren't taking this lying down, and a strong and steadfast call for change is growing by the day.

At the grassroots level, professional artists and illustrators have been staging protests by filling

community sites like [DevinatArt](#) and [ArtStation](#) with a tidal wave of protest placards. In addition, many of the companies that produce the software and sites creatives use, have found themselves needing to clarify their stance in the face of widespread unrest. Where companies have always claimed to be supportive of artists, now they're being asked to prove it through their actions - by for example, banning AI art from their sites.



Simultaneously, New-York based illustrator Molly Crabapple has posted [an open letter](#) and petition, calling for book and magazine publishers to take a stand with artists and refuse to use

AI art. Letters like this have become focal points for thousands of supporters, and Crabapple's includes signatures from high profile people such as author Naomi Klein and actor Jon Cusack.

But some artists have been brave enough to take their challenge all the way to court. After discovering that her work has been used to train AI models without her consent, concept artist [Karla Ortiz](#) - along with two other artists - is driving a class-action lawsuit against StabilityAI, Midjourney and DeviantArt. Their legal challenge cites copyright infringement, unfair competition and reputational harm. Karla's fight has helped bring public attention to the plight of artists and raise awareness about the real human impact of AI.

"If we're going to talk about what really stops people from pursuing art, it's those economic issues. It's issues like a lack of universal health care, few meaningful grants for artists, and the fact that most of us are just a missed paycheck away from homelessness and hunger. This doesn't solve that."

— Karla Ortiz



Illustration by Karla Ortiz. Source: <https://shorturl.at/avCX1>

At the time of writing, big players such as [The New York Times](#) and [Getty Images](#) have also launched their own lawsuits, adding significant weight behind the call for justice. With all this mounting pressure, it seems unlikely that image generators will be able to continue as they have been for much longer.

Last year, an art competition at the Colorado State Fair was unwittingly won by an Midjourney-generated image. Jason Allen, the man behind it, only admitted using AI after he had won the competition, and was subsequently disqualified. Nevertheless, he contested the decision amid heated debate over what constituted both 'art' and 'artist.'

Around the same time, a man named Stephen Thaler tried to get US copyright for an image he produced with AI which he called 'A Recent Entrance to Paradise.' In court, he argued that an AI model should be granted a copyright because we've been using mechanical devices like cameras for years. He didn't convince the federal judge however, who ruled that "human authorship is an essential part of a valid copyright claim."

I firmly agree with that judge. Whilst artists regularly use different devices to produce art, they are always the source of imagination and skill, and always directly controlling the tools. With AI art, the user is no longer in any real control of what comes out, and so should not be considered its author.



'Théâtre D'opéra Spatial' - made by Midjourney from a prompt by Jason Allen - which contraversially won first prize in the 2023 Colorado State Fair art competition. Source: <https://shorturl.at/fhkow>



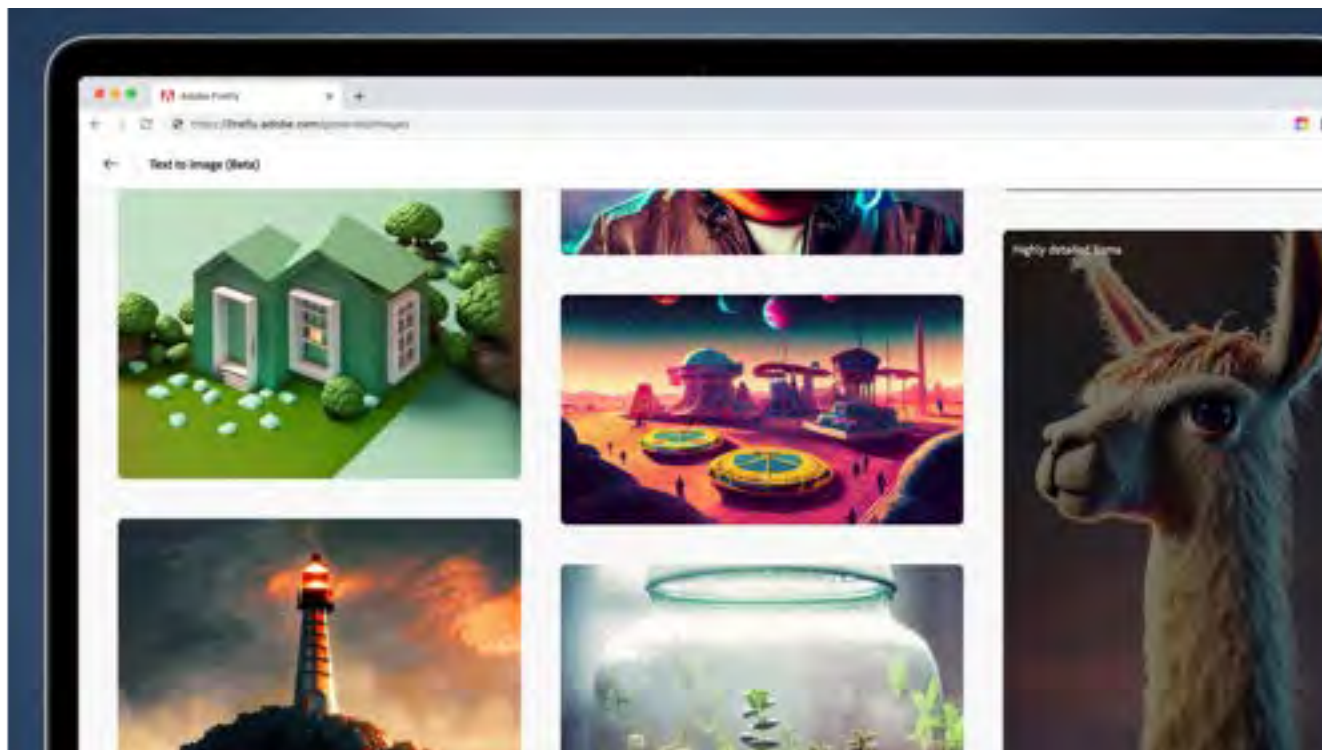
'A Recent Entrance to Paradise' - made by AI from a prompt by Stephen Thaler - who tried and failed to win a copyright issue for the machine as an author. Source: <https://shorturl.at/eoALO>

Copyright law, design registration and similar legal frameworks exist to protect the livelihoods of creatives. They're old and far from perfect, but nevertheless deserve to be protected.

Copyright law reinforces the value of their work and ensures artists - many of whom are self-employed - get some small amount of pay when sick or something in retirement. If AI can just ignore copyright law, then it can take away what little income protection artists have.

Art software giant Adobe are now attempting a more ethical take on generative AI with [Firefly](#). Targeted at creatives, Firefly claims to retain an evidence trail linking results to the original input images, though how well this will work is still unclear. Adobe and others are also trying to establish some governance with a universal 'Do Not Train' tag which might protect artists' work from being scraped by AI models. One has to ask though, why the onus should be on the

artists to enforce protection, rather than reigning in AI in the first place.



Firefly - Adobe's first attempt at a more ethical image generator. Source: <https://t.ly/mxNZq>

Another way artists can protect their work is with [Glaze](#). A software tool created by a team at the University of Chicago, Glaze allows artists to invisibly encrypt their digital artworks to prevent AI models using them.

Whilst the images remain unchanged to the human eye, they will now completely confuse AI models attempting to train on them. It's an active digital watermark that just might afford creatives some genuine protection and peace of mind whilst the legal and commercial worlds struggle to catch up.



Glaze offers artists protection against unwanted scraping by AI models

Source: <https://shorturl.at/apvL3>

Constructive criticism

I'm never one to just complain without offering a better suggestion, so the last thing to address is - what's a more positive way forward?

Ideally, I would love to see AI image generators disappear in their current form. I'm a designer, and the reason I became one in the first place was to make technology revolve around people. If we don't know how something will benefit people, then why should we bring it into the world in the first place? If it actively harms people, then we should stop immediately, and rethink what we're doing. That's why I'd like to see these tools rethought by people with a stronger understanding of the impact on working artists.

The money and resources put behind AI tools today could be redirected towards goals that really benefit society. It would be encouraging to see a human-centric exploration of how AI could be put to better use and to have the outcomes thought through, prototypes trialled and communities' feedback sought, before anything is launched.

I would like to see copyright law updated and reinforced to tackle this new type of problem, with a focus on protecting the livelihoods of working artists, and recognition of the value their

art brings to society. Hopefully the lawsuits happening at the time of writing will set a precedent for this and lead to greater legal protections.

I learned recently that Stable Diffusion allows people to train their own custom AI models using their own training datasets. If people were legally required to own or have permission to use all the data they put into models, that would certainly be a step in the right direction - perhaps by requiring companies to keep copies of the original datasets - though this alone would not undo the damage that's already been caused.



Photo by Elena Mozhvilo. Source: <https://shorturl.at/sxGN8>

A wonderful legacy

But above all else, perhaps this drama could remind us all of the value of art and how it enriches our lives. We're coming precariously close to destroying something that in no small part makes the world a place worth living in, and enables society to function through expression, reflection and connection. We're lost without art, and we cannot afford to take it

for granted.

What if we all decided to say no to AI, and instead worked up the courage to try and draw, or paint, or sculpt by ourselves? What if the strange, artificial taste of 'AI art' left us dissatisfied, and curious enough about creativity to start doodling with a biro in our notebook? It might feel good, and it might spark a flicker of creative confidence. After all, we all drew as children, but as we've got older, we've forgotten how.

What a wonderful legacy AI art might have, if it disappeared and made all of us into artists. Artists who need nothing more than pencil, paper and our own feelings as inspiration. We had everything we needed, all along.

Further reading & listening

Podcasts

- [Why AI is a Threat to Artists w/ Molly Crabapple](#) : Tech Won't Save Us Podcast, Episode 174
- [AI Criticism has a Decades-Long History w/ Ben Tarnoff](#) : Tech Won't Save Us Podcast, Episode 182
- [The Human Side of the AI Underclass w/ Joanne McNeil](#) : Tech Won't Save Us Podcast, Episode 196
- [The Future of the State w/ James Plunkett](#) : Jon Richardson & The Futureonauts, Season 3, Episode 4

Books

- [Blood in the Machine: The Origins of the Rebellion Against Big Tech](#) by Brian Merchant
- [Resisting AI: An Anti-Fascist Approach to Artificial Intelligence](#) by Dan McQuillan
- Hello World: Being Human in the Age of Algorithms – Hannah Fry

Articles

- [AI-Generated Art Controversy: The Future of Creativity or a Replacement for Human Talent?](#) by Adam Hencz for Artland
- [Restrict AI from Publishing: An Open Letter](#) by Molly Crabapple for The Center for Artistic Inquiry and Reporting
- [Independent Artists Are Fighting Back Against A.I. Image Generators With Innovative Online Protests](#) by Richard Whiddington for Artnet
- [Molly Crabapple Has Posted an Open Letter by 1,000 Cultural Luminaries Urging Publishers to Restrict the Use of 'Vampirical' A.I.-Generated Images](#) by Jo Lawson-Tancred for Artnet
- [No, teaching AI to copy an artist's style isn't 'democratization.' It's theft](#) by Soleil Ho for The San Francisco Chronicle
- [Stability AI swerves copyright infringement allegations in response to Getty lawsuit](#) by Tim Smith and Kai Nicol-Schwarz for Sifted
- [Glaze protects art from prying AIs](#) by Natasha Lomas for TechCrunch
- [US judge: Art created solely by artificial intelligence cannot be copyrighted](#) by Jon Brodtkin for ArsTechnica
- [The US Copyright Office says an AI can't copyright its art](#) by Adi Robertson for The Verge
- [Weizenbaum's Nightmares: How the inventor of the first chatbot turned against AI](#) by Ben Tarnoff for The Guardian
- [AI Machines aren't 'hallucinating,' but their makers are](#) by Naomi Klein for The Guardian
- [How Adobe is managing the AI copyright dilemma, with general counsel Dana Raoby Nilay Patel](#) for The Verge
- [Why Generative AI Angers Artists but Not Writers](#) by Alberto Romero for The Algorithmic Bridge
- [AI: Digital artist's work copied more times than Picasso](#) by Clare Hutchinson & Phil John for BBC News
- [This artist is dominating AI-generated art. And he's not happy about it](#) by Melissa Heikkiläarchive page for MIT Technology Review

- [Artists slam Marvel over AI-generated credits in Secret Invasion](#) by Mark Sellman for The Times
- [Courts are using AI to sentence criminals. That must stop now](#) by Jason Tashea for Wired
- [The New York Times is suing OpenAI and Microsoft for copyright infringement](#) by Emma Roth for The Verge
- [Getty gets tough on London-based AI firm](#) by William Charrington and Hoi-Yee Roper for Farrer & Co

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Matt Corral



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WINTER STORM 2021

ERCOT overcharged power companies \$16 billion for electricity during winter freeze, firm says

An independent market monitor for the Public Utility Commission of Texas wrote in a letter that the power grid operator kept market prices too high for nearly two days after widespread outages ended.

BY **ERIN DOUGLAS** AND **MITCHELL FERMAN** MARCH 4, 2021 UPDATED: 8 PM CENTRAL **SHARE**

The Electric Reliability Council of Texas made a \$16 billion error in pricing during the week of the winter storm that caused power outages across the state, according to a [filing by its market monitor](#).

Potomac Economics, the independent market monitor for the Public Utility Commission of Texas, which oversees ERCOT, wrote in a letter to the Public Utility Commission that ERCOT kept market prices for power too high for nearly two days after widespread outages ended late the night of Feb. 17. It should have reset the prices the following day.

That decision to keep prices high, the market monitor claimed, resulted in \$16 billion in additional costs to Texas power companies. The news of the overcharging was first reported by [Bloomberg](#).

Some of the providers that were charged during the high price period could pass the costs to customers, depending on the type of contract they have, according to Detlef Hallermann, director of the Reliant Energy Trade Center at Texas A&M University.

In Texas, wholesale power prices are determined by supply and demand: When demand is high, ERCOT allows prices to go up. During the storm, PUC directed the grid operator to set wholesale power prices at \$9,000 per megawatt hour — the maximum price. Raising prices is intended to incentivize power generators in the state to add more power to the grid. Companies then buy power from the wholesale market to deliver to consumers, which they are contractually obligated to do.

Because ERCOT failed to bring prices back down on time, companies had to buy power in the market at inflated prices.

The error will likely result in higher levels of defaults, wrote Carrie Bivens, a vice president of Potomac Economics, the firm that monitors the grid operator. She said the PUC should direct ERCOT to remove the pricing interventions that occurred after outages ended, and allowing them to remain would result in “substantial and unjustified” economic harm.

At least \$1.5 billion could be passed on to retail electric providers and their customers. Some retail providers have already begun to file for bankruptcy.

“They’re going to suffer the most,” Hallermann said.

Retail power providers have been in financial distress across Texas since the storm; many were forced to buy power on the wholesale market at extremely high prices.

Brazos Electric Power Cooperative Inc., Texas’ largest power cooperative, has already filed for bankruptcy protection after incurring \$2.1 billion in combined charges owed to ERCOT, according to court documents filed Monday.

Many retail power providers complained in filings to regulators that generators of electricity, which were unable to produce enough power during the storm, profited and left retail companies scrambling.

“The ERCOT market was not designed to deal with an emergency of this scale,” wrote Patrick Woodson, CEO of ATG Clean Energy Holdings, a retail power provider based in Austin, to the Public Utility Commission. The pricing failure, he wrote, “has pushed the entire market to the brink of collapse.”

Bivens wrote that while she recognizes that retroactively revising the prices is “not ideal,” correcting the error will reflect the accurate supply and demand for power during the period after the outages.

Cathy Webking of the Texas Energy Association for Marketers told lawmakers during a Texas Senate Committee on Business and Commerce meeting Thursday that prices should be set back to what the market value would have been.

“There are more defaults imminent. Immediate action is required,” Webking said.

A spokesperson for ERCOT declined to comment on the matter.

Kenan Ogelman, the ERCOT vice president of commercial operations, who testified during a Texas Senate committee hearing Thursday, was not asked by state senators about ERCOT’s \$16 billion mistake. Sen. Kelly Hancock, R-North Richland Hills, who chairs the Business

and Commerce committee, did not indicate what action he or other senators would take on the various financial ripple effects from the winter storm.

“There are financial concerns — let’s put it that way — that we have to address,” Hancock said.

Reese Oxner and Shannon Najmabadi contributed to this report.



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How Texas' Drive for Energy Independence Set It Up for Disaster

Texas has refused to join interstate electrical grids and railed against energy regulation. Now it's having to answer to millions of residents who were left without power in last week's snowstorm.



By **Clifford Krauss, Manny Fernandez, Ivan Penn and Rick Rojas**

Published Feb. 21, 2021 Updated May 13, 2021

HOUSTON — Across the plains of West Texas, the pump jacks that resemble giant bobbing hammers define not just the landscape but the state itself: Texas has been built on the oil-and-gas business for the last 120 years, ever since the discovery of oil on Spindletop Hill near Beaumont in 1901.

Texas, the nation's leading energy-producing state, seemed like the last place on Earth that could run out of energy.

Then last week, it did.

The crisis could be traced to that other defining Texas trait: independence, both from big government and from the rest of the country. The dominance of the energy industry and the “Republic of Texas” ethos became a devastating liability when energy stopped flowing to millions of Texans who shivered and struggled through a snowstorm that paralyzed much of the state.

Part of the responsibility for the near-collapse of the state's electrical grid can be traced to the decision in 1999 to embark on the nation's most extensive experiment in electrical deregulation, handing control of the state's entire electricity delivery system to a market-based patchwork of private generators, transmission companies and energy retailers.

The energy industry wanted it. The people wanted it. Both parties supported it.

“Competition in the electric industry will benefit Texans by reducing monthly rates and offering consumers more choices about the power they use,” George W. Bush, then the governor, said as he signed the top-to-bottom deregulation legislation.

Mr. Bush's prediction of lower-cost power generally came true, and the dream of a free-market electrical grid worked reasonably well most of the time, in large part because Texas had so much cheap natural gas as well as abundant wind to power renewable energy. But the newly deregulated system came with few safeguards and even fewer enforced rules.

With so many cost-conscious utilities competing for budget-shopping consumers, there was little financial incentive to invest in weather protection and maintenance. Wind turbines are not equipped with the de-icing equipment routinely installed in the colder climes of the Dakotas and power lines have little insulation. The possibility of more frequent cold-weather events was never built into infrastructure plans in a state where climate change remains an exotic, disputed concept.

"Deregulation was something akin to abolishing the speed limit on an interstate highway," said Ed Hirs, an energy fellow at the University of Houston. "That opens up shortcuts that cause disasters."

The state's entire energy infrastructure was walloped with glacial temperatures that even under the strongest of regulations might have frozen gas wells and downed power lines.

But what went wrong was far broader: Deregulation meant that critical rules of the road for power were set not by law, but rather by a dizzying array of energy competitors.

Utility regulation is intended to compensate for the natural monopolies that occur when a single electrical provider serves an area; it keeps prices down while protecting public safety and guaranteeing fair treatment to customers. Yet many states have flirted with deregulation as a way of giving consumers more choices and encouraging new providers, especially alternative energy producers.

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California, one of the early deregulators in the 1990s, scaled back its initial foray after market manipulation led to skyrocketing prices and rolling blackouts.

States like Maryland allow customers to pick from a menu of producers. In some states, competing private companies offer varied packages like discounts for cheaper power at night. But no state has gone as far as Texas, which has not only turned over the keys to

the free market but has also isolated itself from the national grid, limiting the state's ability to import power when its own generators are foundering.

Consumers themselves got a direct shock last week when customers who had chosen variable-rate electricity contracts found themselves with power bills of \$5,000 or more. While they were expecting extra-low monthly rates, many may now face huge bills as a result of the upswing in wholesale electricity prices during the cold wave. Gov. Greg Abbott on Sunday said the state's Public Utility Commission has issued a moratorium on customer disconnections for non-payment and will temporarily restrict providers from issuing invoices.

A family in Austin, Texas, kept warm by a fire outside their apartment on Wednesday. They lost power early Monday morning. Tamir Kalifa for The New York Times

There is regulation in the Texas system, but it is hardly robust. One nonprofit agency, the Electric Reliability Council of Texas, or ERCOT, was formed to manage the wholesale market. It is supervised by the Public Utility Commission, which also oversees the transmission companies that offer customers an exhaustive array of contract choices laced with more fine print than a credit card agreement.

But both agencies are nearly unaccountable and toothless compared to regulators in other regions, where many utilities have stronger consumer protections and submit an annual planning report to ensure adequate electricity supply. Texas energy companies are given wide latitude in their planning for catastrophic events.

Into a snowstorm with no reserves

One example of how Texas has gone it alone is its refusal to enforce a “reserve margin” of extra power available above expected demand, unlike all other power systems around North America. With no mandate, there is little incentive to invest in precautions for events, such as a Southern snowstorm, that are rare. Any company that took such precautions would put itself at a competitive disadvantage.

A surplus supply of natural gas, the dominant power fuel in Texas, near power plants might have helped avoid the cascade of failures in which power went off, forcing natural gas production and transmission offline, which in turn led to further power shortages.

In the aftermath of the dayslong outages, ERCOT has been criticized by both Democratic and Republican residents, lawmakers and business executives, a rare display of unity in a fiercely partisan and Republican-dominated state. Mr. Abbott said he supported calls for the agency’s leadership to resign and made ERCOT reform a priority for the Legislature. The reckoning has been swift — this week, lawmakers will hold hearings in Austin to investigate the agency’s handling of the storm and the rolling outages.

For ERCOT operators, the storm’s arrival was swift and fierce, but they had anticipated it and knew it would strain their system. They asked power customers across the state to conserve, warning that outages were likely.

But late on Sunday, Feb. 14, it rapidly became clear that the storm was far worse than they had expected: Sleet and snow fell, and temperatures plunged. In the council’s command center outside Austin, a room dominated by screens flashing with maps, graphics and data tracking the flow of electricity to 26 million people in Texas, workers quickly found themselves fending off a crisis. As weather worsened into Monday morning, residents cranked up their heaters and demand surged.

Power plants began falling offline in rapid succession as they were overcome by the frigid weather or ran out of fuel to burn. Within hours, 40 percent of the power supply had been lost.

The entire grid — carrying 90 percent of the electric load in Texas — was barreling toward a collapse.

Much of Austin lost power last week due to rolling blackouts. Tamir Kalifa for The New York Times

In the electricity business, supply and demand need to be in balance. Imbalances lead to catastrophic blackouts. Recovering from a total blackout would be an agonizing and tedious process, known as a “black start,” that could take weeks, or possibly months.

And in the early-morning hours last Monday, the Texas grid was “seconds and minutes”

away from such a collapse, said Bill Magness, the president and chief executive of the Electric Reliability Council.

“If we had allowed a catastrophic blackout to happen, we wouldn’t be talking today about hopefully getting most customers their power back,” Mr. Magness said. “We’d be talking about how many months it might be before you get your power back.”

Earlier warnings of trouble

The outages and the cold weather touched off an avalanche of failures, but there had been warnings long before last week’s storm.

After a heavy snowstorm in February 2011 caused statewide rolling blackouts and left millions of Texans in the dark, federal authorities warned the state that its power infrastructure had inadequate “winterization” protection. But 10 years later, pipelines remained inadequately insulated and heaters that might have kept instruments from freezing were never installed.

During heat waves, when demand has soared during several recent summers, the system in Texas has also strained to keep up, raising questions about lack of reserve capacity on the unregulated grid.

And aside from the weather, there have been periodic signs that the system can run into trouble delivering sufficient energy, in some cases because of equipment failures, in others because of what critics called an attempt to drive up prices, according to Mr. Hirs of the University of Houston, as well as several energy consultants.

Another potential safeguard might have been far stronger connections to the two interstate power-sharing networks, East and West, that allow states to link their electrical grids and obtain power from thousands of miles away when needed to hold down costs and offset their own shortfalls.

But Texas, reluctant to submit to the federal regulation that is part of the regional power grids, made decisions as far back as the early 20th century to become the only state in the continental United States to operate its own grid — a plan that leaves it able to borrow only from a few close neighbors.

The border city of El Paso survived the freeze much better than Dallas or Houston because it was not part of the Texas grid but connected to the much larger grid covering many Western states.

But the problems that began with last Monday’s storm went beyond an isolated electrical

grid. The entire ecosystem of how Texas generates, transmits and uses power stalled, as millions of Texans shivered in darkened, unheated homes.

A surplus supply of natural gas, the dominant power fuel in Texas, near power plants might have helped avoid the cascade of failures. Eddie Seal/Bloomberg

Texans love to brag about natural gas, which state officials often call the cleanest-burning fossil fuel. No state produces more, and gas-fired power plants produce nearly half the state's electricity.

"We are struggling to come to grips with the reality that gas came up short and let us down when we needed it most," said Michael E. Webber, a professor of mechanical engineering at the University of Texas at Austin.

The cold was so severe that the enormous oil and natural gas fields of West Texas froze up, or could not get sufficient power to operate. Though a few plants had stored gas reserves, there was insufficient electricity to pump it.

The leaders of ERCOT defended the organization, its lack of mandated reserves and the state's isolation from larger regional grids, and said the blame for the power crisis lies with the weather, not the overall deregulated system in Texas.

"The historic, just about unprecedented, storm was the heart of the problem," Mr. Magness, the council's chief executive, said, adding: "We've found that this market structure works. It demands reliability. I don't think there's a silver-bullet market structure that could have managed the extreme lows and generation outages that we were facing Sunday night."

In Texas, energy regulation is as much a matter of philosophy as policy. Its independent power grid is a point of pride that has been an applause line in Texas political speeches for decades.

Deregulation is a hot topic among Texas energy experts, and there has been no shortage of predictions that the grid could fail under stress. But there has not been widespread public dissatisfaction with the system, although many are now wondering if they are being well served.

"I believe there is great value in Texas being on its own grid and I believe we can do so safely and securely and confidently going forward," said State Representative Jeff Leach, a Republican from Plano who has called for an investigation into what went wrong. "But it's going to take new investment and some new strategic decisions to make sure we're protected from this ever happening again."

Steven D. Wolens, a former Democratic lawmaker from Dallas and a principal architect of the 1999 deregulation legislation, said deregulation was meant to spur more generation, including from renewable energy sources, and to encourage the mothballing of older plants that were spewing pollution. "We were successful," said Mr. Wolens, who left the Legislature in 2005.

But the 1999 legislation was intended as a first iteration that would evolve along with the needs of the state, he said. "They can focus on it now and they can fix it now," he said. "The buck stops with the Texas Legislature and they are in a perfect position to determine the basis of the failure, to correct it and make sure it never happens again."

Clifford Krauss reported from Houston, Manny Fernandez and Ivan Penn from Los Angeles, and Rick Rojas from Nashville. David Montgomery contributed reporting from Austin, Texas.

Clifford Krauss is a national energy business correspondent based in Houston. He joined The Times in 1990 and has been the bureau chief in Buenos Aires and Toronto. He is the author of “Inside Central America: Its People, Politics, and History.” [More about Clifford Krauss](#)

Manny Fernandez is the Los Angeles bureau chief. He spent more than nine years covering Texas as the Houston bureau chief. He joined The Times as a Metro reporter in 2005, covering the Bronx and housing. [More about Manny Fernandez](#)

Ivan Penn is a Los Angeles-based reporter covering alternative energy. Before coming to The Times in 2018 he covered utility and energy issues at The Tampa Bay Times and The Los Angeles Times. [More about Ivan Penn](#)

Rick Rojas is a national correspondent covering the American South. He has been a staff reporter for The Times since 2014. [More about Rick Rojas](#)

A version of this article appears in print on , Section A, Page 1 of the New York edition with the headline: Texas Officials Had Few Rules For Power Grid

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Illustration by Tyler Comrie

Anthony Mecke had drifted to sleep in the break room when a loud knock roused him at 1:23 a.m. “We just got the call,” a coworker said.

Mecke, a moonfaced 45-year-old, is the manager of systems operation training at CPS Energy, the city-owned electricity provider that serves San Antonio. He started at the company not long after high school, working at one point as a cable splicer, a job he performed in hot tunnels beneath the sidewalks of San Antonio. He thought he’d seen it all. But when he hustled from the break room, where he’d sneaked in a power nap after an all-day shift, into the company’s cavernous control room, housed in a tornado-proof building on the city’s East Side, what he witnessed unsettled him.

This was Monday, February 15, 2021. A winter storm had brought unusually frigid temperatures to the entire middle swath of the United States, from the Canadian border to the Rio Grande. In San Antonio, it dropped to 9 degrees. In Fort Worth, the storm’s icy arrival a few days earlier had led to a 133-vehicle pileup that left 6 dead. Abilene and Pflugerville had advised residents to boil their water, the first of thousands of such warnings that would eventually affect 17 million Texans. Across the state, families hunkered down and did anything they could to stay warm. The overwhelming majority of Texas homes are outfitted with electric heaters that are the technological equivalent of a toaster oven. During the most severe cold fronts, residents crank up those inefficient units, and some even turn on and open electric ovens and use hair dryers.





The control room at CPS Energy, in San Antonio.

Photograph by Jeff Wilson

Mecke could track the spiking energy use in real time. One wall of the control room is covered in enormous computer monitors displaying maps and data. He scanned for one particular piece of information. The state's electricity reserves, which are tapped to prevent emergencies, were already depleted. The problem wasn't just surging demand. Power plants all across the grid were shutting off, incapacitated by frozen equipment and a dearth of natural gas, the primary source of fuel.

The Texas power grid was, at that moment, like an airplane low on fuel that needed to jettison cargo to stay aloft. That's what the call had been about. The state's grid operator, the Electric Reliability Council of Texas, or ERCOT, had just told CPS Energy and fifteen of the state's other electric utility companies to immediately begin turning off power for portions of their

companies to immediately begin turning on power for portions of their service areas. The result would be blackouts.

Nobody yet knew just how widespread the blackouts would become—that they would spread across almost the entire state, leave an unprecedented 11 million Texans freezing in the dark for as long as three days, and result in as many as seven hundred deaths. But neither could the governor, legislators, and regulators who are supposed to oversee the state's electric grid claim to be surprised. They had been warned repeatedly, by experts and by previous calamities—including a major blackout in 2011—that the grid was uniquely vulnerable to cold weather.

Unlike most other states that safely endured the February 2021 storm, Texas had stubbornly declined to require winterization of its power plants and, just as critically, its natural gas facilities. In large part, that's because the state's politicians and the regulators they appoint are often captive to the oil and gas industry, which lavishes them with millions of dollars a year in campaign contributions. During the February freeze, the gas industry failed to deliver critically needed fuel, and while Texans of all stripes suffered, the gas industry scored windfall profits of about \$11 billion—creating debts that residents and businesses will pay for at least the next decade.

Since last February, the state has appointed new regulators and tweaked some of its statutes. But despite the misery, death, economic disruption, and embarrassment that Texas suffered, little has changed. The state remains susceptible to the threat that another winter storm could inflict blackouts as bad as—or even worse than—last year's catastrophe. Despite promises from public officials to rectify these problems, we remain largely defenseless and can only hope we aren't thrashed by another Arctic blast. Even as forecasters predict a relatively warm winter on average, there is compelling evidence that such extreme weather phenomena are becoming more common. To understand the danger, it's worth examining how close the Texas grid came last year to a meltdown that could have left much of the state without power for several weeks, or even months.



Aerial view of Dallas's Highland Meadows neighborhood on February 15, 2021.

Isaac Murray/Getty

Two days before Mecke was awakened in his office, ERCOT had held an emergency conference call to warn the state's utilities and rural electric cooperatives that blackouts were likely. ERCOT officials said the grid might have to shed as much as 7,500 megawatts—effectively darkening roughly one of every eight homes in the state. That's nearly twice as much as the last controlled load shed, in 2011, when rolling blackouts had lasted as long as eight hours, which in turn was four times longer than the previous large-scale blackout, in 2006.

The worst-case scenario ERCOT had gamed out, what it called “extreme winter,” contemplated a record-setting demand of 67.2 gigawatts. Electricity consumption blew past that mark at 7 p.m. on February 14. Meanwhile, electricity supply continued to dwindle as underinsulated power plants went down, one after another.

For the grid to function properly, the supply of electricity must always match

demand; this equilibrium is reflected in the grid's frequency, which usually remains steady at 60 hertz. Power plants across the state are tuned in to the frequency, and they automatically increase or decrease generation to maintain equilibrium. The grid is like a giant synchronized machine, its components linked across hundreds of miles, from Midland to Houston, from Amarillo to Brownsville. On this night, as demand drastically outpaced supply, the frequency dropped and the vast machine began churning faster. But eventually it couldn't compensate on its own.

By 1:23 a.m., ERCOT could no longer delay action. An operator in its control room picked up the hotline phone, which was wired to sixteen of the state's utility companies, and ordered a thousand-megawatt load shed statewide. "You practice for this for years," Mecke said. "You hope it never happens."

In fact, a few hours earlier, he'd run his coworkers through a simulation of a nearly identical load shed. When the time came to carry out the operation for real, there were no hiccups. "It was surprisingly calm," he said. "It was smooth." Within seconds, electricity in parts of San Antonio began to blink off. Mecke, hopeful that the grid would stabilize, breathed a sigh of relief. The calm was short-lived.

The frequency should have risen after the load shed, but instead it kept falling. It was "nerve-racking," said Mecke.

At 1:47 a.m., the hotline phone rang again. Everyone in the CPS control center stopped what they were doing. ERCOT needed another thousand megawatts cut. Because of coronavirus precautions, CPS executives weren't in the control room. Rudy Garza, the chief customer officer, tracked the frequency's dangerous decline on his phone, texting back and forth with industry friends and former coworkers from across the state. "We were scared," he said.

CenterPoint Energy, a utility in Houston, runs a control room similar to that of CPS. Eric Easton, CenterPoint's vice president of real-time operations, was hastening to execute the second round of blackouts when the hotline phone rang for the third time. At 1:51 a.m., ERCOT ordered another three

phone rang for the third time, at 1:51 a.m. ERCOT ordered another three thousand megawatts—more than the first two combined. “Calls started coming in so fast that they were overlapping,” said Easton. “When are we going to stop shedding load?” he wondered.

But the situation was only growing more dire. At the precise time of the third call, the frequency reached a critical threshold: 59.4 hertz. The Texas grid, which has been around in some form since World War II, had only once in its history fallen this low. Automated turbines across the state began spinning even faster to produce more electricity, but when the frequency dips below 59.4 hertz, the turbines reach speeds and pressures that can cause catastrophic damage to them, requiring that they be repaired or replaced. This scenario was unlikely because, to prevent it, the grid automatically triggers a nine-minute countdown when it strikes 59.4 hertz. If the frequency did not rise in time, power plants would shut down and the grid would begin turning itself off completely. This would leave all 26 million Texans who relied on the ERCOT grid without power for weeks or months.

A few more minutes ticked by. The frequency kept falling, touching 59.302 hertz, yet another alarming precipice. At 59.3 hertz, human operators are taken out of the equation: they are too slow to make the urgent adjustments that are needed to stabilize the grid. The system is programmed to automatically start blacking out as many areas as are necessary to balance power supply and demand. But in this scenario, that fail-safe may not have worked because so many areas had already been manually cut off. “We were on the very edge,” said Easton.

In a last-ditch effort to prevent the grid’s collapse, ERCOT placed a fourth hotline call, at 1:55 a.m., and ordered another 3,500 megawatts. All across Texas, grid operators were moving as quickly as they could, blacking out more and more neighborhoods, but they were running out of options. As the countdown approached zero, the frequency suddenly shot back up. The immediate crisis was over—the last-second load shed had worked—but for most of the following day, the grid remained dangerously unstable.





Bill Magness, former CEO of ERCOT, testifying at the Texas House.
Eric Gay/AP

It is hard to fathom the devastation a total shutdown would have wreaked. Bill Magness, then the CEO of ERCOT, would explain as much to the Texas Senate ten days later. Magness is a lawyer with a buzz cut and ramrod-straight posture who spent time in the nineties and aughts as a practicing Buddhist. “What my team and the folks at the utilities in Texas would be doing is an exercise called ‘black start,’” he said. A black start would have required carefully rebooting a few power plants at a time and using them to jump-start others, thereby restoring the grid piece by piece. It’s not a matter of flipping switches. The steps required for a black start are numerous, complex, and delicate. No one knows how long that process would take, because no one has ever needed to do it. Magness said it would have been weeks at least.

Most of the state’s residents would have been without heat, potable water, or light, as would almost all of the businesses on which they depend. Traffic lights wouldn’t have worked. Caravans of trucks, likely escorted by the National Guard, would have delivered fuel to generators to keep hospitals (many of which were nearly at max capacity because of COVID-19), fire departments, and other emergency services operating. When the freeze

departments, and other emergency services operating. When the freeze lifted and the roads thawed, many would have attempted an exodus into neighboring states—all of which, with a few brief exceptions, kept power—but even that would have proved difficult because gas pumps run on electricity. Magness looked grimly around the Senate chamber as he described the doomsday scenario. “Imagine: the suffering that we saw [would have been] compounded.”

An Oncor power substation in Waco on February 18, 2021.

Photograph by Matthew Busch

In her ground-floor apartment on Uvalde Road, a busy commercial thoroughfare in the Cloverleaf community, just east of Houston, Mary Gee liked to sit by the window, watching the people and cars passing by. Across the way were an auto-parts store, a car wash, and a Tex-Mex restaurant. There was always something happening. But the snow and ice in February brought Uvalde to a standstill.

The neighborhood lost power early Monday morning, February 15. After the sun rose, a few neighbors ventured out. Word passed in Gee’s complex, the Havenwood, that a nearby Burger King was open—that would have meant not only food but warmth. Some decided to check it out. This wasn’t an option for Gee. She was relatively healthy, but at 84, walking more than a mile on slippery sidewalks was out of the question.

“It kind of felt like the end of the world,” said Christion Jones, who lived a few doors down from Gee. Other residents sat in cars to warm up, their engines idling, the exhaust forming small clouds in the frigid air. But Gee had stopped driving years before and had given the last car she owned to a god-granddaughter.

Gee had grown up with eight brothers and sisters, and much of her childhood was spent in a rural house with a wood-burning heater in the small town of Normangee, between Houston and Waco. She had worked as a nurse for more than twenty years at Houston Methodist hospital. On weekends, she

and her husband, Herman, had kept a shed at a local flea market selling clothes, LPs, purses, electronics—a little bit of everything. Gee liked to chat up the regulars.

Herman died in 2018, and she tragically lost her only child, Michael, the next year. But Gee wasn't alone. Most of her siblings had moved to Bryan and Houston, which meant she was surrounded by nieces and nephews. She often spoke with them on the phone, calls that would stretch at least half an hour as Gee asked about relatives one by one. If any were struggling, she would pray for them.

The day Gee lost power, her niece Zona Amerson tried to call her, but no one picked up. Amerson, who's 64, was concerned, but there was no way to drive across town to check on her. The roads were impassable. Then one of Amerson's pipes burst, with no plumbers available to fix it. She was distracted by her own crisis.

Vehicles at a standstill on Interstate 35 near Temple on February 18, 2021.

Joe Raedle/Getty

Power lines surrounded by snow in Dallas on February 18, 2021, as power was still out for many.

Photograph by Nitashia Johnson

On Thursday, Amerson heard from a relative that her aunt had died. A Harris County medical examiner ruled that the cause was hypothermia. Gee was one of hundreds of Texans who died because of the lack of electricity. (The state recently updated the death toll to 246, **a number that falls far short of the total that experts on mortality say is the true measure of the cost in lives of this disaster**, which accounts for those who, for example, had a heart attack and couldn't get to a hospital.) Others included a centenarian in a senior living community in Houston who'd also succumbed to hypothermia; she'd received a college degree in the thirties and had taught elementary school in a single-room schoolhouse in Wisconsin. An 87-year-old Austin woman died of a fast-moving urinary tract infection after her catheter froze. Two men in Garland are believed to have died of carbon monoxide poisoning —neighbors said they were running a gas-powered generator inside an apartment unit. In Sugar Land, southwest of Houston, a family used their

replace to stay warm. The house caught fire, and a grandmother and three of her grandchildren died. The mother survived. “Most of all, I think, what I will miss is just seeing them grow into these amazing human beings that I knew that they would be,” she told the *Houston Chronicle*.

Of the millions of Texans who lost electricity during the blackouts, which lasted from Monday through Thursday, most experienced it as a week of compounding problems. Millions either lost water or needed to boil water. When the water finally came back on, burst pipes began to flow, causing billions of dollars in damage. Plumbers were so overwhelmed with calls that some homeowners had to wait months for repairs. Economists at the Dallas Federal Reserve estimated that the blackouts cost the state’s economy somewhere in the \$80 to \$130 billion range, potentially making it the most expensive disaster in state history.

Of the millions of Texans who lost electricity during the blackouts, most experienced it as a week of compounding problems.

Even Texas newcomer Elon Musk, the chief executive of Tesla and the world’s richest person, was affected. “I was actually in Austin for that snowstorm in a house with no electric, no lights, no power, no heating, no internet—couldn’t actually even get to a food store,” he said at an investor meeting in October.

Dan Meador, an engineering manager at Austin tech firm Anaconda, also lost power. He and his pregnant wife bedded down in their living room in front of their fireplace. When they woke in the morning, it was 7 degrees outside with a windchill factor of -8. He used the fire to boil cowboy coffee and cook

sausages in a cast-iron pan. The following days were devoted solely to meeting basic needs: finding firewood and preparing meals. When a neighbor's cedar tree splintered and fell under the weight of ice, he fetched his chain saw—before remembering it was electric. Meador, a former linebacker for the University of Arkansas football team, used a hacksaw instead. When I spoke to him eight months later, he was still shaken by the experience. “You turn your water faucet on, water comes out,” he said. “There's a lot of faith that we have in this stuff just showing up.”

A family outside their powerless Austin apartment, warming up next to a fire made by burning a discarded armoire.

Photograph by Tamir Kalifa

The Texas Legislature was still in the early stages of its biennial gathering in Austin when the blackouts occurred. Lawmakers and staff were told to stay off the icy roads. This appears to be the first time in state history that winter weather forced legislators to stay home.

Of course, that didn't stop politicians from pointing fingers. Rick Perry, former governor and former U.S. secretary of energy, tried to preempt calls to increase federal oversight of the state's grid. In a striking display of insensitivity to the families who were grieving the loss of loved ones, he claimed that Texans were willing to forgo power “for longer than three days to keep the federal government out of their business.” Lieutenant Governor Dan Patrick was one of many politicians to blame wind turbines. “Our renewables aren't reliable,” he said on *Good Morning America*. Governor Greg Abbott appeared on Sean Hannity's Fox News show and argued that the blackouts showed how the Green New Deal, which was then a subject of intense debate in Washington, D.C., would be a “deadly deal for the United States.”

Blaming renewables was, of course, a politically convenient lie. Yes, some wind farms in West and South Texas had frozen up—their operators hadn't invested in blades with internal warming coils that allow windmills to

invested in blades with internal warming coils that allow windmills to function perfectly fine in other states and regions, including north of the Arctic Circle in Norway. But many windmills kept working, helping to prevent a worse disaster. Even Abbott admitted, while the blackouts were ongoing, that the biggest culprit was power plants that ran on gas.

As the death toll climbed, the politicians' bluster ebbed. Abbott added a new item to the legislative session: winterizing the state's power system. Patrick promised, "We're going to get to the bottom of this and find out what the hell happened, and we're going to fix it."

Such promises had been made before. A decade earlier, in February 2011, temperatures in Texas plunged into the single digits, and ERCOT instituted rolling blackouts that affected 3.4 million homes and businesses (but for only a matter of hours, rather than days). David Dewhurst, a Republican who was then the lieutenant governor, blamed a lack of "winterization and preparation." Weeks later, the Legislature held a hearing on the blackouts, and Troy Fraser, a Republican state senator representing the Twenty-fourth District, demanded, "How are we going to make sure that doesn't happen again?"

Suzanne Mitchell in her Dallas home after her pipes burst and her roof caved in.

Photograph by Nitashia Johnson

A Fiesta Mart in Houston during the blackout.

Go Nakamura/Getty

The answer came in the form of a bill introduced by Senator Glenn Hegar, a Republican from Katy. It required the Public Utility Commission, which oversees ERCOT and the state's electricity utilities, to review power plants' weatherization plans. If any plan was deemed insufficient, the PUC could request more detail, but it had no enforcement authority. (The bill didn't mention the need to winterize natural gas pipelines, an omission that rendered the measure effectively meaningless, since those power plants, even if fully operational, can't produce electricity without a steady supply of gas.) Craig Estes, a Republican senator from Wichita Falls, tried to put some teeth on the bill with a substitute that required power plants to comply with the state's findings. But a few days later, Hegar's original bill was back, with

Estes's changes stripped out. Hegar, who later left the Senate and was elected state comptroller in 2014, ensured the PUC was little more than a glorified paper collector.

DeAnn Walker reiterated as much when, on February 25 of last year, the Senate business and commerce committee held a fourteen-hour hearing to determine what had happened this time around. Walker, the chair of the PUC, testified that her agency's job was simply to gather and warehouse the plans. "I don't believe we, as the PUC, have authority to require weatherization," she said.

Representative Chris Paddie looking on as Greg Abbott signs a pair of ERCOT reform bills on June 8, 2021.

Montinique Monroe/Getty

For many of the lawmakers, the lengthy hearing was a crash course in the labyrinthine mechanics and bureaucracy of the state's grid. The federal government regulates all of the country's regional grids except for ERCOT, which operates wholly inside Texas. (When regional grids experience blackouts, they are able to import power from neighboring grids; because the Texas grid is an island unto itself, with only a few small connections to Mexico and other states, importing large amounts of power isn't an option.) The ERCOT grid covers almost all of Texas, though El Paso and parts of East Texas are plugged into other regional grids. ERCOT is overseen by the PUC, whose three commissioners are appointed by the governor. Since the 2011 freeze and blackouts, all the agency's commissioners have been picked by Abbott and Perry. (The PUC was later expanded to include five commissioners.)

A separate body, the Railroad Commission of Texas, regulates the state's oil and gas industry—or at least it's supposed to. In practice, it seldom does. Its three commissioners are elected, and their campaign coffers are filled by oil and gas industry executives. Following the 2021 blackout, the commissioners expressed little interest in learning why the February storm caused statewide outages only in Texas, not in neighboring states and states far to

statewide outages only in Texas, not in neighboring states and states far to the north. They instead aggressively defended the industry they're supposed to regulate, arguing publicly that the state's failure to require winterization of natural gas providers played no role in the disaster. At the February committee hearing, Christi Craddick, then the Railroad Commission chair, tried to pin the blame on electric power producers, claiming that the gas industry was hamstrung by lack of electricity, not the other way around. "The oil field simply cannot run without power," she testified.

"Yes, it can happen again." That's what Curt Morgan, chief executive of the power company Vistra, told me.

That claim, however, doesn't withstand scrutiny. Craddick was well aware of problems with the gas supply before the blackouts began, something I discovered while reviewing records of dozens of phone calls, emails, and texts among those responsible for keeping the lights on. Five days before the blackouts began, Walker, the PUC chair, received an unwelcome call from an executive at Vistra, an Irving-based company that is the largest power producer in ERCOT. The executive warned that the company would be unable to meet the rising demand for electricity because it would soon face natural gas shortfalls at several of its plants. Texas normally produces about 29 billion cubic feet of gas a day. By February 11, when temperatures hit 22 degrees in Midland, about 915 million cubic feet were already offline, according to a federal report on the blackout. (Six days later, around the peak of the blackouts, 3.7 billion cubic feet were offline. All but 591 million of that was caused by the failure of gas infrastructure.)

That morning, Walker called Craddick. "We are going to have gas problems at our gas plants," Walker said. The next day, the Railroad Commission issued an emergency order intended to help power plants get access to gas

issued an emergency order intended to help power plants get access to gas, but the order added to the growing confusion. There was only so much natural gas to go around, and the Railroad Commission wasn't sure exactly who should get it. On February 13, two days before the blackouts began, 22 gas processing plants had been disrupted by cold weather conditions. Not a single one was disrupted by loss of electric power.

State senator Charles Schwertner in the Texas Capitol.

Photograph by Jeff Wilson

State senator Charles Schwertner, an orthopedic surgeon from Georgetown and a conservative Republican, knew little about the grid when his home's power went out that week. But he was a quick study. A few weeks after the initial February hearing, Lieutenant Governor Patrick asked him to carry the main bill to fix the grid. Schwertner later told me he concluded right away that the PUC commissioners were “derelict” in their oversight duties.

I met him in his Capitol office, which is adorned with prints of the Battle of Gettysburg and the Confederate attack on Fort Sumter. He was proud of the bill he wrote. It created a government body to ensure coordination between the gas and power industries. (As reliant as these industries are upon each other, no such formal body had existed before.) The bill directed the PUC and the Railroad Commission to levy a \$1 million fine each day on power plants, pipelines, and natural gas facilities that failed to winterize, and it allocated an initial \$21 million to the Railroad Commission to hire about a hundred inspectors to verify that the gas industry was preparing for cold weather.

When Schwertner sent his bill to the House, the legislation also created a committee to map the gas-electric supply chain and determine which gas facilities were critical to the operation of power plants. It authorized the Railroad Commission to use its hundred new employees to inspect and, if necessary, fine gas companies. When the bill returned from the House, though, the language had been revised: only companies “prepared to operate

during a weather emergency” were considered critical. This created a troubling loophole. Once the bill had passed, the Railroad Commission was responsible for implementing it, and the agency proposed a rule allowing gas companies to exempt themselves from winterizing simply by paying a \$150 filing fee and claiming that a facility wasn’t prepared to stay operational—a dizzying bit of circular reasoning.

Schwertner told me that requiring winterization for one part of the grid (the electric power providers) but not another (those who provide gas to the electricity providers) reflected the political power of the gas industry. “There was some pushback by industry,” he said, citing natural gas producers and pipeline operators. He said he didn’t like the House changes, especially the weakening of “weatherization requirements of natural gas.” Some of his colleagues were less diplomatic.

During a Senate committee hearing in September, Lois Kolkhorst, a Republican senator from Brenham, reamed out Railroad Commission executive director Wei Wang for not effectively implementing the law. Kolkhorst called the \$150 opt-out fee “disturbing.” At the same hearing, Senator John Whitmire, a Houston Democrat, offered Wang a compliment of sorts. “You’ve unified this body—let me just thank you for that. You’ve brought the family together here . . . Your rule-making proposal sucks.” Schwertner wrapped up the conversation by demanding change.

The Railroad Commission didn’t budge, and it was roundly condemned. Then, in late November, it appeared to reverse course, at least rhetorically. It announced that most pipelines and gas processing plants, along with many wells, would be required to winterize. But thus far the commission has engaged in delay tactics. These rules won’t be finalized until sometime later this year—after the winter. Perhaps the rules will be potent enough to compel real change. But if past is prologue, the new rules are likely to be ineffectual—a repeat of what happened in 2011.

Savannah and Sam Peyton, huddled in their Austin home after more than three days without power, on February 18, 2021.

Photograph by Tamir Kalifa

The week of the blackout produced staggering, hard-to-fathom energy bills Texans will be paying for years. That's because the state's electricity market broke sometime around midday on Monday, February 15. In the hours after the blackouts, ERCOT tried to shore up electricity reserves to stabilize the grid. The computer system that runs the market, though, interpreted this as an oversupply (in the middle of blackouts!) and dropped prices. When ERCOT and the PUC realized what was happening, officials decided to bypass the market and, on Monday evening, manually set prices at the maximum of \$9,000 per megawatt hour. (By comparison, the average hourly price in 2020 was \$25.73.) For fear that restarting the market and letting prices fluctuate in the midst of blackouts would lead to instability, officials kept prices at that artificially inflated level until Friday.

As a result, Texans spent an exorbitant amount on electricity during a week in which most of them couldn't get much electricity. For the entirety of 2020, Texans paid \$9.8 billion to keep the juice flowing. On February 16 alone, they spent roughly \$10.3 billion. Costs for the month of February totaled more than \$50 billion.

The bill for this pricing disaster is coming due. The Legislature approved the issuance of what will likely end up being about \$5 to \$6 billion in bonds to pay back some of these costs. That form of borrowing creates an obligation of about \$200 for every adult and child in Texas.

Of the 2,500 participants in the ERCOT market—power plant owners, electricity marketers, electric cooperatives, creditors, and traders—many are privately held and don't disclose their profits and losses. But some of the big shareholder-owned electricity generators were stuck with major losses because, while electricity prices were astronomical, natural gas prices were even higher.

As a result, anyone who had natural gas to sell came away a winner. Large

Dallas-based pipeline owner Energy Transfer posted a net profit of \$3.29 billion for the first three months of 2021; it had never posted even a \$1 billion quarterly margin before. The company chalked up its profits to preparation—it had forked over the money to winterize parts of its facilities, so they remained up and running during the storm. Kinder Morgan made \$1.41 billion, its best quarter ever. British oil giant BP, which supplies more gas in the U.S. than any other company, was coy. “It was a very exceptional quarter in gas trading,” CEO Bernard Looney told Bloomberg, which pointed to an estimate suggesting that the firm reaped \$1 billion during that stretch. Gas producer Comstock Resources president Roland Burns put it much more plainly, saying it was “like hitting the jackpot.”

Houstonians lining up to refill propane tanks on February 16, 2021.

Brett Coomer/Houston Chronicle via AP

Pastor Jessie Prince and his son Josiah handing out water at their Plano church.

Tony Gutierrez/AP

According to Bloomberg, about \$8.1 billion was spent on gas burned to generate electricity during that week. Another \$3.3 billion went for gas sold directly to homeowners, a figure that's publicly available only because the Railroad Commission approved the issuance of bonds to compensate three gas utilities that paid exorbitant prices for fuel that week. These bonds will be paid off through extra charges on customers' monthly bills, though it's not yet clear for how long. Even more was spent by city-owned gas companies, many of which have tacked on additional charges to customers' bills to pay off the enormous costs the companies ran up over a few days.

It's possible that some of the massive profits made by gas companies were illicit. The Federal Energy Regulatory Commission is looking into potential market manipulation by Texas pipeline companies, which are subject to the least regulation and oversight of any pipelines in the country. Those companies operate in a regulatory penumbra. For their pipelines operating only in Texas, they're generally exempt from reporting tariffs and other market information the federal government requires of interstate pipelines. This makes it difficult to determine whether gas prices were manipulated. In September, FERC chair Richard Glick told Congress, "We have found a

number of anomalies.” FERC later disclosed that two cases of possible natural gas market manipulation were being investigated, though it wouldn’t identify the two companies involved. (Disclosure: *Texas Monthly*’s chairman is Randa Duncan Williams, who is also chairman of the general partner of Enterprise Products Partners, a major pipeline company whose gas pipelines are located entirely within Texas. Company executives say they’ve received no inquiries from FERC.)

At the same September hearing, Kansas senator Roger Marshall asked a panel of FERC commissioners, “Was there price gouging, and who made the money?” None of the four commissioners came up with an answer.

Most Texas politicians have shown little interest in even asking this question. The chief executive of every major power plant in Texas testified repeatedly before state lawmakers. But Kelcy Warren, the chair of Energy Transfer, never appeared. Warren then gave a \$1 million campaign contribution to Abbott on June 23, shortly after the legislative session ended—a session in which Abbott, despite his initial calls to fix the grid, resisted muscular new regulation of the gas industry. (Oil and gas executives, employees, and political action committees contributed about \$16.6 million of the \$79 million that Abbott raised from 2017 through 2020, according to an analysis by Texans for Public Justice.)

For his part, Texas attorney general Ken Paxton, who’s awaiting trial on felony securities fraud charges, hasn’t announced that his office is investigating any energy price gouging. I recently asked PUC chair Peter Lake, whom Abbott appointed in the spring, after DeAnn Walker resigned, to assess who profited from the disaster. “I’m not sure anybody has a full picture of the complexity of all these financial transactions,” he said cautiously. It’s telling that Lake, who was supposedly brought in to fix ERCOT and the electricity market, dodged a question about who stood to gain and lose the most from maintaining the status quo.

Though it may be hard to believe today

Though it may be hard to believe today, Texas's grid became a pioneer in the world of electricity generation and distribution two decades ago.

The most prominent Texan who appears to be itching for a confrontation with gas companies is Paula Gold-Williams, the longtime head of CPS Energy, the city-owned utility in San Antonio. Days after the crisis ended, two gas suppliers owned by Energy Transfer sent CPS an email asking for \$317.5 million. “Due to the unprecedented weather event over the past 10 days, the price of natural gas rose dramatically,” the email said. It demanded that CPS pay cash or provide a letter of credit.

CPS refused to pay. Instead, it filed a lawsuit claiming price gouging. In its downtown San Antonio offices, Gold-Williams had watched electricity and gas prices carefully before and during the debacle. A native of San Antonio’s East Side, she trained as an accountant and worked as a regional controller for Time Warner and as a vice president at Luby’s before joining CPS, where she worked her way up to CEO. “We will pay every legitimate price and cost,” she promised, but not a dollar more. Working with her staff, she determined that about \$40 per unit of gas was reasonable. Anything above that? The \$500 price Energy Transfer had demanded? That was unconscionable. When I reached out to Energy Transfer about the financial tug-of-war, the company sent a statement saying CPS was responsible for the costs because it didn’t prepare for the storm. “CPS is trying to play politics and place blame on others,” the statement said.

But when I talked to Gold-Williams, she was resolute. “I am absolutely focused on getting to the bottom of this,” she said. In October, she announced her retirement, but she promised to stay engaged with the company—specifically to assist with the lawsuit.

Idle drilling rigs near Midland.

Matthew Busch/Bloomberg via Getty

Yes, it can happen again.” That’s what Curt Morgan, chief executive of the power company Vistra, told me when I asked about the potential for another electricity crisis. Vistra lost about \$2 billion during the storm, and it plans to spend more than \$80 million by the end of this year to prepare its plants for the next Arctic blast.

In November, I visited one of those plants, in Odessa. During the February storm, ice had accumulated and clogged the air-intake system, so Vistra is investing \$2.5 million to make sure that doesn’t happen again. From atop any of the plant’s four 10-story boilers, which produce the high-pressure steam that’s converted into electricity, you could look out toward the horizon and see a landscape dotted with pipelines, pump jacks, and flare stacks. The irony was stark: the plant sits in the heart of one of the world’s largest oil and gas fields, yet when blanketed by extreme temperatures, it couldn’t get the gas it needed to stay operational.

Though Vistra is ensuring its own plant will be able to sustain such conditions, the same can’t be said for its neighboring gas producers, which means its own investment may be futile. “I worry about the gas system,” Morgan told me. “The area that I’m most concerned about is the Railroad Commission oversight.” He’s not alone.

You might think that the natural gas industry, having scored a multibillion-dollar windfall at the expense of other Texans, might show some magnanimity in victory and agree to take steps to ensure against future blackouts. But you would be wrong. The gas industry continues to fight ferociously to avoid the kinds of regulations that are commonplace in other states. It has boosted by millions of dollars its campaign contributions to friendly politicians, including the three officials leading the Railroad Commission.

Meanwhile. Governor Abbott promised in November that “**everything that**

needed to be done was done to fix the power grid.” The Texas Tribune reported that in December, after the blackouts became an issue in his reelection campaign, Abbott went a step further by **enlisting officials at ERCOT and the PUC** to launch an optimistic public relations offensive. But when I interviewed nearly a dozen experts in natural gas and electricity, the consensus was that little has been done to secure our electric grid. ERCOT itself has admitted we could face blackouts this winter. Just before the new year, the agency released a report in which it suggested it had enough power generation to easily manage “normal” winter weather. Doug Lewin, an Austin-based energy consultant, blasted this conclusion on Twitter: “To say we have enough power in normal weather is not helpful.”

Days later, on January 2, a cold front passed through West Texas. The temperature in Midland hit a low of 14 degrees before rebounding to 56 the next day. During that brief spell, the gas infrastructure faltered, with production falling by 25 percent, according to market intelligence firm S&P Global. Still, the approach of our governor and legislators and regulators boils down to hoping we don’t see extreme temperatures anytime soon.

Indeed, forecasters predicted a relatively warm winter this year. Some might reason that if the planet is warming, Arctic storms are less likely. There is growing evidence, however, that the opposite is true. Judah Cohen, a visiting scientist at Massachusetts Institute of Technology, has published two influential papers on the topic, the first in 2018 and **another this past September**. The second paper, which appeared in *Science*, a prestigious peer-reviewed publication, explained that as the earth warms, conditions are occurring more frequently that enable a blast of Arctic air to push far into North America, even all the way down to Texas. In other words, the overall warming of the planet disrupts weather systems in ways that increase the chances for occasional extreme-cold events.

Cohen told me it all has to do with the polar vortex, an atmospheric river that circles around the Arctic at an average of 90 miles an hour. Typically, it traps the cold air in the far northern latitudes, but as Arctic Sea ice melts and the world warms, the polar vortex is more likely to wobble and stretch. In

January and February of 2021, a warm mass of air from Eurasia banged into the vortex, causing it to dip southward and push cold air as far down as the Rio Grande Valley. “Where the polar vortex goes, so goes the cold air,” Cohen explained.

So what would it cost to winterize all the wells in Texas, as most other states do, and ensure the electricity flows the next time an Arctic blast hits the Lone Star State? Dallas Federal Reserve economists cite a 2011 estimate that it would cost each gas power plant \$50,000 to \$500,000 to winterize. Statewide, it would cost between \$85 and \$200 million annually—the rough equivalent of one or two days of revenue from the Texas gas industry, and less than one-fiftieth the cost that the industry charged during the February disaster.

It’s worth noting that much of the cost of winterization would remain in the Texas economy. One of the world’s leading manufacturers of heat-tracing equipment, Thermon Group Holdings, is based in Austin and operates a major factory in San Marcos. A few years ago, it winterized an oil complex on Russia’s Sakhalin Island—where the average low temperature in January is 3 degrees Fahrenheit—for \$12 million. “All of this technology exists,” said Thermon CEO Bruce Thames. “We just haven’t invested in it in the state of Texas.”

Power lines in Austin.

Photograph by Jeff Wilson

This is particularly shameful to hear for anyone versed in Texas’s history as an energy leader.

Though it may be hard to believe today, Texas’s grid became a pioneer in the world of electricity generation and distribution two decades ago. Under Governor Perry, Texas spent \$6.9 billion on an ambitious program to build 3,600 miles of new high-voltage transmission lines. One set of lines stretched from Dallas to the Panhandle, forming a looping figure that

came to be known colloquially as “the norenead.” Other lines began in Central Texas and headed west, reaching toward Midland and Odessa and into the windy counties where the Chihuahuan Desert meets the Great Plains. In other states, the construction of comparable transmission lines often gets delayed for years, mired in bureaucratic morasses and landowner lawsuits. Texas completed its entire network in a relatively brisk nine years.

These lines were a boon to renewable energy developers, connecting the large population centers along Interstate 35 (and east to Houston) to western parts of the state, where land is cheap, landowners are welcoming, and wind and sun are plentiful. In 2020 Texas generated more renewable electricity than any other state by far. California was a distant second. By all accounts, Texas, long famed for being a global powerhouse in oil and gas, had cemented its leadership in the next generation of energy.

And then came the February blackouts. Our folly was laid bare: it’s as if we’d built a powerful, expensive car and then tried to pinch pennies by not buying antifreeze for it.

Despite this embarrassment, Texas still enjoys unmatched expertise in energy engineering, financing, and manufacturing. Some of the technology and gear developed to frack oil and gas is now being repurposed to tap renewable energy. Shipyards that once made vessels to install offshore oil rigs are now adapting for offshore wind turbines. Taking advantage of these resources would create tens of thousands of good jobs, including for workers displaced as oil and gas exploration inevitably declines.

Low-carbon grids are the future, and Texas has a multiyear head start. But before this opportunity can be grasped, the state needs political leaders and regulators who are focused on the jobs and well-being of average Texans rather than on the narrower incumbent interests of owners and executives of fossil fuel companies.

*This article originally appeared in the February 2022 issue of Texas Monthly with the headline “It Could Happen Again.” **Subscribe today.***

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Ireland's data center
energy usage was
equal to
53%
of its renewable
energy supply
in 2022

AI IS ALREADY WREAKING HAVOC ON GLOBAL POWER SYSTEMS

Technology | The Big Take

June 21, 2024



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Like much of Northern Virginia, Loudoun County was once known for its horse farms and Civil War battle sites.

But over the past 15 years, many of this community's fields and forests have been cleared away to build the data centers that form the backbone of our digital lives.

The rise of artificial intelligence is now turbocharging demand for bigger data centers, transforming the landscape even more and taxing the region's energy grids.

On a crisp afternoon this spring, the newest facility was nearing completion. When it's done, this 200,000-square-foot building could use as much energy as 30,000 homes in the US.

But first, it needs to get enough power...

The energy supply can't come soon enough for DataBank, the data center provider that owns the Virginia facility. An unnamed "big tech" client leased the entire facility and was so eager to tap into the complex to access computing resources for AI applications that it had servers ready in the building before DataBank was scheduled to have electricity for them.

"That's the thing with AI. They need a lot of power and as soon as you have it, they want it right away," said James Mathes, who manages some DataBank facilities. "Right now, it's like a blank check for AI."

The almost overnight surge in electricity demand from data centers is now outstripping the available power supply in many parts of the world, according to interviews with data center operators, energy providers and tech executives. That dynamic is leading to years-long waits for businesses to access the grid as well as growing concerns of outages and price increases for those living in the densest data center markets.

The dramatic increase in power demands from Silicon Valley's growth-at-all-costs approach to AI also threatens to upend the energy transition plans of entire nations and the clean energy goals of trillion-dollar tech companies. In some countries, including Saudi Arabia, Ireland and Malaysia, the energy required to run all the data centers they plan to build at full capacity exceeds the available supply of renewable energy, according to a Bloomberg analysis of the latest available data.

By one official estimate, Sweden could see power demand from data centers roughly double over the course of this decade — and then double again by 2040. In the UK, AI is expected to suck up 500% more energy over the next decade. And in the US, data centers are projected to use 8% of total power by 2030, up from 3% in 2022, according to Goldman Sachs, which described it as "the kind of electricity growth that hasn't been seen in a generation."

Globally, there are more than 7,000 data centers built or in various stages of development, up from 3,600 in 2015.

These data centers have the capacity to consume a combined 508 terawatt hours of electricity per year if they were to run constantly. That's greater than the total

annual electricity production for Italy or Australia.

By 2034, global energy consumption by data centers is expected to top 1,580 TWh, about as much as is used by all of India.

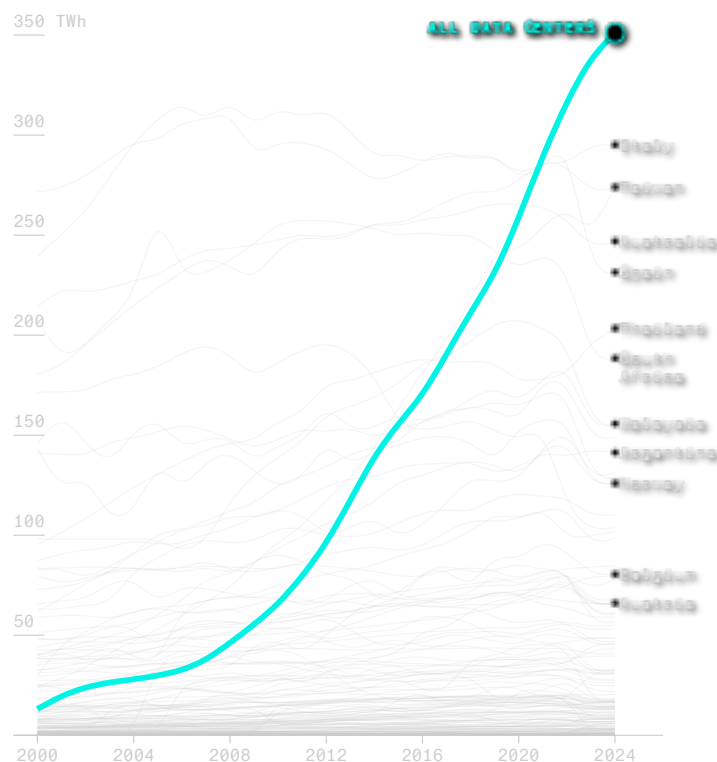
These are only estimates and there remains a high degree of uncertainty about how the current AI frenzy will play out. There's also a difference between the projections for how much electricity data center developers want and how much generation actually gets built.

While tech companies are quick to point out that data centers account for less than 2% of global energy use even with all the expansion, an April report from Goldman Sachs estimates that figure could rise to 4% by the end of the decade. Any percentage point increase is monumental, given overall electricity demand has remained almost flat for years — if not declining in some regions.

In the US, power demand is expected to grow by 40% over the next two decades, compared with just 9% growth over the past 20 years, according to John Ketchum, chief executive officer at NextEra Energy Inc., the world's biggest builder of wind and solar energy that isn't backed by a government. Data centers are the biggest reason for that demand boom, Ketchum said, citing electrification and manufacturing as other factors. Asked why data centers were suddenly sucking up so much power, his answer was blunt: "It's AI," he said, citing the energy needs for training models and also the inference process by which AI draws conclusions from data it hasn't seen before. "It's 10 to 15 times the amount of electricity."

Altogether, data centers use more electricity than most countries

Only 16 nations, including the US and China, consume more



Sources: Bloomberg analysis of BloombergNEF and DC Byte data

Note: Data center energy consumption through Q1 2024. National energy consumption levels are actual through 2022 and projected for 2023 and 2024.

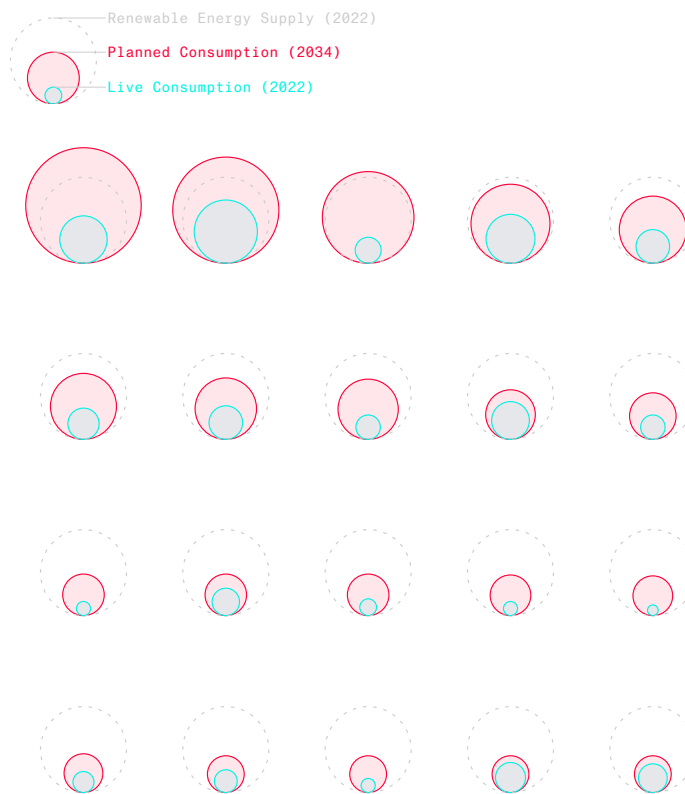
The biggest cloud services providers, Amazon.com Inc., Microsoft Corp. and Alphabet Inc.'s Google, have announced goals to run their data centers entirely on green energy — Amazon by next year, Google and Microsoft by 2030. All three say they are working on technological methods to use less power or balance the demand on the grid more efficiently. That can include wringing more efficiency from chips and servers, laying out equipment in ways that require less cooling and shifting loads to different areas based on where energy — particularly green energy — is available.

But some tech leaders like OpenAI CEO Sam Altman have said an energy breakthrough — likely from nuclear power — is needed to adapt to this new picture. Microsoft and Amazon are also betting on nuclear energy, with Amazon recently buying a nuclear-powered data center in Pennsylvania and indicating it's open to more. For now, the path forward remains unclear. Microsoft recently admitted that its AI push is jeopardizing its long-held goal to be carbon negative by 2030.

"We need terawatts and terawatts more of traditional green energy, whether it's wind or solar, and that's across the globe," said Amanda Peterson Corio, Google's global head of data center energy, speaking broadly rather than solely about AI demands. For context, a single terawatt is equivalent to the output of about 1,000 nuclear power plants. "Of course we want to decarbonize ourselves, but if we're just decarbonizing ourselves and not the whole grid, what's the point?"

Global renewable energy supplies under pressure

As new data centers are built, their energy usage could equal or exceed some countries' renewable energy supply



Sources: Bloomberg analysis of BloombergNEF and DC Byte data

Note: These 20 countries have the highest energy consumption from data centers relative to their renewable energy supply. The latest available data for renewable electricity supply is for 2022.

In today's data centers, you might find thousands of Nvidia Corp.'s coveted H100 chips — the engine of the generative AI boom — each of which draws as much as 700 watts, or nearly eight times the power used by a typical 60-inch flat screen TV. Data centers built for training AI models require even more. Microsoft, for example, strung together tens of thousands of Nvidia processors inside a facility used to develop OpenAI's technology. These are fed by networking and other types of chips which, combined with machinery in data centers used to prevent the gear from overheating, saps up even more power.

And the conventional wisdom in Silicon Valley is that the amount of energy needed will only go up. Nvidia's newest chip, the B100, can consume nearly twice as much power as the H100. Nvidia contends companies will be able to do more with fewer chips, but Ian Buck, the company's head of accelerated computing, admits it's likely AI deployments will increase. "People like to fill their data centers," he said.

AI development is evolving fast, too, with a feverish push toward developing ever larger artificial intelligence models. The Microsoft supercomputer built in 2020 that trained OpenAI's GPT-3 system used 10,000 of what was then the latest AI chip. A November 2023 Microsoft supercomputer relied on 14,400 of the top of the line Nvidia H100 chips and the next one, which is already being designed, will be 30 times more powerful, according to a May slide presentation by Microsoft Azure Chief Technology Officer Mark Russinovich.

Meanwhile, Nvidia CEO Jensen Huang said many nations will push to build their own "sovereign" AI systems to stay competitive and process data

locally. The global battle for AI supremacy may well depend on which countries have enough data centers and power to support the technology. If so, Loudoun County is a vision of what's to come for the rest of the world — and the challenges other countries will face keeping up.

The data center capitals of the world

Over the past five years, Dominion Energy Inc., the power company that services Loudoun County, also known as “data center alley,” has connected 94 data centers that consume about four gigawatts of electricity, combined. Now it's fielding requests for data centers campuses that would consume multiple gigawatts — enough to power hundreds of thousands of homes — two or three of which could use as much electricity combined as all the facilities hooked up since 2019.

The surge in demand is causing a backlog. Data center developers now have to wait longer to hook their projects up to the electric grid. “It could be as quick as two years, it could be four years depending on what needs to be built,” Dominion Energy Virginia president Edward Baine said in an interview.

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Dominion is trying to build out the infrastructure to support it. New power lines hang from massive metal towers and run along roads and over creeks to feed electricity to these towering, windowless data centers. The company is building a large new wind farm off the coast and a lot of solar farms, but coal and gas powered plants could also stay online longer.

In late 2022, Dominion filed a previously unreported letter to its regulators asking for permission to build new substations and power lines to serve “unprecedented” load growth. In the letter, Dominion said it experienced 18 load relief warnings in the spring of that year. These warnings occur when the grid operator tells the company that it might need to shed load, the technical term for the controlled interruption of power to customers, which could include rotating outages.

“This is far outside of the normal, safe operating protocol,” Dominion told regulators.





A DataBank location in Ashburn, Va. *Photographer: Moriah Ratner/Bloomberg*

Virginia is not alone in struggling to keep pace with demand. In West London, historically considered a hub for data centers, new facilities have to wait until 2030 to connect to the grid, according to David Bloom, chairman of UK-based data center operator Kao Data. "We are now being pushed into new locations," he said. Likewise, in the south of Sweden, a strong market for renewables, there is so much demand to get connected that businesses may have to wait years. "We have one queue, and you need to get your ticket," said Peter Hjalmar, German utility E.ON SE's regional manager for southern Sweden. And across the US, many new AI data centers are expected to consume 100 megawatts each, according to a recent Morgan Stanley analysis, prompting the analysts to wonder "how all of the proposed data center projects will be powered in a timely manner." Demand is so high that large tech companies are having bidding wars over data center sites with ready access to power, according to NextEra.

Data center growth, as it's being forecast, may also run up against the limits of how much power can be carried through transmission lines, said Ali Farhadi, CEO of the Allen Institute for AI. "I don't think we can move that much electricity around the globe, forget about generating it," he said.

Data centers will get more efficient over time, energy analysts say, but they're also getting significantly bigger. The average size of data center facilities worldwide is now 412,000 square feet, an almost fivefold increase from what it was in 2010, according to data from DC Byte, a market intelligence firm.

More powerful data centers require more land

Building Area (square feet)

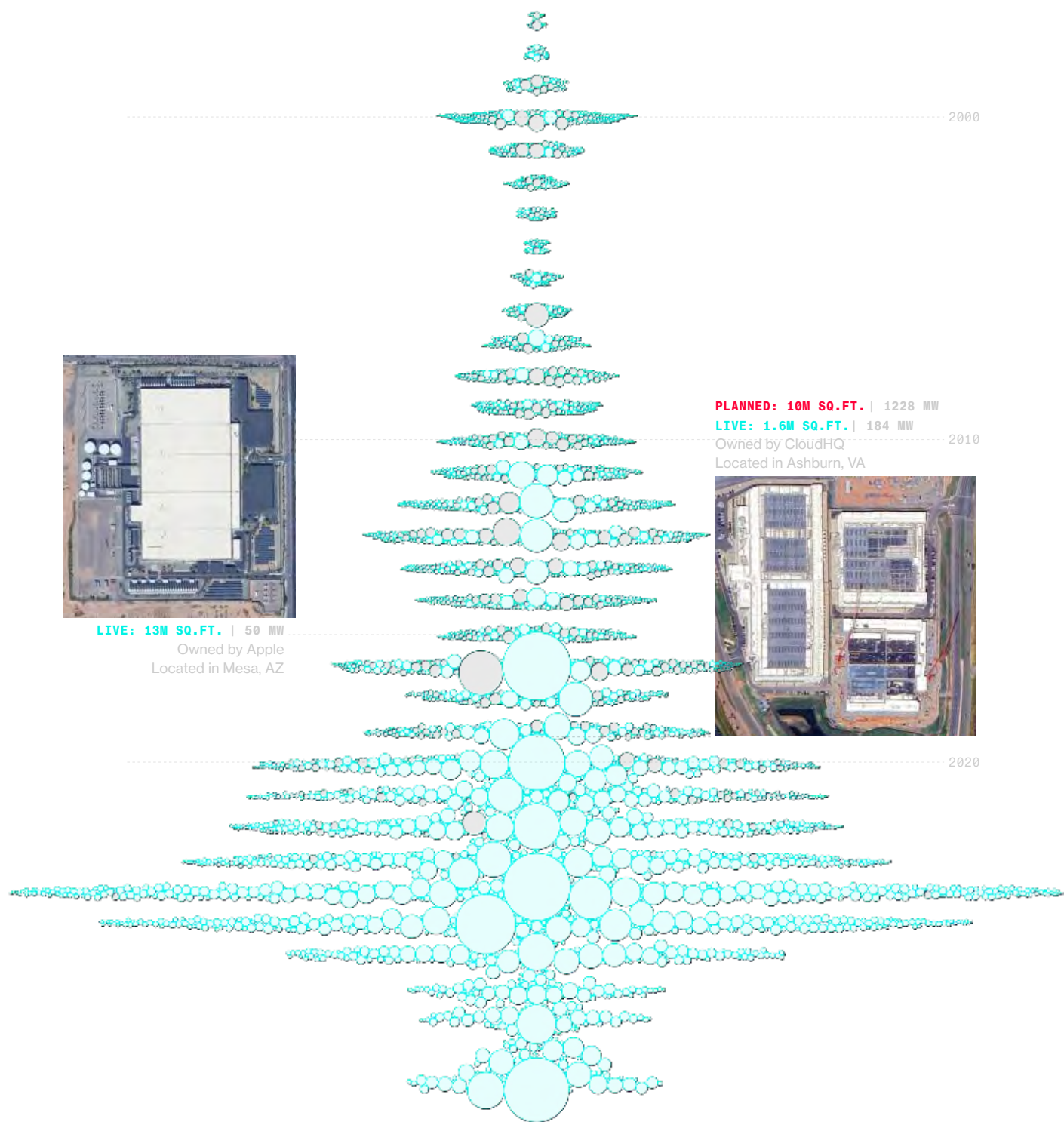


○ Includes building area in development stages



LIVE: 11K SQ.FT. | 1 MW
Owned by TIM
Located in Palermo, Italy





Source: Bloomberg analysis of DC Byte data

Note: Values are shown on a logarithmic scale. Includes all data centers built since 1930.

The surge in data center demand, combined with heavy investments from power companies like Dominion on new substations, transmission lines and other infrastructure to support it, are also increasing the likelihood customers will see their energy prices go up, experts say. The cost of some upgrades are typically allocated among electricity customers in an entire region, showing up as a line item on everyone's monthly utility bill.

Goldman Sachs estimates that US utility companies will have to invest roughly \$50 billion in new power generation capacity to support data centers. "That's going to raise energy prices for both wholesale energy and retail rates," said power market analyst Patrick Finn of energy consultancy Wood Mackenzie.

Costs including grid improvements are divided among each customer class, from residential to industrial, based on how much it actually costs to serve each, according to a Dominion representative. As a result, residential customers have seen their share of transmission costs drop in recent years while data centers have seen their portion rise, the representative said.

In Ireland, another heavily saturated market, there are some early signs of rate increases. Blessed with a moderate climate, Ireland has attracted so many data centers from Microsoft, Amazon and others that these facilities are forecast to consume a third of the country's energy by 2026, up from 18% in 2022.

Wholesale power prices in Ireland have been a third higher on average this year than the rest of Europe. Other factors, including the country's geography, play a role, but Sarah Nolan, senior modeler at Cornwall Insight, said growing data center demand can contribute — and that's in a country that took strong steps to limit buildout just before the AI craze kicked off.

To manage energy constraints, Ireland's state-owned electricity operator imposed a moratorium in Dublin in early 2022 and set out conditions to connect new data centers to the grid, including a preference for those

generating their own electricity. The operator has “comprehensive” plans to build out the grid, said Donal Travers, the head of technology, consumer and business services for IDA Ireland, the state agency tasked with attracting foreign direct investment. But he said restrictions on large new connections are expected to continue “probably until 2028 or thereabouts.”

In the meantime, other regions are all too eager to open their doors.



A new data center under construction in Gainesville, Va. on March 20, 2024. *Photographer: Moriah Ratner/Bloomberg*

The next Virginia

When Rangu Salgame looks at Malaysia, he sees the next Virginia “in the making.” Johor, the southernmost state in peninsular Malaysia, has a policy to speed up clearances for data centers. Crucially, it’s also a short drive to Singapore, a longtime data center hub that imposed a moratorium for several years on new facilities to manage energy growth on the tiny island.

Once a sleepy fishing village, the suburbs of the city of Johor Bahru are now marked by vast construction sites. Microsoft and Amazon are investing in the region, as is Salgame’s company, Princeton Digital Group (PDG). At Sedenak Tech Park, a sprawling complex about 40 miles south of Johor Bahru’s city center, towering cranes dot the sky. PDG’s new 150 megawatt data center occupies one corner of the park, across from similar facilities from other providers.

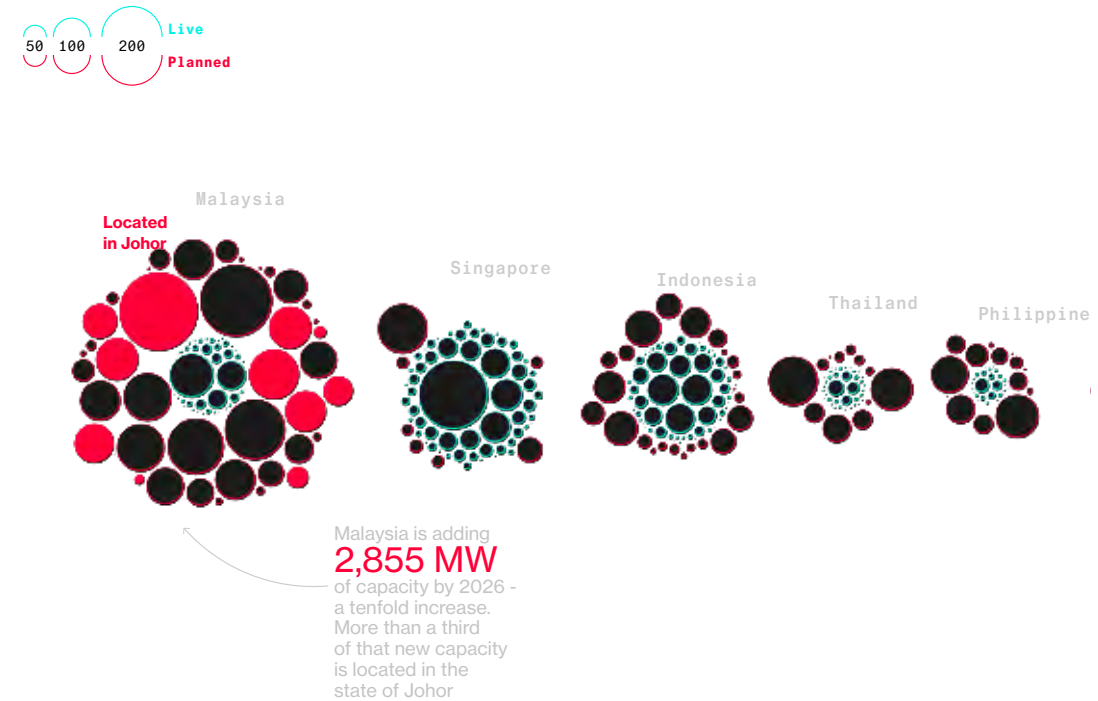
“We have gone from shovel to production in 12 months,” said Salgame, who expects to have 300 megawatts of capacity in Johor within two years. “Not all parts of the world can execute at this speed and scale.”

But even markets eager to streamline data center buildout face constraints. What’s missing in Johor, especially for an industry like tech that is known for its climate pledges, is renewable energy. The power supply at Sedenak comes from Tenaga Nasional Berhad, which uses coal or gas-fired plants. While Malaysia has ambitious goals to bolster renewables, including plans to build a 500-megawatt solar farm in Johor, today it relies on coal for more than a third of its generation. Most of Malaysia’s data center capacity is not in use yet, but factoring in everything under construction, the amount of electricity used just by data centers would exceed the country’s total renewable output in 2022, the latest year for which data is available, a Bloomberg analysis found.

A data rush in Southeast Asia

Driven by Malaysia, the region has 153 new data centers that could potentially be built in the near future, adding a potential 5,419 MW of capacity

Total Energy Capacity (MW)



Source: Bloomberg analysis of DC Byte data

Like Malaysia, Texas has emerged as one of the fastest growing data center markets in the US, thanks in part to the promise of shorter wait times on its independent and deregulated grid. Texas sites can get on the grid in the one to two years it takes to build data centers, said Bobby Hollis, the vice president of energy at Microsoft, which is the largest player in Texas by megawatt, according to DC Byte.

Texas offers plentiful solar power and, in the state's panhandle, some of the best access to wind power in the world, Hollis said. But a hotter climate and strained water supplies are pushing Microsoft and Google to try weaning their data center cooling gear off water. Alternate approaches, however, require more energy — about 5% more on average, according to Microsoft.

While power in Texas looks plentiful, there are limits. Solar panels and other gear needed for clean power are starting to see some supply constraints, Hollis said. The Electric Reliability Council of Texas also recently cautioned that it has underestimated demand in the San Antonio area, where Microsoft's big data center campus is located, potentially causing cascading outages statewide in the future.

Back in Virginia, opposition to data centers is growing. At a March supervisors meeting in Prince William County, frustrated residents spoke out against a zoning change that would allow data centers on a specific plot of land to be built about twice as tall. One woman told officials that data centers were turning her quiet neighborhood into a "dystopian nightmare."

A 48-year-old homemaker named Rachel Ellis spoke remotely to say the change would mean more demand on a grid that's already strained. "It's

reckless governing to continue to approve data centers without knowing the full impacts of where this electricity will come from,” she said.

After hearing a dozen people speak against the zoning change, the supervisors voted. The bigger data centers were approved.

Related tickers:
MSFT (Microsoft Corp.)
AMZN (Amazon.com Inc.)
NVDA (NVIDIA Corp.)
GOOGL (Alphabet Inc.)
D (Dominion Energy Inc.)
NEE (NextEra Energy Inc.)

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With assistance from: Olivia Solon, Eamon Farhat, Lars Paulsson, Dan Murtaugh, Matt Day, Seohee Song, Matthias Kimmel and Rodrigo Quintero for Bloomberg Technology
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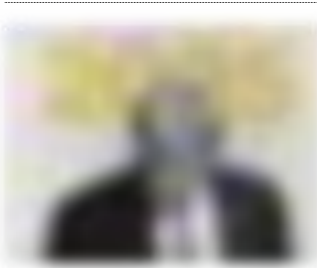
Methodology

Data collection
Bloomberg collected data on the energy capacity of data centers globally, including active facilities and those under construction, from DC Byte, a market intelligence firm. Bloomberg also relied on DC Byte data to determine the location and size of thousands of facilities around the world. Using BloombergNEF, a research service, Bloomberg gathered data on electricity consumption and generation by fuel type for every country. Bloomberg excluded data centers intended for cryptocurrency mining from the analysis.

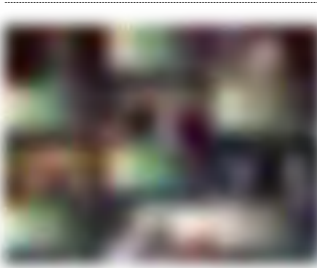
Calculations
Bloomberg converted data center capacity values to energy consumption estimates using the following formula: MWh = (capacity) * (hours in a year) * (utilization rate) * (Power Usage Effectiveness) where capacity is the installed IT capacity, utilization rate is 70% and Power Usage Effectiveness (PUE) is equal to an average of 1.5. This calculation assumes that data centers are running 70% of the time and that their PUE -- a ratio to determine a data center's efficiency -- is 1.5 on average. These numbers can vary from facility to facility, but Bloomberg had energy experts review these calculations.

Renewables
Bloomberg calculated the ratio of available renewable electricity (as of 2022, per latest data) to data center electricity consumption (estimate) for each country. In some countries, the data center electricity consumption estimate is nearly equal to, or greater than, the supply of renewable electricity.





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"OPEN-AND-SHUT" CASE?

Meta mocked for raising “Bob Dylan defense” of torrenting in AI copyright fight

Meta fights to keep leeching evidence out of AI copyright battle.

ASHLEY BELANGER – MAR 12, 2025 1:01 PM | 124



Credit: Pgiam | E+



TEXT SETTINGS

Authors think that Meta's admitted torrenting of a pirated books data set used to train its AI models is evidence enough to win their copyright fight—which previously hinged on a court ruling that AI training on copyrighted works isn't fair use.

Moving for summary judgment on a direct copyright infringement claim on Monday in a US district court in California, the authors alleged that "whatever the merits of generative artificial intelligence, or GenAI, stealing copyrighted works off the Internet for one's own benefit has always been unlawful."

In their filing, the authors accused Meta of brazenly deciding to torrent terabytes of pirated book data after attempts to download pirated books one by one "posed an immense strain on Meta's networks

and proceeded very slowly."

Knowing that such activity has been deemed infringing for more than two decades, the authors alleged, Meta took a risk, seemingly hoping to evade detection while struggling to catch up in the AI race and needing speedier access to large chunks of data. To cover its tracks, the social media company allegedly deviated from usual practices and attempted to conceal the torrenting by using Amazon Web Services.

"In most cases, and in this case too, users who download via torrent also upload the same file they are downloading to reap the benefits of faster file sharing," the authors alleged.

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In February, authors argued that Meta's torrenting of the pirated books was infringing, even if Meta limited seeding when the downloads were completed, as the company claims it does. They explained that Meta's leeching during the download process (allowing other users to download partial files before the download was completed) is allegedly evidence enough that Meta shared pirated books with others.

"There is no genuine dispute that Meta made widely available and even reuploaded to other online pirates at least some quantity of the pirated data as part of the peer-to-peer (P2P) sharing process," the authors alleged. "Meta's response in this case seems to be that a powerful technology corporation should not be held to the same standard as everyone else for illegal conduct."

The authors mocked Meta for raising what they call "the Bob Dylan defense" of its torrenting, citing song lyrics from "Sweetheart Like You" that say, "Steal a little and they throw you in jail / Steal a lot

and they make you king."

Meta opposes requests for leeching evidence

Meta does not want the court to weigh these leeching claims. Last week, Meta argued that authors should not be allowed to do more discovery on Meta's alleged leeching or introduce a new expert to potentially discuss why the leeching may have clinched the case for the authors.

While resisting introducing new evidence on leeching, Meta simultaneously argued that the authors' motion for summary judgment based on the leeching theory is inappropriate because Meta has not had a chance to defend against the claims.

"They intend to move for summary judgment on torrenting issues, presumably in reliance on this new theory in a new expert report from a new expert, to which Meta has not had an opportunity to investigate or respond," Meta's letter said.

On May 1, Judge Vince Chhabria will weigh these arguments at a hearing where Meta will get a chance to respond to the leeching claims. Last week, Chhabria wrote in an order that consideration will be given to whether "it would be unfair to Meta" to rule on the summary judgment at this stage.

The authors, however, think that torrenting pirated works is so notoriously illegal that they now have an "open-and-shut case" of copyright infringement.

"Meta's reproduction of Plaintiffs' Copyrighted Books without permission, including through peer-to-peer file sharing, is not fair use," the authors alleged, citing a major court ruling against Napster and insisting that "Meta infringed each of their copyrights, full stop."

Chhabria may be curious to learn more about leeching, though. Last month, he admitted at a hearing that the term was foreign to him, Meta's letter said in a footnote.

"I don't remember hearing it before," Chhabria said.

The authors are hoping to make Meta pay after Meta allegedly shirked offers to license their data for a fee.

"Meta plainly attributed significant value to the copyrighted works it took for free: a windfall to Meta, but not for authors, who were paid nothing," the authors alleged. Further, "Whether another user actually downloaded the content that Meta made available" through torrenting "is irrelevant," the authors alleged. "Meta 'reproduced' the works as soon as it made them available to other peers."

Meta resists request to depose Zuckerberg

The authors want Chhabria to agree that Meta's alleged leeching is key to winning their case. Their filing even pointed out that Meta's pirating included copies of books written by at least 10 Supreme Court justices, seemingly hoping the judge will see that Meta's activity harms more than just authors.

To further their case, the authors had asked for additional discovery requiring Meta to provide written answers about their torrenting and leeching. They also sought to depose both Meta employees who previously testified, including Mark Zuckerberg, as well as those whose roles in Meta's torrenting, they suggested, were only recently clarified in unsealed emails.

"That Meta knew taking copyrighted works from pirated databases could expose the company to enormous risk is beyond dispute: it triggered an escalation to Mark Zuckerberg and other Meta executives for approval," the authors argued. "Their gamble should not pay off."

Meta said the authors' new discovery requests were "unnecessary, unwarranted, and infeasible." The company would only agree to allow six employees to be deposed ahead of the May hearing, including Nikolay Bashlykov, a software engineer who sent an internal message at Meta saying, "Torrenting from a corporate laptop doesn't feel right."

However, the authors "have made no showing to justify additional deposition time with Meta's CEO Mr. Zuckerberg," Meta claimed, offering instead two senior-level employees "who can speak to executive decision-making."

Piracy can never be fair use, authors say

The authors claimed there are gaps in the court's understanding about Meta's torrenting, pointing out that Meta's expert failed to replicate the company's torrenting in her analysis, leaving it unclear "how much data Meta uploaded and/or seeded." Meta's expert also allegedly ignored that "BitTorrent's default configuration provides for continuous uploading during the 'leeching' phase—simultaneous to downloading."

Although the authors expect their leeching theory may be a winning one, they noted that fair-use findings typically come from juries, not from judges at the summary judgment stage. They also acknowledged that the court may decide "that the fair use analysis applies to Meta's unmitigated piracy and use of torrenting."

But "it should nevertheless grant summary judgment under the four fair use factors regarding Meta's decision to make available to other P2P pirates millions of copyrighted books in exchange for faster download speed," they argued.

Considering that Meta hasn't found a single case where a court determined downloading or uploading pirated works on P2P networks is fair use, the authors warned, "The use of piracy to further piracy can never be 'fair use.'"

**ASHLEY BELANGER** SENIOR POLICY REPORTER

Ashley is a senior policy reporter for Ars Technica, dedicated to tracking social impacts of emerging policies and new technologies. She is a Chicago-based journalist with 20 years of experience.

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Making AI Less “Thirsty”: Uncovering and Addressing the Secret Water Footprint of AI Models

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Abstract

The growing carbon footprint of artificial intelligence (AI) has been undergoing public scrutiny. Nonetheless, the equally important water (withdrawal and consumption) footprint of AI has largely remained under the radar. For example, training the GPT-3 language model in Microsoft’s state-of-the-art U.S. data centers can directly evaporate 700,000 liters of clean freshwater, but such information has been kept a secret. More critically, the global AI demand is projected to account for 4.2 – 6.6 billion cubic meters of water withdrawal in 2027, which is more than the total annual water withdrawal of 4 – 6 Denmark or half of the United Kingdom. This is concerning, as freshwater scarcity has become one of the most pressing challenges. To respond to the global water challenges, AI can, and also must, take social responsibility and lead by example by addressing its own water footprint. In this paper, we provide a principled methodology to estimate the water footprint of AI, and also discuss the unique spatial-temporal diversities of AI’s runtime water efficiency. Finally, we highlight the necessity of holistically addressing water footprint along with carbon footprint to enable truly sustainable AI.

1 Introduction

- “Water is a finite resource, and every drop matters.” — *Facebook (now Meta) Sustainability Report 2020* [1].
- “Fresh, clean water is one of the most precious resources on Earth ... Now we’re taking urgent action to support water security and healthy ecosystems.” — *Google’s Water Commitment 2023* [2].
- “Water is a human right and the common development denominator to shape a better future. But water is in deep trouble.” — *U.N. Secretary-General António Guterres at the U.N. Water Conference 2023* [3].
- “Historic droughts threaten our supply of water ... As the source of both life and livelihoods, water security is central to human and national security.” — *The U.S. White House Action Plan on Global Water Security 2022* [4].

Artificial intelligence (AI) has enabled remarkable breakthroughs in numerous areas of critical importance, including tackling global challenges such as climate change. On the other hand, many AI models, especially large generative ones like GPT-4, are trained and deployed on energy-hungry servers in warehouse-scale data centers, accelerating the data center energy consumption at an unprecedented rate [5]. As a result, AI’s carbon footprint has been undergoing scrutiny, driving the recent progress in AI carbon efficiency [6,7]. However, AI’s water footprint — many millions of liters of freshwater consumed for cooling the servers and for electricity generation — has largely remained under the radar and keeps escalating. If not properly addressed, AI’s water footprint can potentially become a major roadblock to sustainability and create social conflicts as freshwater resources suitable for human use are extremely limited and unevenly distributed.

As acknowledged in Google’s sustainability report [8] and the recent U.S. Department of Energy study [5], the expansion of AI products and services is a key driver of the rapid increase in data center water consumption. Even excluding the water usage in leased third-party colocation facilities, Google’s self-owned data centers alone directly withdrew 29 billion liters and consumed (i.e., evaporated) more than 23 billion liters of freshwater for on-site cooling in 2023, nearly 80% of which was potable water [8].² This amount of annual water consumption even rivals that of a major household-name beverage company [9]. Importantly, Google’s data center water consumption increased by ~20% from 2021 to 2022 and by ~17% from 2022 to 2023 [8], and Microsoft saw ~34% and ~22% increases over the same periods, respectively [10]. Furthermore, according to the U.S. Department of Energy, the total annual on-site water consumption by U.S. data

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²The detailed difference between water withdrawal and water consumption is presented in Section 2.1.

centers in 2028 could double or even quadruple the 2023 level, reaching approximately 150 – 280 billion liters and further stressing the water infrastructures [5].

AI represents the fastest expanding workloads in data centers [5,8]. For example, a recent study suggests that the global AI could consume 85 – 134 TWh of electricity in 2027 [11], whereas a more aggressive projection by the U.S. Department of Energy predicts that AI servers’ electricity consumption in the U.S. alone will surpass 150 – 300 TWh in 2028 [5]. Even considering the lower estimate, the combined scope-1 and scope-2 water withdrawal of global AI is projected to reach **4.2 – 6.6 billion cubic meters** in 2027, which is more than the total annual water withdrawal of 4 – 6 Denmark or half of the United Kingdom.³ Simultaneously, a total of 0.38 – 0.60 billion cubic meters of water will be evaporated and considered “consumption” due to the global AI demand in 2027. Moreover, these global estimates will be exceeded by the total water withdrawal and consumption attributed to AI in the U.S. alone in 2028 if the U.S. Department of Energy’s projection comes to fruition.

Despite its profound environmental and societal impact, the increasing water footprint of AI has received disproportionately less attention from the AI community as well as the general public. For example, while the scope-2 carbon emissions are routinely included as part of AI model cards, even scope-1 direct water usage (either withdrawal or consumption) is missing, let alone scope-2 water usage. This may impede innovations to enable water sustainability and build truly sustainable AI. Crucially, water and carbon footprints are complementary to, not substitutable of, each other for understanding the environmental impacts. Indeed, optimizing for carbon efficiency does not necessarily result in, and may even worsen, water efficiency, which varies with the fuel mixes for electricity generation and outside weather in a unique way [5,13].

To ensure that the growth in AI does not exacerbate the global water stresses or outweigh the environmental benefits it provides, it is critical to uncover and address AI’s hidden water footprint amid the increasingly severe freshwater scarcity crisis, worsened extended droughts, and quickly aging public water infrastructure. The urgency can also be reflected in part by the recent commitment to “Water Positive by 2030” from industry leaders, including Google [8] and Microsoft [10], by policy guidelines and legislative efforts to mitigate AI’s water consumption [14,15], and by the inclusion of water consumption as a key environmental metric into the first international standard on sustainable AI to be published by the ISO/IEC [16]. Moreover, setting targets for minimizing water consumption is included as part of the recent U.S. executive order for advancing AI data center infrastructures [17].

In this paper, we advocate for a holistic approach to sustainable AI that extends beyond the carbon footprint to also address the water footprint. Specifically, we present a principled methodology to estimate AI’s total water footprint, including both operational water and embodied water. By taking the GPT-3 model with 175 billion parameters as an example [18], we show that training GPT-3 in Microsoft’s U.S. data centers can consume a total of 5.4 million liters of water, including **700,000 liters** of scope-1 on-site water consumption. Additionally, GPT-3 needs to “drink” (i.e., consume) a **500ml bottle of water** for roughly 10 – 50 medium-length responses, depending on when and where it is deployed. Note that our estimate of inference water consumption for GPT-3 is conservative, and the actual water consumption could be several times higher.

Next, we emphasize the need for increasing transparency of AI’ water footprint, including disclosing more information about operational data and keeping users informed of the runtime water efficiency. We show that WUE (Water Usage Effectiveness, a measure of water efficiency) varies both spatially and temporally, suggesting that “when” and “where” to perform AI training can significantly affect the water footprint. Finally, we highlight the necessity of holistically addressing the water footprint along with the carbon footprint to enable truly sustainable AI — *the water footprint of AI can no longer stay under the radar*.

2 Background

2.1 Water Withdrawal vs. Water Consumption

There are two related but different concepts — water withdrawal and water consumption, both of which are important for understanding the impacts on water stress and availability [19,20].

³The scope definition of water usage [12] is in line with that of carbon emissions and is discussed in Section 2.2. Our scope-2 water withdrawal (and consumption when applicable) is for location-based electricity generation throughout the paper. Large data centers often adopt sustainability programs (e.g., renewable purchasing agreements) to offset their location-based electricity usage and thus may have lower market-based carbon and water footprints.

- **Water withdrawal:** It refers to freshwater taken from the ground or surface water sources, either temporarily or permanently, and then used for agricultural, industrial, or municipal uses (normally excluding water used for hydroelectricity generation) [19]. As water is a finite shared resource, water withdrawal indicates the level of competition as well as dependence on water resources among different sectors.

- **Water consumption:** It is defined as “water withdrawal minus water discharge”, and means the amount of water “evaporated, transpired, incorporated into products or crops, or otherwise removed from the immediate water environment” [20]. Water consumption reflects the impact on downstream water availability and is crucial for assessing watershed-level scarcity [19].

These two types of water usage correspond to two different water footprints, i.e., water withdrawal footprint (WWF) [12,21] and water consumption footprint (WCF), respectively [22]. By default, water footprint refers to the water consumption footprint unless otherwise specified.

2.2 How Does AI Use Water?

AI’s water usage spans three scopes: on-site water for data center cooling (scope 1), off-site water for electricity generation (scope 2), and supply-chain water for server manufacturing (scope 3).

2.2.1 Scope-1 Water Usage

Nearly all the server energy is converted into heat, which must then be removed from the data center server room to avoid overheating. This process involves two sequential stages: server-level cooling followed by facility-level cooling.

In the server-level cooling stage, heat is transferred from the servers to the facility or a heat exchanger, typically using either air or liquid cooling methods (e.g., direct-to-chip cooling or immersion cooling), which do not evaporate or consume water. In general, new data centers dedicated to AI training often rely on liquid cooling due to the high server power densities.

In the facility-level cooling stage, heat is rejected from the data center facility to the outside environment. While there are various cooling methods, water-intensive cooling towers and water evaporation-assisted air cooling are two common approaches used in many data centers, including those operated by major technology companies [5,8].

Cooling tower. As illustrated in Figure 1, some water is evaporated (i.e., “consumed”) in the cooling tower to dissipate heat into the environment, while the remaining water moves along an open loop to the heat exchanger to further absorb the server heat. Additionally, non-evaporated water can be recycled only a few times (typically 3–10 cycles, depending on water quality) before discharge, requiring continuous clean freshwater replenishment to prevent mineral and salt buildup. Thus, to keep the cooling tower working, new water must be constantly added to make up for the evaporated water and discharged water. Importantly, clean freshwater (potable water in many cases [8]) is needed to avoid pipe clogs and/or bacterial growth.

For cooling towers, water withdrawal refers to the amount of added water, including both evaporated water and discharged water, while water consumption exclusively indicates the amount of evaporated water. With good water quality, roughly 80% of water withdrawal is evaporated and considered “consumption” [8]. On average, depending on the weather conditions and operational settings, data centers can evaporate approximately 1 – 9 liters per kWh of server energy: 1 L/kWh for Google’s annualized global on-site water efficiency [8] and 9 L/kWh for a large commercial data center during the summer in Arizona [23].

Air cooling with water evaporation assistance. When the climate condition is appropriate, data centers may use “free” outside air to directly reject the heat to the outside environment. Nonetheless, water evaporation is still needed when the outside air is too hot (e.g., higher than 85 degrees Fahrenheit); additionally,

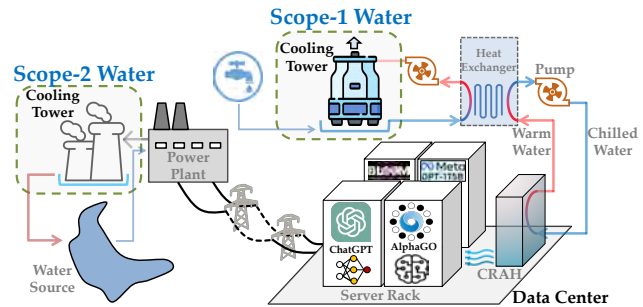


Figure 1: An example of data center’s operational water usage: on-site scope-1 water usage for data center cooling (via cooling towers in the example), and off-site scope-2 water usage for electricity generation. The icons for AI models are only for illustration purposes.

water is also needed for humidity control when the outside air is too dry [24]. The added water is considered “withdrawal”, out of which about 70% is consumed based on Meta’s report [25]. Generally, outside air cooling is more water-efficient than cooling towers on average. However, hot weather raises the evaporative water demand and maximum water consumption, potentially stressing local water supplies during peak demand on hot days. Additionally, the application of outside air cooling may have challenges in hot regions and/or for many colocation facilities that are located in business districts.

Some data centers may opt for dry coolers, which consume no on-site water year-round [26]. However, this approach typically increases cooling energy consumption compared to water-based cooling methods, potentially exacerbating the overall stress on water resources due to higher scope-2 water consumption.

2.2.2 Scope-2 Water Usage

In many countries, thermoelectric power is among the top sectors in terms of water withdrawal and water consumption [12]. Thus, similarly to scope-2 carbon emissions, data centers are accountable for off-site scope-2 water usage associated with electricity consumption, which forms part of the “true water cost of data centers,” as highlighted by the U.S. Department of Energy [5].

Different power plants use different amounts of water for each kWh generation, depending on the cooling techniques. Typically, water withdrawal due to hydropower generation is excluded, but water consumption due to increased water evaporation rates from hydropower generation is included [5]. For electricity generation, the U.S. national average water withdrawal and consumption are estimated at about 43.8 L/kWh [27] and 3.1 L/kWh [12], respectively. Meta’s self-reported scope-2 water consumption for its global data center fleet was 3.7 L/kWh (i.e., 55,475 megaliters divided by 14,975,435 MWh) in 2023 [25].

2.2.3 Scope-3 Water Usage

AI chip and server manufacturing uses a huge amount of water [28,29]. For example, ultrapure water is needed for wafer fabrication and water is also needed for keeping semiconductor plants cool. Importantly, the discharged water may contain toxic chemicals and/or hazardous wastes. While water recycling at semiconductor plants can effectively reduce water withdrawal, the recycling rate in many cases remains low, e.g., the average recycling rate for wafer plants and semiconductor plants in Singapore are 45% and 23%, respectively [29]. Although largely obscure, scope-3 water usage is likely significant [28]. For instance, Apple reports that its supply chain accounts for 99% of its total water footprint [30].

It is important to recognize that, unlike agriculture whose water footprint is mostly green (i.e., water stored in soil and used by plants), the majority of AI’s water footprint is blue water extracted from rivers, lakes, or groundwater, which is directly accessible for human use but often more limited in availability.

3 Estimating AI’s Water Footprint

We present a general methodology for estimating AI’s water consumption footprint. To obtain the water *withdrawal* footprint, we simply replace the WUE with water withdrawal efficiency.

3.1 Operational Water Footprint

We collectively refer to on-site scope-1 water and off-site scope-2 water as the operational water.

- **On-site WUE.** We denote the on-site scope-1 WUE at time t by $\rho_{s1,t}$, which is defined as the ratio of the on-site water consumption to server energy consumption and varies over time depending on the outside temperature (see [13] for an example of on-site WUE based on cooling towers). Concretely, $\rho_{s1,t}$ increases significantly for cooling towers when the outside wet bulb temperature increases, and increases for outside air cooling when the outside dry bulb temperature is too hot or the humidity is too low.

- **Off-site WUE.** We denote the off-site scope-2 WUE at time t as $\rho_{s2,t}$, which is defined as the ratio of off-site water consumption for each kWh of electricity consumption and measures the electricity water intensity factor (EWIF). While there are different methods to estimate $\rho_{s2,t}$, a common one is weighted averaging: $\rho_{s2,t} = \frac{\sum_k b_{k,t} \times EWIF_k}{\sum_k b_{k,t}}$ where $b_{k,t}$ denotes the amount of electricity generated from fuel type k at time t for the grid serving the data center under consideration, and $EWIF_k$ is the EWIF for fuel type k [31,32]. Thus, variations in energy fuel mixes of electricity generation result in temporal variations of the off-site WUE. Moreover, the off-site WUE also varies across regions due to different energy fuel mixes [5,12].

Table 1: Estimate of GPT-3’s operational water consumption footprint. “*” denotes data centers under construction as of July 2023, whose PUE and WUE are projected by Microsoft.

Location	PUE	On-site WUE (L/kWh)	Off-site EWIF (L/kWh)	Water for Training (million L)			Water for Each Request (mL)			# of Requests for 500ml Water
				On-site Water	Off-site Water	Total Water	On-site Water	Off-site Water	Total Water	
U.S. Average	1.170	0.550	3.142	0.708	4.731	5.439	2.200	14.704	16.904	29.6
Arizona	1.180	1.630	4.959	2.098	7.531	9.629	6.520	23.406	29.926	16.7
Georgia*	1.120	0.060	2.309	0.077	3.328	3.406	0.240	10.345	10.585	47.2
Illinois	1.350	0.740	2.233	0.952	3.880	4.833	2.960	12.060	15.020	33.3
Iowa	1.160	0.140	3.104	0.180	4.634	4.814	0.560	14.403	14.963	33.4
Texas	1.280	0.250	1.287	0.322	2.120	2.442	1.000	6.590	7.590	65.9
Virginia	1.140	0.140	2.385	0.180	3.499	3.679	0.560	10.875	11.435	43.7
Washington	1.150	0.950	9.501	1.223	14.063	15.285	3.800	43.706	47.506	10.5
Wyoming	1.110	0.130	2.574	0.167	3.677	3.845	0.520	11.429	11.949	41.8
Australia*	1.120	0.012	4.259	0.015	6.138	6.154	0.048	19.078	19.126	26.1
Denmark*	1.160	0.010	3.180	0.013	4.747	4.760	0.040	14.754	14.794	33.8
Finland*	1.120	0.010	4.542	0.013	6.548	6.561	0.040	20.350	20.390	24.5
India*	1.430	0.000	3.445	0.000	6.340	6.340	0.000	19.704	19.704	25.4
Indonesia*	1.320	1.900	2.271	2.445	3.858	6.304	7.600	11.992	19.592	25.5
Ireland	1.190	0.020	1.476	0.026	2.261	2.287	0.080	7.027	7.107	70.4
Mexico*	1.120	0.056	5.300	0.072	7.639	7.711	0.224	23.742	23.966	20.9
Netherlands	1.140	0.060	3.445	0.077	5.054	5.131	0.240	15.708	15.948	31.4
Sweden	1.160	0.090	6.019	0.116	8.986	9.101	0.360	27.927	28.287	17.7

• **Operational water footprint.** Consider a time-slotted model $t = 1, 2, \dots, T$, where the length of each time slot depends on how frequently we want to assess the operational water footprint. At time t , suppose that an AI model uses energy e_t which can be measured using power meters and/or servers’ built-in tools, and the data center hosting the AI model has a power usage effectiveness (PUE) of θ_t that accounts for the non-IT energy overhead. Then, the total operational water footprint of the AI model can be written as $WaterOperational = \sum_{t=1}^T e_t \cdot [\rho_{s1,t} + \theta_t \cdot \rho_{s2,t}]$.

3.2 Embodied Water Footprint

Similar to accounting for the embodied carbon footprint [33], the total scope-3 water footprint is amortized over the lifespan of a server. Specifically, if W represents the total water used to manufacture the AI servers and the servers are expected to operate for a period of T_0 , then the embodied water footprint over a period of T is calculated as $WaterEmbodied = \frac{T \cdot W}{T_0}$.

By adding up the operational and embodied water footprints, we obtain the total water footprint as $WaterTotal = \sum_{t=1}^T e_t \cdot [\rho_{s1,t} + \theta_t \cdot \rho_{s2,t}] + \frac{T \cdot W}{T_0}$. In practice, to obtain a rough estimate, we can use the average values for the annualized WUE and the estimated AI server energy consumption.

3.3 Case Study: Estimating GPT-3’s Operational Water Consumption Footprint

The core of ChatGPT, a popular online service, is a large language model (LLM) based on subsequent versions of GPT-3. We present a case study to estimate the operational water consumption for the full GPT-3 model with 175 billion parameters [18]. We exclude embodied water footprint due to the lack of public data for scope-3 water usage. We choose GPT-3 as Microsoft publishes its location-wise WUE and PUE [34,35]. The results are summarized in Table 1.

3.3.1 Training

GPT-3 was trained and deployed by OpenAI in Microsoft’s data centers, with an estimated training energy of 1287 MWh [36]. In line with the practice of estimating the carbon footprint, we use the most recent annualized average on-site PUE and WUE for each location, as reported by Microsoft [34,35]. For power plant water efficiency, different references may provide different estimates of EWIF. Thus, for consistency across regions, we use the EWIF provided by [12] to estimate scope-2 water consumption, as it employs the same methodology for calculating EWIF across a large number of regions. Moreover, a large number of Microsoft’s data centers are located in the U.S., where the average EWIF provided by [12] is 3.14 L/kWh and significantly lower than 4.35 L/kWh reported by the recent U.S. Department of Energy study [5]. The specific location for training GPT-3 is not public. Thus, we consider Microsoft’s different data center locations, while excluding Singapore and Taiwan since the EWIF data for these regions is not available in [12].

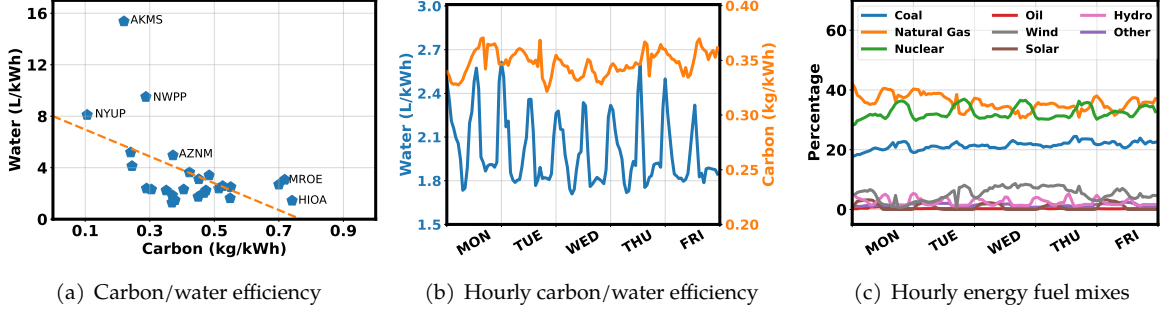


Figure 2: (a) The U.S. eGRID-level scope-2 water consumption intensity factor vs. carbon emission rate [12, 39]. The dashed line represents a linear regression model, showing that the eGRID-level scope-2 carbon emission and water consumption efficiencies are not aligned. (b) A 5-day snapshot of scope-2 carbon emission rate and water consumption intensity in Virginia, starting from April 4, 2022. The values are calculated based on the fuel mixes, carbon emission rate and water consumption intensity for each fuel type [12, 27, 39]. The scope-2 carbon and water efficiencies only have a weak Pearson correlation coefficient of 0.06 in Virginia. (c) A 5-day snapshot of energy fuel mixes serving Virginia, starting from April 4, 2022 [27].

3.3.2 Inference

As a representative usage scenario for an LLM, we consider a conversation task, which typically includes a CPU-intensive prompt phase that processes the user’s input (a.k.a., prompt) and a memory-intensive token phase that produces outputs [37]. More specifically, we consider a medium-sized request, each with approximately ≤ 800 words of input and 150 – 300 words of output [37]. The official estimate shows that GPT-3 consumes an order of 0.4 kWh electricity to generate 100 pages of content (e.g., roughly 0.004 kWh per page) [18]. Thus, we consider 0.004 kWh as the per-request server energy consumption for our conversation task. The PUE, WUE, and EWIF are the same as those used for estimating the training water consumption.

Our estimate of inference water consumption for GPT-3 is on the conservative side, and the actual water consumption could be several times higher. Concretely, when service level objectives (SLOs) for LLM response time are considered in real enterprise-grade Nvidia DGX H100 systems for conversation tasks, the inference server energy consumption for a much smaller model (e.g., Llama-3-70B) to process each medium-sized request is already ~ 0.010 kWh when using a state-of-the-practice LLM inference solution and accounting for the non-GPU server overhead [37]. When considering the Falcon-180B model with a comparable size to GPT-3-175B, the server energy consumption will even reach ~ 0.016 kWh for processing a medium-sized request [37]. Additionally, our EWIF for the U.S. (i.e., 3.14L/kWh on average) is conservatively chosen and significantly lower than 4.35L/kWh recently reported by [5].

While there is no official information on its resource consumption, the subsequent GPT-4 model is expected to consume substantially more energy and water than GPT-3 for processing the same request [38]. Nonetheless, we emphasize that the on-site WUE of Microsoft’s data centers is already among the lowest in the industry. With continued efforts to reduce AI’s computational demand and improve the overall water efficiency, the water consumption per request may decrease in the future, while the total water consumption is likely to rise further as a result of the growing demand for AI services and the increasing scale of AI applications [5].

4 Our Recommendations

We provide our recommendations to address AI’s water footprint from the scheduling and policy perspectives, making future AI more environmentally sustainable.

4.1 More Transparency and Comprehensive Reporting

Despite its growing importance, AI’s water footprint has received relatively less attention. For example, while AI model cards routinely include carbon emissions and serve as an important reporting framework for understanding AI’s environmental impacts, they currently omit information on AI’s water consumption. The lack of transparency may obstruct efforts to drive innovations that enhance water sustainability and

support truly sustainable AI. As an initial step to raise awareness among end users about the water resource impacts of their AI usage, we recommend tracking and reporting AI’s water consumption in AI model cards and/or through cloud dashboards.

Moreover, a comprehensive understanding and reporting of AI’s scope-2 water consumption associated with electricity generation remain limited. Although data centers have increasingly adopted climate-conscious cooling system designs to minimize on-site water consumption [8, 24, 26], these efforts primarily focus on scope-1 water usage while largely overlooking scope-2 impacts. Just as addressing scope-2 carbon emissions is important for mitigating climate change, it is equally crucial to address scope-2 water consumption to reduce AI’s “true water cost”, as noted by the U.S. Department of Energy study [5]. To better reflect the true impacts of data centers on water resources, some technology companies such as Meta have begun to include scope-2 water consumption in their sustainability reports [25]. We recommend the reporting of scope-2 water consumption as a standard practice. This approach makes the off-site water consumption visible to AI model developers as well as end users and can unlock new opportunities for demand-side flexibility, thereby alleviating the overall strain on water resources.

Finally, despite the enormous scope-3 supply-chain water footprint [30], there is limited data available for embodied water usage by chip manufacturing. We recommend further research on scope-3 water consumption to achieve a comprehensive understanding of AI’s overall water footprint and to foster corporate water stewardship.

4.2 “When” and “Where” Matter

Judiciously deciding “when” and “where” to train a large AI model can significantly affect the water footprint. The water efficiency exhibits a spatial-temporal diversity — on-site water efficiency changes due to variations of outside weather conditions, and off-site water efficiency changes due to variations of the grid’s energy fuel mixes to meet time-varying demands (Figure 2). Therefore, we can dynamically schedule AI training and inference in a water-wise manner to cut the water footprint. For example, we may schedule AI training at midnight and/or in a data center with better water efficiency. Likewise, if informed of the real-time water efficiency, some water-conscious users may prefer to use AI inference during water-efficient hours and/or in water-efficient data centers, which can reduce AI’s water footprint by enabling demand-side flexibility.

4.3 “Follow the Sun” or “Unfollow the Sun”

To cut the carbon footprint, it is preferable to “follow the sun” when solar energy is more abundant. Nonetheless, to cut the water footprint, it may be more appealing to “unfollow the sun” to avoid high-temperature hours of a day when WUE is high. This conflict can also be shown in Figure 2(a) and Figure 2(b), where we see misalignment between the scope-2 water consumption intensity factor and carbon emission rate: minimizing one footprint might increase the other footprint. This observation further corroborates the previous finding that the environmental impacts of carbon and water footprints are not substitutable [5, 13]. Therefore, to judiciously achieve a balance between “follow the sun” for carbon efficiency and “unfollow the sun” for water efficiency, we need to reconcile the potential water-carbon conflicts by using holistic approaches that are both carbon-efficient and water-wise.

5 Conclusion

In this paper, we uncover AI’s water usage as a critical concern for socially responsible and environmentally sustainable AI. We present a principled methodology to estimate AI’s water footprint. Then, using GPT-3 as an example, we show that a large AI model can consume millions of liters of water for training. We also discuss that the scope-1 and scope-2 water efficiencies vary spatially and temporally — judiciously deciding “when” and “where” to run a large AI model can significantly cut the water footprint. In addition, we recommend increased transparency and comprehensive reporting of AI’s water footprint, and highlight the necessity of holistically addressing the water footprint along with the carbon footprint to build truly sustainable AI.

AI’s water footprint can no longer stay under the radar and must be addressed as a priority as part of the collective efforts to combat global water challenges.

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Appendix: Operational Water for Global AI in 2027

A recent study suggests that the global AI could consume 85 – 134 TWh of electricity in 2027 based on the GPU shipment [11], whereas a more aggressive projection by the U.S. Department of Energy predicts that AI servers’ electricity consumption in the U.S. alone will surpass 150 – 300 TWh in 2028 [5]. Based on the former and more conservative projection, we estimate the potential water usage for global AI in 2027, while noting that our global estimates will be exceeded by the water usage attributed to AI in the U.S. alone in 2028 if the U.S. Department of Energy’s projection comes to fruition.

Scope-1 water usage. The scope-1 water efficiency depends on a variety of factors, including the cooling system designs, climate conditions, and operational settings. To set the global scope-1 water efficiency, we utilize the annualized water efficiencies reported by two leading data center operators, Google and Equinix, in their latest sustainability reports [8,40]. Specifically, for on-site scope-1 water withdrawal, we assume 1.2 L/kWh, which results in a total scope-1 water withdrawal of 0.11 – 0.16 billion cubic meters. Similarly, assuming 1.0 L/kWh for the global scope-1 water consumption efficiency, we obtain a total on-site scope-1 water consumption of 0.09 – 0.14 billion cubic meters. Note that Google and Equinix both operate data centers globally, but represent two distinct categories of data centers: hyperscale data centers (Google) and multi-tenant colocation data centers (Equinix). According to the U.S. Department of Energy [5], these two types of data centers collectively account for the vast majority of data center energy consumption in the U.S., with colocation data centers consuming slightly more energy than hyperscalers.

Scope-2 water usage. As noted by the U.S. Department of Energy [5], scope-2 water usage is part of the true water cost of data centers. The U.S. average electricity water withdrawal and consumption intensity factors are both lower than the global averages [12]. Thus, in our estimate, we use the U.S. average electricity water withdrawal intensity factor 43.83 L/kWh [41], and electricity water consumption intensity factor 3.14 L/kWh [12], respectively. Note that, since [12] includes hydropower in the calculation, it has a higher electricity water withdrawal factor than the U.S. Energy Information Administration’s calculation (i.e., 386.07 L/kWh vs. 43.83 L/kWh for the U.S.). Moreover, our value of 3.14 L/kWh for the U.S. average water consumption factor is lower than 4.35 L/kWh reported by the U.S. Department of Energy [5], as well as lower than Meta’s global electricity water consumption intensity factor of 3.70 L/kWh in 2024 (i.e., 55,475 megaliters divided by 14,975,435 MWh) [25]. Therefore, our choices of 43.83 L/kWh and 3.14 L/kWh for electricity water withdrawal and consumption intensity factors are both on the conservative side, which can partly absorb potential over-estimates of global AI’s energy demand in 2027 provided by [11].

To account for the data center non-IT energy overheads, we conservatively assume a power usage effectiveness (PUE) of 1.1, which is a fairly low value even for state-of-the-art data center facilities [8]. Thus, AI’s total electricity consumption becomes 93.5 – 147.4 TWh. Thus, after multiplying 43.83 L/kWh and 3.14 L/kWh by 93.5 – 147.4 TWh, we obtain the total scope-2 water withdrawal of 4.10 – 6.46 billion cubic meters and water consumption of 0.29 – 0.46 billion cubic meters, respectively.

Total water usage. By adding up scope-1 and scope-2 water usage together, the total water withdrawal and water consumption of global AI may reach 4.2 – 6.6 billion cubic meters and 0.38 – 0.60 billion cubic meters, respectively. According to the U.S. Central Intelligence Agency [42], the estimated U.S. annual water withdrawals in Denmark and the United Kingdom in 2020 (the latest year available as of January, 2025) were 0.98 billion cubic meters and 8.42 billion cubic meters, respectively. Thus, assuming that the 2027 water withdrawals in these two countries remain similar to their 2020 levels, the total water withdrawal attributed to global AI in 2027 is projected to surpass the equivalent of the total annual water withdrawal of 4 – 6 Denmark or approximately half of the United Kingdom. The U.S. Central Intelligence Agency [42] does not provide the country-wide annual water consumption information, and hence we do not contextualize the total water consumption of global AI in 2027.

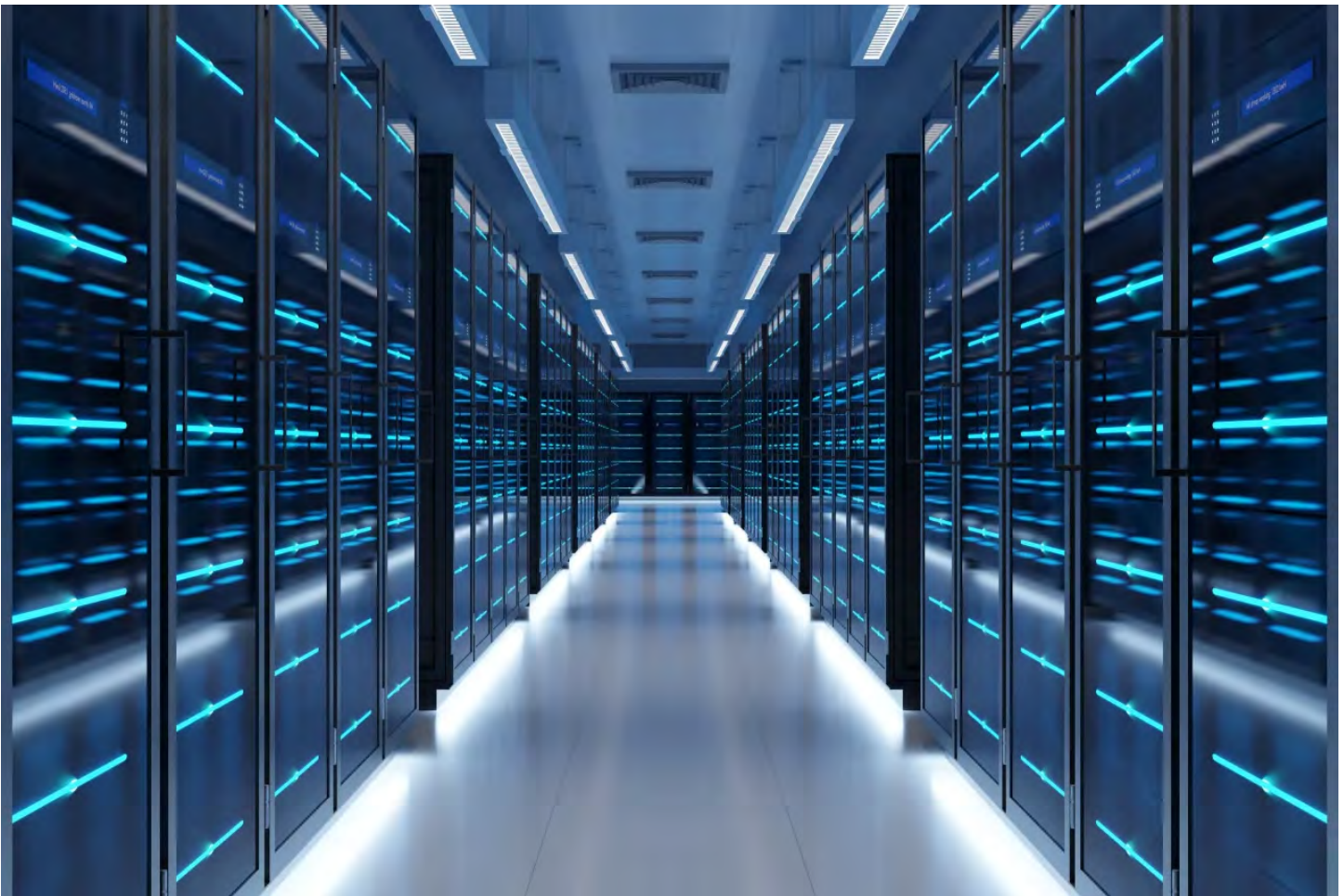
The estimates of global AI’s water usage in 2027 are naturally subject to uncertainties, e.g., the future water efficiency may differ from the current value we use. Nonetheless, we emphasize that our estimates are on the conservative side. For example, considering a higher estimate, the U.S. Department of Energy’s scope-1 water consumption attributed to AI in the U.S. alone could exceed 0.2 billion cubic meters in 2028 [5]. Moreover, based on the reported scope-2 water consumption efficiency, the combined scope-1 and scope-2 water consumption attributed to AI in the U.S. alone is projected to reach up to about 2 billion cubic meters in 2028 [5], which is significantly higher than our estimate of global AI’s total water consumption in 2027.

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California wildfires raise alarm on water-guzzling AI like ChatGPT

BY **JANE THIER** AND **FORTUNE EDITORS**

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If there weren't enough of an argument against AI [from an environmental standpoint](#), a new waterfall of data might push even the most ambivalent consumer over the edge.

Per the [International Energy Agency](#), energy consumption by global data centers could more than double by 2026, "reaching levels that exceed large nations." Ironically, "while we're using AI to solve some of the world's biggest challenges—from climate modeling to health-care breakthroughs—we're also contributing to an environmental crisis of a different kind," Chris Gladwin, a tech founder and CEO, [wrote for Fortune](#) recently.

Related Video

How much water does AI use?

Now, reporting finds that OpenAI's ChatGPT—which uses the GPT-4 language model—consumes 519 milliliters or just over one bottle of water, to write a 100-word email. That's

according to the [Washington Post](#) in a research collaboration with the University of California, Riverside.

In order to shoot off one email per week for a year, ChatGPT would use up 27 liters of water, or about one-and-a-half jugs. Zooming out, WaPo wrote, that means if one in 10 U.S. residents—16 million people—asked ChatGPT to write an email a week, it'd cost more than 435 million liters of water.

While much has been made about the power usage each ChatGPT prompt immediately necessitates, the water conversation has gained additional steam in recent months.

As WaPo explained, every prompt a user enters into ChatGPT is quickly turned into code, and “flows through a server that runs thousands of calculations to determine the best words to use in a response.” All those calculations go through real, physical servers which are housed in enormous data centers around the world. Spitting out an answer—or answering a command—makes the servers heat up, like an under-duress old laptop.

Why does AI use water?

This is where water comes in; to keep those ever-important servers from overheating and breaking down, the data centers rely on cooling mechanisms, often via “cooling towers” that themselves require water. Each facility, depending on the climate where it's based, uses a different amount of water and electricity. West Des Moines, Iowa, is [quickly becoming a popular destination](#), owing to a temperate climate that calls for fewer cooling interventions.

“We haven't come to the point yet where AI has tangibly taken away our most essential

natural water resources,” [wrote](#) Shaolei Ren, an associate professor of engineering at UC Riverside who has been trying for years to quantify AI’s climate impact. Nonetheless, Ren called AI’s increasing water usage “definitely concerning.”

Amid rapid population growth and a changing climate, “depleting water resources and aging water infrastructures” are some of the most preeminent challenges, he wrote in November. “The concern is not only about the absolute amount of AI models’ water usage, but also about how AI model developers respond to the shared global challenge of water shortage.”

How are AI companies addressing water and energy use?

Droughts, he noted, are among the most immediate consequences of climate change, and it’s incumbent upon businesses to address water usage in their operations—and tech firms using generative AI top that list. “We already see heated tensions over water usage between AI data centers and local communities,” Ren wrote. “If AI models keep on guzzling water, these tensions will become more frequent and could lead to social turbulence.”

Google and Microsoft report rising water consumption

In Microsoft’s [sustainability report](#) last year, the company said its global water consumption had spiked 34% between 2021 and 2022. Over the same period, Google’s water usage [rose 20%](#), it wrote in its own report. “It’s fair to say” that the majority of that growth at both companies “is due to AI,” Ren [told the AP](#) at the time. (Microsoft’s data center used up 700,000 liters of water in training GPT-3, WaPo reported.)

Holly Alpine, who was once Microsoft’s senior program manager of Datacenter Community Environmental Sustainability, resigned from the company earlier this year on principle, she [wrote for Fortune](#), due to the company’s ecologically irresponsible AI development.

“Analyst reports suggest that advanced technologies—such as AI or machine learning—have the potential to increase fossil fuel yield [by 15%](#), contributing to a resurgence of oil and potentially delaying the global transition to renewable energy,” Alpine wrote. “The real-world impacts are staggering: A single such deal between Microsoft and [ExxonMobil](#) could generate emissions that exceed Microsoft’s 2020 annual carbon removal commitments by over 600%.”

When she was a Microsoft employee, she wrote, she witnessed “dozens” of such deals.

An earlier version of this story published on [Fortune.com](https://fortune.com) on Sept. 23, 2024.

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Latest in Tech

Data centers are booming in Texas. What does that mean for the grid?

As energy demand surges, largely due to crypto mining facilities, data centers and industrial electrification, Texas officials are looking at how to increase supply and shore up the grid.

BY **KAYLA GUO** JAN. 24, 2025 5 AM CENTRAL

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Sign up for [The Brief](#), The Texas Tribune's daily newsletter that keeps readers up to speed on the most essential Texas news.

The rise of artificial intelligence, the digitization of the economy and everyday life's growing computing needs have turbocharged the expansion of data centers, driving up a surge in electricity demand in Texas and across the country.

Texas' main grid operator predicts power demand will [nearly double](#) by 2030, in part due to more requests to plug into the grid from large users like data centers, crypto mining facilities, hydrogen production plants and oil and gas companies.

On Tuesday, President Donald Trump [announced Stargate](#), a joint venture between OpenAI, SoftBank and Oracle that will invest up to \$500 billion in AI-related infrastructure.

Texas will serve as ground zero, with 10 data centers by the venture already under construction in the state, 10 more on the way and the first project based in Abilene, [Oracle CEO Larry Ellison said](#). Each building will occupy half a million square feet.

The announcement reflected the hunger for data centers across industries and a yearslong push to increase data capacity. Ellison noted that the partnership had been in the works for years. He said the new data centers could offer services like maintaining electronic health care records and helping hospitals share medical knowledge.

"The demand for digital services continues to increase and continues to be necessary to build out our capabilities for the 21st century economy," Dan Diorio, senior director of state policy at the Data Center Coalition, an industry trade group, said in an interview. "Texas is uniquely poised to benefit from that."

That expansion — in addition to other large energy users and factors such as population growth and extreme weather — will stretch the grid over the next decade, raising questions about how Texas can meet the skyrocketing demand for power while ensuring affordability and reliability for everyday consumers.

Data centers in Texas

Data centers — which house servers that provide computing power and the fans and cooling units needed to keep the equipment from overheating — are energy-intensive facilities that operate 24/7.

Large data centers can require 100 MW or more each, consuming the same amount of power per year as 350,000 to 400,000 electric cars, [according to](#) the International Energy Agency. Put another way, a larger facility can use as much electricity as a medium-sized power plant, the U.S. Energy Information Administration estimates.

Texas has seen a rapid increase in data capacity thanks to the state's relatively cheap energy prices, the ease with which facilities can connect to the grid and its overall business-friendly tax and regulatory environment.

Companies generally employ around 50 to 150 or more employees in each data center, in addition to an array of building and maintenance contractors, according to the Data Center Coalition, which estimates that each job in a data center supported six jobs elsewhere in the economy.

The state had 279 data centers as of September, [according to the Texas Comptroller](#). The Dallas-Fort Worth area has about 141 of those.

That translated to 591 MW of power leased by data centers in Dallas and Fort Worth last year — the second most in the country — and nearly 190 MW in Austin and San Antonio, according to a [CBRE report](#).

The Electric Reliability Council of Texas, the state's primary grid operator, estimates that 1 MW of electricity can power roughly 200 homes.

What do more data centers mean for the grid?

In Texas, the U.S. Energy Information Administration [predicted that demand](#) from large users — including but not limited to data centers — would grow by 60% this year, making up around 10% of the total forecast demand on the state's main grid.

Large users requiring 5,496 MW of power have been [approved by ERCOT](#) to connect to the grid, according to a September report. The EIA expects that by the end of this year, ERCOT will have approved 9,500 MW in total large-user demand — a 73% increase.

That includes data centers and other large users like crypto mining facilities, which represent the biggest share of large users looking to connect to the grid, according to ERCOT.

Several other large-load projects — which would use up to 56,458 MW a year — were awaiting ERCOT consideration as of September.

Some large users, primarily crypto mining facilities, have committed to temporarily lowering their energy usage in periods of [grid strain](#) — an agreement that [earned some crypto mining companies millions of dollars](#) while many Texans' saw their power bills surge.

Data centers, on the other hand, generally require an uninterrupted supply of power and typically do not participate in ERCOT's high-demand response programs, according to a recent [report](#) from the Texas Senate Business and Commerce Committee.

Nationally, data centers are expected to consume between 11% and 12% of total U.S. power demand by 2030 — up from around 3% and 4% of demand today, according to an [analysis by McKinsey](#).

Is the grid prepared?

How to meet soaring power demand is set to drive the discussion around the grid during this year's legislative session.

Texas lawmakers have sought to boost the state's supply of natural gas through the Texas Energy Fund, which will offer companies up to \$10 billion in low-interest loans to build gas-fueled power plants. State regulators are currently vetting loan applications, but new plants will not be operational for years.

“Data centers are going to provide a very essential product for consumers that underpins the functions of our life,” Mark Bell, president of the Association of Electric Companies of Texas, said. “As an industry, we are ready to step up to the challenges that we face with this type of large load.”

Bell added that the projected demand “provides forward signals in the market” that encourage companies to invest in new power generation.

ERCOT's demand forecast, which reflected a sharp increase from previous years, also raised questions among lawmakers about whether large users needed more state oversight.

"I think we need to rise to the challenge of getting the needed generation onto the grid," state Sen. [Charles Schwertner](#), chair of the Business and Commerce Committee, told The Texas Tribune in June. "But there is eventually a prioritization that could be discussed, and obviously Texans — their families, their homes, their businesses — are the most important individuals, the most important clients for electricity."

On social media, Lt. Gov. [Dan Patrick](#) said in June that the Legislature needed to "take a close look" at data centers and crypto mining facilities. "We want data centers, but it can't be the Wild Wild West of data centers and crypto miners crashing our grid and turning the lights off," [he wrote](#).

Patrick said in a Thursday statement to The Texas Tribune that he supported Stargate and believed Texas should be the "world leader in AI, data center and crypto. The key is to ensure they have the power they need without a major impact to our electrical grid. The industries understand that and they are working on solutions."

Some companies are [building generation locally or on site](#) to help lessen their impact on the grid and lock in their own power supply. Building their facilities near existing generation sites can also help alleviate grid congestion. Lawmakers this session will likely consider whether companies should be forced to do so, with the Texas Senate Business and Commerce Committee [recommending](#) that large loads be required to "offset their impact on the grid by adding on-site power systems or participating in programs to curtail electricity usage during peak demand periods."

Judging whether data centers and other large projects might actually build in Texas after requesting ERCOT consideration remains difficult, experts testified to lawmakers last year, making ERCOT's demand prediction less certain. Companies looking to build data centers may submit requests in multiple prospective locations.

In order to help firm up that forecast, the Texas Senate Business and Commerce Committee [recommended](#) that the state ensure regulators have enough information about how large users might operate, such as by asking companies to submit more detailed information about their proposed projects.

The Public Utility Commission [approved a rule](#) in November requiring crypto mining facilities connected to the ERCOT grid to register their power usage with regulators.

The projected growth in usage also means the grid will need [more transmission lines](#), ERCOT CEO Pablo Vegas said in April.

“The forecasted pace of load growth could exceed the pace at which transmission capacity can be built to support it,” [Vegas’ presentation](#) said. “A new era of transmission system planning is necessary to manage the large amount of prospective load.”

Typically, the costs of building out transmission and distribution infrastructure are spread across a utility’s customers. But the major investments needed to support demand driven by large industrial users raised the question of who should foot the bill.

Lawmakers have [signaled interest](#) in limiting the costs passed onto small energy consumers “by ensuring that industries with significant electricity demands bear a fair portion of their actual costs.”

Diorio, of the Data Center Coalition, emphasized that the industry was “fully committed to paying our full cost of service.”

“We do not want residential customers subsidizing data centers,” he said. “We have a strong stake in helping Texas build out appropriately, and we’re leaning in to do that.”

Disclosure: Association of Electric Companies of Texas (AECT) has been a financial supporter of The Texas Tribune, a nonprofit, nonpartisan news organization that is funded in part by donations from members, foundations and corporate sponsors. Financial supporters play no role in the Tribune's journalism. Find a complete [list of them here](#).

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ON-PREM

How datacenters use water – and why kicking the habit is nearly impossible

If they're not consuming H₂O directly, the power plant almost certainly is

 [Tobias Mann](#)

Sat 4 Jan 2025 // 18:30 UTC

FEATURE The explosive growth of datacenters that followed ChatGPT's debut in 2022 has shone a spotlight on the environmental impact of these power-hungry facilities.

But it's not just power we have to worry about. These facilities are capable of sucking down prodigious quantities of water.

In the US, datacenters can consume anywhere between 300,000 and four million gallons of water a day to keep the compute housed within them cool, Austin Shelnutt of Texas-based Strategic Thermal Labs explained in a presentation at SC24 in Atlanta this fall.

We'll get to why some datacenters use more water than others in a bit, but in some regions rates of consumption are as high as 25 percent of the municipality's water supply.

This level of water consumption, understandably, has led to concerns over water scarcity and desertification, which were already problematic due to climate change, and have only been exacerbated by the proliferation of generative AI. Today, the AI datacenters built to train these models often require tens of thousands of GPUs, each capable of generating 1,200 watts of power and heat.

However, over the next few years, hyperscalers, cloud providers, and model builders plan to deploy millions of GPUs and other AI accelerators requiring gigawatts of energy, and that means even higher rates of water consumption.

According to researchers at UC Riverside and the University of Texas Arlington, by 2027 global AI demand could account for the withdrawal of 4.2-6.6 billion cubic meters of water annually. That's roughly the equivalent of half the UK's water withdrawal over the course of a year.

However, mitigating datacenter water consumption isn't as simple as ditching evaporative cooling towers for waterless alternatives.

The datacenter water cycle

Datacenters consume water in a couple of ways. The first, and the area we'll focus most of our attention on, is direct water consumption. This is water that's pulled from local sources including water and wastewater treatment plants.

This water is pumped into cooling towers, where it evaporates, transferring heat to the air. If you've ever used a swamp cooler to chill your home or apartment, cooling towers work in a similar manner.

Evaporative cooling has become popular among datacenter operators for a couple of reasons, but the big one is they're really good at getting rid of heat and don't require a ton of electricity to do it.

According to Shelnutt, evaporating ten gallons a minute is enough to cool roughly 1.5 megawatts of compute.

When we talk about "consumption," we're referring to water that's been evaporated. It isn't actually consumed so much as it's removed from the local watershed by the prevailing winds. This can be problematic given evaporative coolers are most effective in arid climates where water scarcity is commonly a problem.

According to researchers, about 70-80 percent of the water that enters a cooling tower is actually consumed [PDF], the rest is used to flush out mineral deposits similar to those found when cleaning a humidifier. The brine that's left behind is recycled through the system until it exceeds a certain concentration, at which point it's flushed away to a holding pond or treatment plant run onsite or by the local municipality before it's returned to the local watershed.

For this to work, the wastewater treatment plant needs to be sized correctly to handle the volume and concentration of brine generated by datacenters in the region. Things can get complicated pretty quickly when this isn't done, as was the case for Microsoft's campus in Goodyear, Arizona.

Why datacenters' drinking habit is so hard to quit

One of the reasons that datacenter operators have gravitated toward evaporative coolers is because they're so cheap to operate compared to alternative technologies.

"It is always of a higher coefficient of performance (COP), meaning less energy required, to evaporate water, regardless of what cooling medium is being utilized," Shelnutt said.

In fact, COP, which refers to the amount of heat removed for a given amount of power, for evaporative cooling comes in at 1,230 while dry coolers and chillers manage a COP of about 12 and 4, respectively, he explained.

In terms of energy consumption, this makes an evaporatively cooled datacenter far more energy efficient than one that doesn't consume water, and that translates to a lower operating cost.

The challenge is that not every location and climate is well suited to evaporative cooling. In hotter climates where water is either scarce or places with high humidity where evaporative coolers are ineffective, chillers, which function similar to your AC unit, may be used instead.

In cooler climates such as the Nordic regions, datacenters often make use of free cooling and dry coolers, which take advantage of the lower ambient air temperature to eject heat into the atmosphere without consuming any water.

Whether or not evaporating cooling is used is highly dependent on location and climate, Digital Realty CTO Chris Sharp told *The Register*.

"You have to understand water is a scarce resource. Everybody has to start at that base point," he explained. "You have to be good stewards of that resource just to ensure that you're utilizing it effectively."

The colocation giant operates more than 300 bit barns around the globe, and uses a variety of designs based on predicted capacity requirements and environmental factors. The company's standard datacenter design, Sharp says, doesn't consume any water at all, instead relying on chillers to pull energy from the facility. However, in some locations, evaporative cooling and dry coolers are employed instead.

Most datacenter water isn't consumed onsite

While dry coolers and chillers may not consume water onsite, they aren't without compromise. These technologies consume substantially more power from the local grid and potentially result in higher indirect water consumption.

According to the US Energy Information Administration, the US sources roughly 89 percent of its power from natural gas, nuclear, and coal plants. Many of these plants employ steam turbines to generate power, which consumes a lot of water in the process.

Ironically, while evaporative coolers are why datacenters consume so much water onsite, the same technology is commonly employed to reduce the amount of water lost to steam. Even

still the amount of water consumed through energy generation far exceeds that of modern datacenters.

A 2016 [study](#) [PDF] by Lawrence Berkeley National Lab (LBL) found that roughly 83 percent of water consumption attributable to datacenters could be attributed to power generation. As a result, reducing onsite water consumption at the expense of higher power draw could lead to an increase in the amount of water consumed.

However, just because power plants may pull more water than datacenters, that doesn't mean they're pulling the same water, Shaolei Ren, associate professor of electrical and computer engineering at UC Riverside, told *The Register*, adding that many power plants get their water from sources like rivers and lakes that may not be suitable for datacenters.

Ren and his team have been studying the datacenter's environmental impact on water consumption and [air quality](#).

This, again, is highly dependent on location and the grid mix. For example, datacenters located in regions with an abundance of hydroelectric, solar, or wind power will have lower indirect water consumption than one powered by fossil fuels or combustion.

What can be done to curb datacenter water consumption?

Understanding that datacenters are, with few exceptions, always going to use some amount of water, there are still plenty of ways operators are looking to reduce direct and indirect consumption.

One of the most obvious is matching water flow rates to facility load and utilizing free cooling wherever possible. Using a combination of sensors and software automation to monitor pumps and filters at facilities utilizing evaporative cooling, Sharp says Digital Realty has observed a 15 percent reduction in overall water usage.

"That equates to about 126 million gallons of avoided withdrawal from the system because we're just running it more efficiently," he said.

We're also seeing datacenters built in colder climates that can take advantage of free cooling most of the year. Better yet, in many Nordic countries, large quantities of hydroelectric power mean that even if auxiliary dry coolers or chillers are required, indirect water consumption isn't as much of an issue.

We've also seen heat generated by datacenters used to warm local offices, support district heating grids, or even greenhouses to grow produce year round.

In locations where free cooling and heat reuse aren't practical, shifting to direct-to-chip and immersion liquid cooling (DLC) for AI clusters, which, by the way, is a closed loop that doesn't really consume water, can facilitate the use of dry coolers. While dry coolers are still more energy-intensive than evaporative coolers, the substantially lower and therefore better power use effectiveness (PUE) of liquid cooling could make up the difference.

If you're not familiar, PUE describes how much power consumed by datacenters goes toward compute, storage, or networking equipment – stuff that makes money – versus things like facility cooling, which don't. The closer the PUE is to 1.0, the more efficient the facility.

This is possible because a sizable chunk, upward of 20 percent, of the energy used by air-cooled AI systems goes to chassis fans. On top of that, water is a much better conductor of heat. Shifting to DLC, something that's already happening with Nvidia's top-specced Blackwell parts, has the potential to drop PUE from 1.69-1.44 to around 1.1 or lower.

However, as Shelnutt noted in his SC24 presentation, this balancing act depends heavily on the power saved by DLC not being reallocated to support additional compute.

Water-aware computing

While many of these water-saving technologies require changes to facility infrastructure to implement, another approach might be to change the way workloads are distributed across datacenters.

The idea here isn't that different from carbon-aware computing, where workloads are routed to different locations based on the time and carbon-intensity of the grid, Ren explained. "They can do something similar based on the water stress level and real-time water efficiency, because this water evaporation rate does change over time - an hourly noon time versus the night time."

This, he admits, isn't something that the cloud providers and hyperscalers will have an easier time achieving as they maintain a tight grip on the orchestration of their infrastructure. "Colocation providers have more challenges due to limited control over the servers and workloads."

This approach may also not be appropriate for latency-sensitive workloads, like AI inferencing, where proximity to users is imperative for real-time data processing. However, workloads like AI training don't have these same limitations. One can imagine an AI training workload, which might run for weeks or months, could be queued up to run in a far-flung datacenter located in the polar regions that can take advantage of free cooling.

Fine-tuning workloads, which involve changing the behavior of a pre-trained model, are far less computationally intensive. Depending on the size of the base model and the dataset used, a fine-tuning job may only require a few hours to complete. In this case, the job could be scheduled to run at night when temperatures are lower and less water is lost to evaporation.

Is water the new oil?

While datacenter water consumption remains a topic of concern, particularly in drought-prone areas, Shelnutt argues the bigger issue is where the water used by these facilities is coming from.

"Planet Earth has no shortage of water. What planet Earth has a shortage of, in some cases, is regional drinkable water, and there is a water distribution scarcity issue in certain parts of the world," he said.

To address these concerns, Shelnutt suggests datacenter operators should be investing in desalination plants, water distribution networks, on-premises wastewater treatment facilities, and non-potable storage to support broader adoption of evaporative coolers.

While the idea of first desalinating and then shipping water by pipeline or train might sound cost-prohibitive, many hyperscalers have already committed hundreds of millions of dollars to securing onsite nuclear power over the next few years. As such, investing in water desalination and transportation may not be so far fetched.

More importantly, Shelnutt claims that desalinating and shipping water from the coasts is still more efficient than using dry coolers or refrigerant-based cooling tech.

"Desalinate ocean water right now at three kilowatt [hours] per cubic meter – that's an average over the last ten years; there are many installations of desalination plants that are down below one kilowatt hour per cubic meter – that's a COP of 222," he said.

Ship that 1,000 miles by pipeline and Shelnutt says the COP drops to 132. Shipped by train, the COP falls yet further to 38, far less than evaporating water sourced from a municipal treatment plant, but still far more efficient than using dry coolers. ®

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FEATURE

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Part of: The future of AI hardware in the data center

How the rise in AI impacts data centers and the environment

AI's impact on data centers raises environmental concerns as rising energy demands from technologies such as ChatGPT strain resources and challenge sustainability.

By **Jacob Roundy**

Published: **25 Nov 2024**

Since OpenAI launched ChatGPT in late 2022, there has been an AI boom across all tech industries that has greatly increased data center electricity consumption and demand.

Generative AI (GenAI) chatbots, like ChatGPT, use [natural language processing](#) technology to interpret prompts conversationally, which greatly lowers the user adoption barrier. This led to ChatGPT becoming an instant viral sensation, and in

3 of 4



This rapid growth is often credited as the acceleration point for public-facing AI projects. Since ChatGPT's launch, other tech giants, including Google, Microsoft and Meta, have launched their own [large language model](#) chatbots, garnering even more users on a global scale. The electricity consumption of these technologies is extremely high, raising concerns about the environmental impact of AI and the overall energy use in data centers.

Take a deeper look at how the AI boom affects the environment, including how it uses energy, real-world impacts on the world and potential ways data centers can balance AI workloads while [mitigating climate impact](#).

How AI uses so much power within the data center

The [International Energy Agency](#) (IEA) found that data centers and data transmission networks each account for 1% to 1.5% of global electricity consumption and 1% of energy-related greenhouse gas emissions. The energy demand strains electricity grids in many regions, and the resulting emissions [harm the environment](#) in various ways.

According to a report published in May 2024 by the [Electric Power Research Institute](#) (EPRI), electricity consumption by large data centers more than doubled between 2017 and 2021 -- before the AI boom. Much of this growth was driven by commercially available digital services, like video streaming and communications applications. Now, the proliferation of AI is further fueling data center load growth.

AI workloads are more energy-intensive than other digital technologies. For example, the EPRI report estimated that traditional Google queries only use about 0.3 [watt-hours](#) each, while ChatGPT requests consume around 2.9 watt-hours each. That's nearly 10 times the amount of electricity consumption. GenAI models that create images, audio and videos consume even more electricity per request.

According to EPRI estimates, AI workloads use 10% to 20% of data center electricity. These statistics raise concerns as AI rapidly

AI must process vast volumes of data and conduct complex



developed future scenarios for data center load growth with this in mind. They projected that data centers may consume 4.6% to 9.1% of U.S. electricity generation annually by 2030 versus an estimated 4% as of 2024.

why it consumes so much more electricity than other digital technologies.

A primer on AI chip design

To put this into perspective, developers are currently building new data centers with capacities reaching up to 1,000 megawatts, enough to power 800,000 homes, according to the EPRI report. EPRI identified three main factors that contribute to the high energy consumption of AI workloads:

1. **Model development.** AI models must be developed and fine-tuned before training. This process uses approximately 10% of their energy footprint, according to EPRI.
2. **Model training.** An AI algorithm must process large amounts of data to train a model. This process requires "substantial computation efforts and high energy expenditure for extended periods," using about 30% of the energy footprint, according to EPRI.
3. **Utilization.** Deploying and using a fully developed and trained AI model in real-world applications requires intensive computations, which, according to EPRI, accounts for around 60% of their energy footprint.

AI must process vast volumes of data and conduct complex computational workloads, which is why it consumes so much more electricity than other digital technologies. As these technologies mature and proliferate, they'll likely grow more complex and demand more energy.

Examples of AI's impact on the environment

IEA created the Net Zero Emissions by 2050 Scenario, which details a pathway for a global transition to clean energy that should limit global warming by 1.5 degrees Celsius.

The Intergovernmental Panel on Climate Change's "[Sixth Assessment Report](#)" outlines the risks. According to IPCC, these risks include frequent extreme weather events, the loss of some entire ecosystems, exceptional heatwaves and more intense tropical cyclones. An increase in severe weather conditions will lead to extreme droughts, increasing flood hazards, and impacts on water and resource availability.

Increase in carbon emissions

A [study](#) by researchers at the University of Massachusetts Amherst estimated that training a large AI model could produce over 626,000 pounds of carbon dioxide equivalent. According to the university's researchers, this is more than five times a car's emissions over its entire lifetime.

Use of nonrenewable resources

According to a United Nations Environment Programme [report](#), critical minerals and

environmentally destructive ways. The electronic waste they produce may also contain hazardous substances.

Increase in water usage

Data centers consume water to liquid-cool the hardware that runs AI applications. According to an [article](#) in *Yale Environment 360*, a user who engages with ChatGPT between 10 and 50 times causes a data center to consume half a liter of water. ChatGPT has millions of users, which means total water consumption can amount to hundreds of millions of gallons of water just to cool the equipment running AI.

These are just a few examples of the strain AI is placing on the environment. A careful approach and thoughtful strategies are required to keep these [environmental impacts](#) in check and to build toward a more sustainable future in the industry.

Potential future stats and scenarios

EPRI created scenarios that project the potential growth rate of data center electricity consumption.

The first scenario starts with a baseline of the average data center load in 2023, which equates to a little more than 150,000,000 megawatt-hours (MWh). In the highest growth rate scenario, EPRI projected an average data center load of more than 400,000,000 MWh by 2030, a staggering 166% change in growth. Conversely, the lowest growth rate scenario has a projected average data center load of less than 200,000,000 MWh by 2030.

While the higher growth rate scenario is intimidating, these are just projections, and much can change in the next decade. Some factors are out of our control, like consumer demand for AI technologies, but others can be controlled.

EPRI guidelines to mitigate AI's negative impact

EPRI offers a few areas to focus on for data centers to curb their rising energy usage, keep load levels toward the lower end of the projected growth rate scenarios and mitigate the [environmental impacts of AI workloads](#).

A comprehensive strategy is necessary to meet rising energy demands and limit emissions growth. Strategies include investing in more energy-efficient processors and server architectures, leaning on [virtualization](#) to improve resource flexibility, adopting more effective cooling technologies, and using continuous monitoring and analytics to ensure optimal operational efficiency and better adaptability.

Collaboration through a shared energy economy model

Electric companies must balance resources between regular customers and data centers with accelerating and unpredictable load growth. To better handle this situation, data centers can collaborate more closely with electric companies to create a [shared energy economy](#). For example, electric companies can utilize data center backup generators as a grid reliability resource, offering a more symbiotic relationship.

Load growth forecasting and modeling

With more accurate forecasting and modeling tools, data centers and electric companies can better collaborate to anticipate interconnection requests. This can help electric companies understand the full power demand that data centers require over time. By doing so, they can avoid stressing the energy grid and introduce flexibility into operational bandwidth and resource management.

Upgrades to the data center

To address the increasing demands of AI in data centers, administrators should consider adopting more flexible computational strategies and efficient server management tools. It's essential to utilize advanced computational hardware, such as tensor processing units, field-programmable gate arrays and GPUs.

Admins should implement resource-efficient algorithmic techniques, like pruning and quantization. It is also crucial to transition to carbon-free and low-carbon electricity sources and adopt cleaner power systems.

Achieving the Net Zero Emissions by 2050 Scenario is still possible despite the massive energy demand to fuel AI workloads. However, the path ahead is narrow and will require global cooperation to ensure data centers can limit energy usage

landscape.

Jacob Roundy is a freelance writer and editor with more than a decade of experience with specializing in a variety of technology topics, such as data centers, business intelligence, AI/ML, climate change and sustainability. His writing focuses on demystifying tech, tracking trends in the industry, and providing practical guidance to IT leaders and administrators.

Next Steps

[6 sustainable resources to power data centers](#)

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AI

What we can VERIFY about AI and its environmental impact

The boom in AI has led to a boom in AI data centers. Several readers asked us to VERIFY how these hubs use water and electricity.

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Author: Emery Winter
Published: 2:37 PM EST January 17, 2025
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The use and prevalence of generative artificial intelligence (AI) technology has ballooned over the past few years. This includes the growth of chatbots like ChatGPT and image generators like Midjourney.

As AI has become ubiquitous, people have raised concerns about the environmental impacts of the technology. One of the more common criticisms is that it requires more water and power than older technology.

Some people have drawn links between this resource use, climate change and the wildfires in Los Angeles. A viral post from Instagram that has since been [reshared many times](#) claimed that a single interaction with ChatGPT uses 10 times the amount of energy as a Google search.

Readers Olive and Dean also asked us to VERIFY the impact artificial intelligence has on water and power usage.

THE SOURCES

- [International Energy Agency \(IEA\)](#)
- [University of Illinois Urbana-Champaign's Center for Secure Water](#)
- [Article on AI energy crisis published by Nature](#)
- [Google's 2024 Environmental Report](#)
- [2023 study by engineering researchers from the University of California, Riverside](#)
- [2023 study by researcher with Digiconomist](#), a research company focused on unintended

consequences of digital trends

- [Sunbird](#), a company that makes data center management software



WHAT WE FOUND

Our current online world relies on vast amounts of computers and data centers to operate. These centers power everything we do online, from conducting internet searches to streaming movies.

Artificial intelligence is more sophisticated than a regular web search or movie stream. It requires exponentially more computing power to complete what may seem like simple tasks.

The AI boom has thus led to a rise in new data centers, too. These new data centers that support the additional computing power required are the source of AI's outsized environmental impact.

How AI uses electricity

An AI tool like ChatGPT relies on large amounts of data and equally large amounts of computer processing power to provide a result. Tech companies keep computer systems to store this data and run programs to process it in physical locations called [data centers](#).

When someone gives an AI program a prompt, it uses computational power to sift through and process all of that data, Katherine Bourzac, a science writer for Nature journals, [wrote in a 2024 article](#). The more computational power used, the more electricity is needed.

How AI uses water

It's not just electricity data centers need more of when they use more computational power; they also need more water, according to the [University of Illinois Urbana-Champaign's Center for Secure Water](#).

The more power a computer uses, the more heat it generates. If a computer gets too hot, it'll start running into problems. That's why laptops and personal computers have fans inside of them that spin faster when the computer works harder.

Many data centers use industrial-sized fans to do the same thing on a large scale. However, traditional air cooling isn't always enough to dissipate the amount of heat generated by all of the computer power AI uses, according to [Sunbird](#), a data center management software company. So AI data centers use liquid coolants, which absorb and transfer heat better than air does.

When data centers use water as their liquid coolant, the water is pumped through pipes surrounding the center's equipment, where it absorbs excess heat and is then typically pumped back out through a [heat exchanger to a coolant tower](#), where the water evaporates. That means these data centers need a constant source of water to run through their systems.

AI resource usage by the numbers

The average electricity demand of a typical Google search without AI is 0.3 Wh (watt-hours) of electricity, while the average electricity demand of a ChatGPT request is 2.9 Wh, according to the [International Energy Agency \(IEA\)](#).

In 2023, John Hennessy, chairman of Google parent company Alphabet, [told Reuters](#) that he predicted an exchange with an AI chatbot would likely be 10 times more energy intensive than a standard Google search without AI.

While 0.3 to 2.9 Wh might not seem like much ([a toaster typically uses 10 to 160 Wh per use](#)), those numbers add up. In 2021, before Google began integrating AI overviews into its search engine, Google consumed more than 18 trillion watt-hours of electricity, according to [a study by a researcher with Digiconomist](#). At that time, AI accounted for 10-15% of the total electricity Google used.

Various estimates within that study estimated that Google search integrated with AI could use between 6.9 and 8.9 Wh per search. Google didn't include the total amount of electricity it consumed in its [most recent environmental report](#), but Google did say that in 2023 it released 37% more emissions from using electricity than it did in 2022.

Google said the increase in emissions was primarily because its increasing demand for electricity for its data centers outpaced its ability to bring more carbon-free energy projects online.

In its most recent environmental report, Google reported it consumed 14% more water in 2023 than it did in 2022. This is "primarily due to water cooling needs" at Google's data centers, "which experienced increased electricity consumption year-over-year."

The exact amount of water used to cool the machines in a data center can depend on the data center's design and location; data centers in hotter locations need more water for cooling.

On average, data centers can consume approximately 1-9 liters of water per kWh of server energy, according to an [estimate from engineering researchers at the University of California, Riverside](#).

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Inside the Guian Data Center of China Unicom, which uses artificial intelligence in its operations. TAO LIANG / XINHUA VIA GETTY IMAGES

As Use of A.I. Soars, So Does the Energy and Water It Requires

Generative artificial intelligence uses massive amounts of energy for computation and data storage and millions of gallons of water to cool the equipment at data centers. Now, legislators and regulators – in the U.S. and the EU – are starting to demand accountability.

BY DAVID BERREBY • FEBRUARY 6, 2024

Two months after its release in November 2022, OpenAI’s ChatGPT had 100 million active users, and suddenly tech corporations were racing to offer the public more “generative A.I.” Pundits compared the new technology’s impact to the Internet, or electrification, or the Industrial Revolution – or the discovery of fire.

Time will sort hype from reality, but one consequence of the explosion of artificial intelligence is clear: this technology’s environmental footprint is large and growing.

A.I. use is directly responsible for carbon emissions from non-renewable electricity and for the consumption of millions of gallons of fresh water, and it indirectly boosts impacts from building and maintaining the power-hungry equipment on which A.I. runs. As tech companies seek to embed high-intensity A.I. into everything from resume-writing to kidney transplant medicine and from choosing dog food to climate modeling, they cite many ways A.I. could help *reduce* humanity’s environmental footprint. But legislators, regulators, activists, and

international organizations now want to make sure the benefits aren't outweighed by A.I.'s mounting hazards.

Right now, it's not possible to tell how your A.I. request for homework help will affect carbon emissions or freshwater stocks.

"The development of the next generation of A.I. tools cannot come at the expense of the health of our planet," Massachusetts Senator Edward Markey (D) said last week in Washington, after he and other senators and representatives introduced a bill that would require the federal government to assess A.I.'s current environmental footprint and develop a standardized system for reporting future impacts. Similarly, the European Union's "A.I. Act," approved by member states last week, will require "high-risk A.I. systems" (which include the powerful "foundation models" that power ChatGPT and similar A.I.s) to report their energy consumption, resource use, and other impacts throughout their systems' lifecycle. The EU law takes effect next year.

Meanwhile, the International Organization for Standardization, a global network that develops standards for manufacturers, regulators, and others, says it will issue criteria for "sustainable A.I." later this year. Those will include standards for measuring energy efficiency, raw material use, transportation, and water consumption, as well as practices for reducing A.I. impacts throughout its life cycle, from the process of mining materials and making computer components to the electricity consumed by its calculations. The ISO wants to enable A.I. users to make informed decisions about their A.I. consumption.

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An Amazon data center in a Northern Virginia suburb. JAHl CHIKWENDIU / THE WASHINGTON POST VIA GETTY IMAGES

Right now, it's not possible to tell how your A.I. request for homework help or a picture of an astronaut riding a horse will affect carbon emissions or freshwater stocks. This is why 2024's crop of "sustainable A.I." proposals describe ways to get more information about A.I. impacts.

In the absence of standards and regulations, tech companies have been reporting whatever they choose, however they choose, about their A.I. impact, says Shaolei Ren, an associate professor of electrical and computer engineering at UC Riverside, who has been studying the water costs of computation for the past decade. Working from calculations of annual use of water for cooling systems by Microsoft, Ren estimates that a person who engages in a session of questions and answers with GPT-3 (roughly 10 to 50 responses) drives the consumption of a half-liter of fresh water. "It will vary by region, and with a bigger A.I., it could be more." But a great deal remains unrevealed about the millions of gallons of water used to cool computers running A.I., he says.

The same is true of carbon.

"Data scientists today do not have easy or reliable access to measurements of [greenhouse gas impacts from A.I.], which precludes development of actionable tactics," a group of 10 prominent researchers on A.I. impacts wrote in a 2022 conference [paper](#). Since they presented their article, A.I. applications and users have proliferated, but the public is still in the dark about those data, says Jesse Dodge, a research scientist at the Allen Institute for Artificial Intelligence in Seattle, who was one of the paper's coauthors.

Data centers' electricity consumption in 2026 is projected to reach 1,000 terawatts, roughly Japan's total consumption.

A.I. can run on many devices – the simple A.I. that autocorrects text messages will run on a smartphone. But the kind of A.I. people most want to use is too big for most personal devices, Dodge says. "The models that are able to write a poem for you, or draft an email, those are very large," he says. "Size is vital for them to have those capabilities."

Big A.I.s need to run immense numbers of calculations very quickly, usually on specialized Graphical Processing Units – processors originally designed for intense computation to render graphics on computer screens. Compared to other chips,

GPUs are more energy-efficient for A.I., and they're most efficient when they're run in large "cloud data centers" – specialized buildings full of computers equipped with those chips. The larger the data center, the more energy efficient it can be. Improvements in A.I.'s energy efficiency in recent years are partly due to the construction of more "hyperscale data centers," which contain many more computers and can quickly scale up. Where a typical cloud data center occupies about 100,000 square feet, a hyperscale center can be 1 or even 2 million square feet.

Estimates of the number of cloud data centers worldwide range from around 9,000 to nearly 11,000. More are under construction. The International Energy Agency (IEA) projects that data centers' electricity consumption in 2026 will be double that of 2022 – 1,000 terawatts, roughly equivalent to Japan's current total consumption.



A QTS data center under construction in Litchfield Park, Arizona last month. ASH PONDER / BLOOMBERG VIA GETTY IMAGES

However, as an illustration of one problem with the way A.I. impacts are measured, that IEA estimate includes *all* data center activity, which extends beyond A.I. to many aspects of modern life. Running Amazon's store interface, serving up Apple TV's videos, storing millions of people's emails on Gmail, and "mining" Bitcoin are also performed by data centers. (Other IEA reports exclude crypto operations, but still lump all other data-center activity together.)

Most tech firms that run data centers don't reveal what percentage of their energy use processes A.I. The exception is Google, which says "machine learning" – the basis for humanlike A.I. – accounts for somewhat less than 15 percent of its data centers' energy use.

Another complication is the fact that A.I., unlike Bitcoin mining or



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Bitcoin's intensive energy demands are sparking a crypto backlash. [Read more.](#)

online shopping, can be used to *reduce* humanity's impacts. A.I. can improve climate models, find more efficient ways to make digital tech, reduce waste in transport, and otherwise cut carbon and water use. One estimate, for example, found that A.I. -run smart homes could reduce households' CO₂ consumption by up to 40 percent. And a recent Google project found that an A.I. fast-crunching atmospheric data can guide airline pilots to flight paths that will leave the fewest contrails.

Google's data centers used 20 percent more water in 2022 than in 2021, while Microsoft's water use rose by 34 percent.

Because contrails create more than a third of commercial aviation's contribution to global warming, "if the whole aviation industry took advantage of this single A.I. breakthrough," says Dave Patterson, a computer-science professor emeritus at UC Berkeley and a Google researcher, "this single discovery would save more CO₂e (CO₂ and other greenhouse gases) than the CO₂e from all A.I. in 2020."

Patterson's analysis predicts that A.I.'s carbon footprint will soon plateau and then begin to shrink, thanks to improvements in the efficiency with which A.I. software and hardware use energy. One reflection of that efficiency improvement: as A.I. usage has increased since 2019, its percentage of Google data-center energy use has held at less than 15 percent. And while global internet traffic has increased more than twentyfold since 2010, the share of the world's electricity used by data centers and networks increased far less, according to the IEA.

However, data about improving efficiency doesn't convince some skeptics, who cite a social phenomenon called "Jevons paradox": Making a resource less costly sometimes *increases* its consumption in the long run. "It's a rebound effect," Ren says. "You make the freeway wider, people use less fuel because traffic moves faster, but then you get more cars coming in. You get more fuel consumption than before." If home heating is 40 percent more efficient due to A.I., one critic recently wrote, people could end up keeping their homes warmer for more hours of the day.





Ai-Da Robot, an AI-powered robot artist, addressing the British House of Lords, October 11, 2022. ROB PINNEY / GETTY IMAGES

“A.I. is an accelerant for everything,” Dodge says. “It makes whatever you’re developing go faster.” At the Allen Institute, A.I. has helped develop better programs to model the climate, track endangered species, and curb overfishing, he says. But globally A.I. could also support “a lot of applications that could accelerate climate change. This is where you get into ethical questions about what kind of A.I. you want.”

If global electricity use can feel a bit abstract, data centers’ water use is a more local and tangible issue – particularly in drought-afflicted areas. To cool delicate electronics in the clean interiors of the data centers, water has to be free of bacteria and impurities that could gunk up the works. In other words, data centers often compete “for the same water people drink, cook, and wash with,” says Ren.

In 2022, Ren says, Google’s data centers consumed about 5 billion gallons (nearly 20 billion liters) of fresh water for cooling. (“Consumptive use” does not include water that’s run through a building and then returned to its source.) According to a recent study by Ren, Google’s data centers used 20 percent more water in 2022 than they did in 2021, and Microsoft’s water use rose by 34 percent in the same period. (Google data centers host its Bard chatbot and other generative A.I.s; Microsoft servers host ChatGPT as well as its bigger siblings GPT-3 and GPT-4. All three are produced by OpenAI, in which Microsoft is a large investor.)

In Chile and Uruguay, protests have erupted over planned data centers that would tap drinking water reservoirs.

As more data centers are built or expanded, their neighbors have been troubled to find out how much water they take. For example, in The Dalles, Oregon, where Google runs three data centers and plans two more, the city government filed a lawsuit in 2022 to keep Google’s water use a secret from farmers, environmentalists, and Native American tribes who were concerned about its effects on agriculture and on the region’s animals and plants. The city withdrew its suit early last year. The records it then made public showed that Google’s three extant data centers use

more than a quarter of the city’s water supply. And in Chile and Uruguay, protests have erupted over planned Google data centers that would tap into the same reservoirs that supply drinking water.

Most of all, researchers say, what’s needed is a change of culture within the rarefied world of A.I. development. Generative A.I.’s creators need to focus beyond the technical leaps and bounds of their newest creations and be less guarded about the details of the data, software, and hardware they use to create it.

Some day in the future, Dodge says, an A.I. might be able – or be legally obligated – to inform a user about the water and carbon impact of each distinct request she makes. “That would be a fantastic tool that would help the environment,” he says. For now, though, individual users don’t have much information or power to know their A.I. footprint, much less make decisions about it.

“There’s not much individuals can do, unfortunately,” Ren says. Right now, you can “try to use the service judiciously,” he says.



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Correction, February 21, 2024: *An earlier version of this article incorrectly quoted researcher Dave Patterson as referring to CO₂ emissions from global aviation. Patterson was actually referring to CO₂e (“carbon dioxide equivalent”) emissions, a measurement that includes both CO₂ and other greenhouse gases.*



David Berreby writes the *Robots for the Rest of Us* newsletter. His work about AI and robotics has appeared in *The New York Times*, *National Geographic*, *Slate*, and other publications. His other science writing includes *Us and Them: The Science of Identity* and work in *The New Yorker*, *Nature*, and many other publications. [MORE →](#)